Zenodo theorem-object release

Forty-Two

Exhaustion, Closure, and Downstream Consequence

Amos Jay Maley

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Core theorem spine. Sections 1–8 are the local proof home for the manuscript's load-bearing theorem chain.

Downstream posture. Physics-facing consequence layers and the appendix atlas remain consequence-facing only and are removable without altering the foundational arc.

Proposition-bearing, governance-assessable, non-degenerate reasoning

Formalization claims are limited to theorem-level, theorem-family, and manuscript-audit surfaces exactly as documented.

Abstract

This manuscript now functions as the local proof home for the core theorem spine. Sections 1–8 carry the load-bearing chain in-manuscript: the non-derivable kernel of admissibility, standing, reference, and irreversibility; canonical state-transition form with identity persistence, irreversible commitment, and fail-closed bivalence; the AMetric boundary and equality-pattern collapse; standing as reuse-stable admissibility together with no repair, no generators, no carriers, quotient uniqueness, and the unique admissible interior; closure of mathematics by same-scope operator exhaustion; and explicit defeat of same-domain meta-escape, including the non-instantiability of Gödel-style closure threats inside the unique admissible interior. Between the interface and operator layers, the manuscript also makes explicit the fixed-domain closure claim that no alternative same-domain admissibility architecture survives except by illicit structure import, collapse of the base goods, or extensional identity with the unique gate.

Downstream consequence layers are preserved but not allowed to function as hidden proof sources for that spine. Physics closure, two-spectrum synthesis, and the Zenodo appendix atlas are therefore consequence-facing and classificatory under the fixed theorem chain of the main text rather than additional foundational engines. The downstream layer places four consequence manuscripts without theorem backflow: Navier–Stokes and Yang–Mills appear as theorem roadmaps whose claim-class discipline, endpoint hygiene, denial maps, and non-claims are preserved rather than re-authored here; measurement and Standard Model manuscripts appear as compact consequence profiles whose fixed-domain hardening is carried over only at theorem level. A Zenodo-only appendix atlas then extends the consequence layer across measurement-record fixation, Bell-scope entanglement, gauge/constraint rigidity payoffs, admissible interior bounds, baryogenesis, black-hole fixation, standing localization, mind, meta-necessity, and dark-sector elimination. None of those downstream appendix manuscripts functions as a proof source for Sections 1–9, and each may be removed without altering the foundational arc. The appendices therefore expose a controlled consequence atlas together with the dependency register, formalization boundary, attack map, shortest reading path, and final source ledger.

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1 Introduction and scope

This manuscript is meant to be read as one forced theorem chain. Its burden is not to advertise a family resemblance among companion papers, but to state in one place and in strict order how the following claims hang together: the target class of reasoning is forced rather than protectively narrowed; the kernel of admissibility, standing, reference, and irreversibility is non-derivable from below; coherent reasoning therefore acquires a canonical state-transition form; the pre-fixational side is AMetric; the same-scope standing regime has one gate, one standing quotient, and one admissible interior; mathematics and physics are both closed under the same non-generative invariant; same-domain code, semantic, infinitary, and extension escape is exhausted; and the post-closure relation between mathematics and physics is a two-spectrum classification of admissible description rather than a grounding hierarchy.

The manuscript is therefore not a survey and not a mere corpus index. For the core theorem spine carried in Sections 2–8, the present manuscript is the proof home. Companion manuscripts remain relevant only where later sections explicitly keep downstream theorem families imported or consequence-facing. Where a theorem family is imported, the import is typed as downstream support only and is not permitted to backflow into the foundational arc.

For hostile reading, the source posture is simple. Sections 2–8 now carry the core theorem spine locally. Later imported material is downstream only: theorem roadmaps, consequence profiles, or synthesis layers whose removal would not alter the foundational chain. The full source-by-source routing, proof-home allocation, and import boundaries are recorded in the dependency register and final source freeze rather than expanded here in the introduction.

A second editorial rule governs every downstream mention in the capstone. Whenever a source manuscript is cited as a theorem roadmap or consequence profile, the capstone preserves the source-level theorem object, claim class, endpoint hygiene, denial map, and non-claims rather than rewriting them into generic capstone summary language. This matters most for Navier–Stokes and Yang–Mills, where the source manuscripts already state the exact theorem object and method-neutral claim class, and for measurement, where the source manuscript already fixes the target discourse, quotient theorem, and denial-condition matrix.

The theorem arc is strict:

1. kernel necessity, derivational presupposition, non-derivability, and minimality;
2. canonical state-transition form, identity persistence, irreversible commitment, and governance-level bivalence;
3. pre-fixational non-reference, relabeling indifference, equality-pattern completeness, AMetric collapse, and no deeper invariant;
4. gate existence, standing emergence, standing–admissibility equivalence, quotient / factorization closure against same-scope discriminators, no repairs / no generators / no carriers, and the unique admissible interior;
5. partial-algebra envelope discipline, no internal totalization, conservative extension collapse, finite

exhaustion schema, and mathematics closure;

6. licensed metric description, physical non-generativity, and the descriptive role discipline needed for the physical spectrum;
7. inferential-life exhaustion, closure-relevance transmission, code-role exhaustion, no admissible same-domain diagonalization, rescue-route exhaustion, and same-domain Gödel non-instantiability;
8. fixation split, two-spectrum exhaustion, closure parity, mirror classification, the prohibition package, post-closure practice, and descriptive closure;
9. downstream consequence placement for Navier–Stokes, Yang–Mills, measurement, and Standard Model manuscripts; and
10. formalization boundary, axiom-necessity layer, dependency register, hostile-reader denial map, shortest reading path, and source ledger.

In particular, the capstone is not asking the reader to choose among rival interface designs. Section 2 fixes the target class and kernel floor as theorem-level boundaries of proposition-bearing reasoning, and Sections 4–5 then show that, once one remains on the same fixed domain, no alternative admissibility architecture survives except by illicit import, same-side gate multiplication, repair / generator / carrier laundering, scope transport, or extensional collapse back to the unique gate. A hostile reader must therefore deny either the target phenomenon itself or a named theorem in the closure chain, not an allegedly open space of unexamined design options.

Two restrictions govern the whole paper.

1. **No theorem-class drift.** Later sections are not allowed to sound like standard PDE, model-theoretic, or metaphysical papers unless their claim class really is PDE, model-theoretic, or metaphysical. The capstone is a closure theorem and keeps that identity all the way through.
2. **No hidden proof-home inflation.** Companion proof homes may justify compact imported families, but they do not become silent premises merely because they exist. If a burden is not stated as imported, it is not live.

The manuscript closes with two controlled downstream consequence placements. First, it states how the periodic Navier–Stokes and local sharp-local Yang–Mills manuscripts should be read: not as proof sources of the foundational arc, but as theorem roadmaps whose own claim-class discipline has already been hardened at source level. Second, it states how the measurement-reference and Standard Model manuscripts should be read: as compact consequence profiles clarifying what the closure theorem does to two very different application domains without changing the theorem class of the capstone itself.

What the paper is not. It is not a construction diary or a public build log. It is not a list of editorial scaffolding. It is not a claim that admissibility itself manufactures analytic estimates, physical dynamics, or metatheoretic encoding power. Every later section is constrained by that restriction.

2 Kernel necessity, non-derivability, and the mechanization boundary

2.1 Why the capstone begins here

The later sections of the capstone depend on a prior fact: the target class is not the class of arbitrary calculi, encodings, or transition systems, but the class of proposition-bearing, governance-assessable reasoning. Within that class, admissibility, standing, reference, and irreversibility are not optional descriptive conveniences. They are the minimal governance conditions under which reasoning remains assessable as reasoning at all. The capstone therefore begins by fixing that floor and by showing that it is forced rather than protectively chosen.

2.2 Governance-level setting

A regime is *meaningful for the present target* iff its outputs are assessable as proposition-bearing reasoning acts rather than as bare symbol traces, encodings, or bookkeeping artifacts. A derivation is *governance-valid* only when it is itself an admissible, standing-bearing, reference-preserving, irreversibility-governed claim-sequence. References below to "the same meaningful regime" are preservational rather than merely lexical: same regime means preservation of governed target, standing-bearing act-status, the admissibility boundary between licensed continuation and prohibited repair, and therefore the same class of governance-valid derivations up to lawful equivalence. The non-derivability results below are about that target and that preservational sense of sameness.[2, 4]

Definition 2.1 (Governance-valid derivation)**.** A derivation of a governance-level proposition is *governance-valid* iff the derivation is itself an admissible act-sequence: each step is reference-preserving, standing-bearing, and governed by irreversibility with respect to invalidity. In particular, a governance-valid derivation is never merely a syntactic string of transitions; it is a licit reasoning act.

Definition 2.2 (Kernel floor)**.** Write

$$K := \{\mathrm{Adm}, \mathrm{St}, \mathrm{Ref}, \mathrm{Irr}\},$$

where Adm abbreviates admissibility, St standing, Ref determinate reference, and Irr irreversibility. The capstone uses *kernel floor* to mean that no faithful lower generator of this governance work exists within the same meaningful regime.

2.3 Minimal assessability boundary

Theorem 2.3 (Minimal assessability boundary)**.** *Let R be any system whose outputs are treated as propositions and whose transitions are evaluated as reasoning steps. Then determinate reference, claim-bearing standing, and non-repairable invalidity are jointly necessary and minimal for that evaluability. More exactly:*

1. *if determinate reference is withdrawn, proposition identity is lost and the outputs are no longer evaluable as determinate proposition-bearing content;*

2. *if standing is withdrawn, no transition qualifies as a claim-bearing step and the outputs are no longer evaluable as reasoning steps;*
3. *if irreversibility is withdrawn, there is no stable distinction between valid continuation and retrospective rescue, and the outputs are no longer evaluable as derivations rather than retrospective normalizations; and*
4. *if an additional governance condition is imposed independently beyond those three, then the regime is governed more tightly than evaluability itself requires.*

Accordingly, these three conditions mark the minimal assessability boundary for proposition-bearing reasoning as such, and admissibility is the governance boundary under which their joint enforceability is secured on the same act.

Proof. If determinate reference is removed, then no fixed proposition remains for inferential assessment. The same inscription can be read as bearing multiple incompatible contents, or no determinate content at all. What remains is a mark sequence, not a proposition-bearing act. So proposition-bearing assessment fails when reference fails.

If standing is removed, then no transition counts as a licit claim-step. One may still have inscription, computation, or transformation, but not a reasoning step whose inferential role is assessable as a claim-bearing act. So reasoning-step assessment fails when standing fails.

If irreversibility is removed, then invalidity is no longer fixed at act-time. A step may later be retrospectively repaired, redescribed, or normalized into apparent legitimacy without loss. In that case the distinction between licensed continuation and retrospective rescue is not stable, and what is being tracked is no longer derivation in the present sense. So derivational assessment fails when irreversibility fails.

For minimality, suppose some further governance condition Q is imposed independently on every proposition-bearing reasoning act. If Q is already enforced by the work done jointly by reference, standing, and irreversibility, then it is not genuinely additional. If instead Q excludes some acts that already satisfy those three conditions, then Q contributes governance beyond what is required for proposition-bearing assessability as such. Hence reference, standing, and irreversibility are jointly necessary and minimal for evaluability as proposition-bearing reasoning. Once those three are jointly enforceable on the same act, there is already a governance boundary distinguishing licensed from non-licensed continuation and distinguishing continuation from repair. That governance-boundary status is exactly what is here called admissibility.[2] □

Remark 2.4 (Why the restriction is not protective). The capstone does not exclude arbitrary calculi because they would threaten a preferred conclusion. It excludes them because systems that lack stable propositional identity, claim-bearing step-status, or a principled distinction between continuation and repair do not instantiate the target phenomenon in the first place. They may still implement search, update, transformation, or computation, but they are not counterexamples to a theorem about proposition-bearing reasoning.[2]

2.4 Kernel forcing and derivational presupposition

Theorem 2.5 (Kernel forcing)**.** *If a meaningful regime admits non-degenerate reasoning, then the full kernel K is already in force there. Equivalently, no meaningful regime supports non-degenerate reasoning while lacking admissibility, standing, reference, irreversibility, or some lawfully equivalent replacement for one of them.*

Proof. Let R be a meaningful regime admitting non-degenerate reasoning. By Theorem 2.3, any regime whose outputs are assessable as proposition-bearing reasoning already requires determinate reference, claim-bearing standing, and irreversibility as the minimal assessability boundary of that evaluability. Once those three conditions are jointly enforceable on the same acts, admissibility is already in force as the governance boundary of that joint enforceability. Hence R already instantiates Ref, St, Irr, and Adm. So the full kernel K is already in force wherever the target phenomenon exists.[2] □

Proposition 2.6 (Derivation presupposes the kernel)**.** *Every governance-valid derivation already presupposes admissibility, standing, reference, and irreversibility. Derivation therefore reflects the kernel locally; it does not generate the kernel from beneath itself.*

Proof. Let $\mathrm{Der}(P)$ be a governance-valid derivation of a proposition P.

First, such a derivation already presupposes admissibility, because a governance-valid derivation is a licit act-sequence rather than a merely syntactic trail. Without admissibility, the sequence lacks a principled governance boundary distinguishing licensed continuation from inadmissible transition.

Second, it already presupposes standing, because without standing the sequence contains no claim-bearing steps. It may still be a chain of inscriptions or transitions, but it does not count as derivation.

Third, it already presupposes reference, because without determinate reference the sequence does not derive determinate content. It manipulates marks without yielding a fixed proposition.

Fourth, it already presupposes irreversibility, because without irreversibility invalidity can be repaired retrospectively and the distinction between valid continuation and repair collapses. The derivation would therefore lack stable governance status at act-time.

So every governance-valid derivation presupposes the full kernel pointwise. It can display the kernel locally inside derivation, but it cannot first produce that kernel from a lower same-regime basis.[2] □

2.5 Non-derivability, mutual closure, and minimality

Theorem 2.7 (Kernel non-derivability)**.** *No member of K is derivable from below within the same meaningful regime.*

Proof. Take the four kernel members one by one.

Suppose admissibility were derivable from below. Then there would exist a governance-valid derivation of admissibility whose validity did not already presuppose admissibility or a lawfully equivalent condition. But by Proposition 2.6, any governance-valid derivation already presupposes admissibility. So the derivation uses what it claims to produce. That is not derivation from below.

Suppose standing were derivable from below. Then there would exist a governance-valid derivation of standing whose validity did not already presuppose standing. But by Proposition 2.6, any governance-valid derivation already presupposes standing. Again the alleged derivation begins inside the very status it claims to generate.

Suppose reference were derivable from below. Then there would exist a governance-valid derivation of reference whose validity did not already presuppose determinate reference. But by Proposition 2.6, governance-valid derivation already presupposes determinate reference. So the purported derivation fails to come from below.

Suppose irreversibility were derivable from below. Then there would exist a governance-valid derivation of irreversibility whose validity did not already presuppose irreversibility. But by Proposition 2.6, the governance-validity of derivation already relies on the stable distinction between valid continuation and invalid repair. So the derivation again presupposes the condition it purports to derive.

Hence no member of K is derivable from below.[2] □

Corollary 2.8 (No faithful lower generator)**.** *There exists no faithful same-regime proof system, meta-system, governance package, or lower architecture that generates the kernel from below while preserving the same meaningful regime.*

Proof. Assume, for contradiction, that some system or package G generates a member $P \in K$ from a genuinely lower governance basis while preserving the same meaningful regime. By Theorem 2.5, any meaningful regime that supports non-degenerate reasoning already has the full kernel in force before the alleged generation begins. So G does not produce P from below; it operates only inside a regime whose meaningfulness already depends on P. That defeats the claim of faithful lower generation.

Nor does extension help. A faithful same-domain or regime-preserving extension that claims to derive admissibility, standing, reference, or irreversibility for the original acts on the original fixed domain can do so only in one of four ways: by importing fresh same-domain authority, by becoming trivial, by violating the base goods, or by collapsing extensionally back to the gate or kernel work already in force. None of those is a faithful lower derivation. Thus there is no faithful lower generator either internally or by same-domain extension.[2, 4] □

Theorem 2.9 (The kernel floor is derived rather than axiomatic)**.** *In the present capstone the primitive governance terms are not introduced as arbitrary axioms. Their governance work is mathematically forced by the minimal assessability boundary, kernel forcing, and kernel non-derivability. Accordingly, falsifying the kernel floor requires more than proposing an alternative vocabulary or proof format: it requires a faithful lower derivation of admissibility, standing, reference, and irreversibility from below within the same meaningful regime.*

Proof. Theorem 2.3 shows that reference, standing, and irreversibility are the minimal assessability boundary of proposition-bearing reasoning. The same theorem also identifies admissibility as the governance boundary of their joint enforceability. Theorem 2.5 then shows that any meaningful regime admitting non-degenerate reasoning already instantiates that full governance package. Theorem 2.7 and Corollary 2.8 show that no same-regime lower generator exists for any kernel member and that changing proof format, meta-level, or same-domain formal dress does not remove that obstruction.

So the capstone does not posit the kernel as an arbitrary axiom cluster. It records a forced floor. To dislodge that floor, one would have to exhibit exactly what these theorems rule out: a faithful lower derivation of the kernel inside the same meaningful regime. Alternative notation, alternate packaging, or a change of formal clothing does not meet that burden. □

Theorem 2.10 (Architecture exhaustion at the kernel floor)**.** *Within the target class fixed by the minimal assessability boundary, no alternative admissibility architecture survives as a genuinely distinct same-regime basis for proposition-bearing reasoning. More precisely, any purported counterarchitecture does exactly one of the following:*

1. *it changes the target phenomenon by abandoning proposition-bearing assessability, stable reference, claim-bearing standing, or irreversibility;*
2. *it imports, hides, redistributes, or launders the same governance work under another name, level, or same-domain authority source; or*
3. *it collapses extensionally to the same kernel package already fixed above.*

There is no fourth same-regime architecture.

Proof. Let A be a purported alternative same-regime basis for proposition-bearing reasoning.

If A omits any member of K, then Theorem 2.3 applies immediately. If the omitted governance work is genuinely absent, then proposition-bearing assessability fails: one loses determinate reference, claim-bearing standing, or irreversibility, and the target phenomenon changes. If instead the omitted work reappears elsewhere in A, then the omission was only apparent: the same governance burden has been redistributed, renamed, or deferred. So omission yields either phenomenon-change or hidden re-expression of the same governance work.

Now suppose A retains the full governance work of admissibility, standing, reference, and irreversibility but claims to organize it through a genuinely lower or distinct same-regime basis. Then Theorem 2.7 and Corollary 2.8 apply. A same-regime lower basis is impossible because the kernel is already in force wherever the target phenomenon exists. If A tries to rescue the claim by same-domain extension, that route is also closed: the extension either imports new same-domain authority, becomes trivial, violates the base goods, or collapses extensionally back to the original gate or kernel work. None of those yields a genuinely distinct same-regime architecture.

These cases are exhaustive. A purported alternative basis either changes the phenomenon, hides the same governance work under another presentation, or coincides extensionally with the already fixed kernel floor. No fourth same-regime architecture remains.[2, 4] □

Inline proof anchor I: kernel-floor exhaustion

The inlined forcing chain carried by this section has five steps.

1. *Assessability floor.* If determinate reference is withdrawn, no determinate proposition remains; if standing is withdrawn, no step counts as a licit claim-bearing step; and if irreversibility is withdrawn, the distinction between licensed continuation and retrospective rescue is no longer fixed at act-time. These are therefore not optional enrichments of reasoning but the weakest conditions under which proposition-bearing reasoning is assessable as reasoning at all.

2. *Admissibility as joint-enforceability boundary.* Once reference, standing, and irreversibility are jointly enforceable on the same act, there is already a governance boundary distinguishing licensed continuation from prohibited repair. That boundary is exactly admissibility. Admissibility is therefore not an additional fifth ingredient appended after the fact; it is the governance form of the joint enforceability of the other three goods.

3. *Kernel forcing.* Any meaningful regime that genuinely instantiates non-degenerate reasoning already carries $\{\mathrm{Adm}, \mathrm{St}, \mathrm{Ref}, \mathrm{Irr}\}$, or a lawfully equivalent package, before any internal derivation begins. In that sense the kernel is regime-level prior to derivation rather than a theorem derived inside derivation.

4. *Derivational shadow.* Governance-valid derivation cannot sit beneath the kernel, because the validity of such derivation already presupposes admissibility, standing, reference, and irreversibility pointwise. So an alleged lower derivation of the kernel begins inside the very governance structure it claims to generate.

5. *Architecture trichotomy.* Let A be a purported same-regime alternative basis for proposition-bearing reasoning. If A omits some member of K, then either the target phenomenon changes or the omitted governance work is merely recovered elsewhere under another name or level. If A retains the full governance work while claiming to be a genuinely lower basis, kernel forcing and non-derivability immediately collapse that claim: A is at most extensionally identical with the fixed kernel package.

Hence the kernel is not a presentational choice-point. Inside the target class, every same-regime alternative either changes the phenomenon, hides the same governance work under another presentation, or collapses extensionally to the kernel floor. That is the exact proof burden now carried directly by Theorem 2.10.[2, 4]

The restriction to meaningful regimes is not a protective narrowing but an identification of the maximal domain in which proposition-bearing reasoning is assessable as reasoning for the present target; systems outside that domain do not instantiate the target phenomenon at all.

Theorem 2.11 (Scope exhaustion for non-degenerate reasoning)**.** *Let $\mathcal{R}$ be the class of meaningful regimes for the present target, namely those in which inferential outputs are assessable as proposition-bearing reasoning acts under admissibility-invariant use.*

Any purported counterexample to the uniqueness and forced-consequence results of this manuscript must fall into exactly one of the following classes:

1. ***Same-regime counterexample.*** *The system lies within $\mathcal{R}$ and satisfies non-degenerate reasoning. Then, by Theorem 2.5 and the closure results above, the full kernel $K = \{\mathrm{Adm}, \mathrm{St}, \mathrm{Ref}, \mathrm{Irr}\}$ is already in force, and the system falls under the same closure structure. Hence the purported counterexample collapses.*

2. ***Lower-generator construction.*** *The system purports to derive some $P \in K$ from a strictly lower same-regime basis. This is impossible by Theorem 2.7 and Corollary 2.8.*

3. ***Regime-exiting construction.*** *The system fails to instantiate non-degenerate reasoning, i.e. it abandons at least one of determinate reference, standing-bearing act status, or irreversibility. In that case the system no longer lies in $\mathcal{R}$ and does not instantiate the target phenomenon addressed by the present results.*

These cases are exhaustive. Therefore no counterexample exists.

Proof. Let S be any purported counterexample. If S lies within $\mathcal{R}$ and still instantiates non-degenerate reasoning, then Theorem 2.5 places the full kernel in force there already. But once the kernel is in force, the architecture-exhaustion, mutual-closure, and minimality results of this section apply to S exactly as they do to every other same-regime case. So S is not a genuine counterexample but only a same-regime instance of the already fixed closure structure.

If instead S claims to escape by generating some member of K from a strictly lower same-regime basis, then Theorem 2.7 and Corollary 2.8 rule that out directly: no faithful lower generator exists within the same meaningful regime.

The only remaining possibility is that S abandons one or more of determinate reference, standing-bearing act status, or irreversibility. But by Theorem 2.3, those are exactly the minimal conditions under which proposition-bearing reasoning is assessable as reasoning for the present target. So abandoning them exits the target domain rather than producing a counterexample inside it.

These three cases exhaust the possibilities. Hence no counterexample to the uniqueness and forced-consequence results exists. □

Theorem 2.12 (Non-kernel systems collapse or exit)**.** *Let Σ be any system presented as a putative form of reasoning for the present target. If Σ appears to violate one or more members of the kernel $K = \{\mathrm{Adm}, \mathrm{St}, \mathrm{Ref}, \mathrm{Irr}\}$, then exactly one of the following holds:*

1. ***Degeneracy.*** *Σ fails to instantiate non-degenerate reasoning, because it lacks determinate reference, standing-bearing act-status, irreversibility, or the joint enforceability boundary under which those conditions can function together.*

2. ***Extensional collapse.*** *Σ preserves the target phenomenon after all, in which case the apparently non-kernel structure is only presentational and the system is extensionally governed by the same kernel architecture already fixed by Theorems 2.5, 2.10, and 2.11.*

No third possibility exists. In particular, there is no system that both genuinely violates the kernel and still instantiates the target phenomenon as a distinct non-degenerate reasoning regime.

Proof. Let Σ be any putative reasoning system for the present target.

If Σ genuinely lacks one of admissibility, standing, reference, or irreversibility in the sense relevant to the present manuscript, then by Theorem 2.3 it loses the conditions under which proposition-bearing reasoning is assessable as reasoning at all. Such a system may still implement calculation, transformation, update, bookkeeping, or search, but it does not instantiate non-degenerate reasoning for the present target. So we are in case (1).

Assume instead that Σ is still offered as a genuine instance of the same target phenomenon. Then, by Theorem 2.5, the full kernel is already in force wherever non-degenerate reasoning exists. Any apparent non-kernel feature of Σ must therefore either hide the same governance work under another presentation, redistribute it across levels, or collapse extensionally to the same admissibility/standing/reference/irreversibility package already fixed. By Theorem 2.10, no genuinely distinct same-regime admissibility architecture survives; by Theorem 2.11, any apparent counterexample must either collapse inside the same regime or exit the target altogether. So we are in case (2).

These alternatives are exhaustive. Therefore any putative non-kernel system either collapses into degeneracy or is extensionally governed by the same kernel architecture. There is no third non-degenerate, same-target possibility. □

Corollary 2.13 (No distinct non-kernel reasoning architecture)**.** *Any system that appears to violate the kernel either fails to instantiate non-degenerate reasoning or else is extensionally equivalent, for the present target, to a kernel-governed regime. Consequently there is no distinct non-kernel architecture of proposition-bearing reasoning for the phenomenon studied here.*

Proof. Immediate from Theorem 2.12. □

Remark 2.14 (The theorem-level force of the kernel floor)*.* The preceding theorem does not merely record that the capstone begins from a preferred vocabulary. It says that, for the target phenomenon fixed here, no same-regime derivation, weaker governance package, or faithful same-domain extension can generate admissibility, standing, reference, or irreversibility as lower achievements of that very regime. In that sense the kernel floor is a theorem-level boundary, not a temporary stopping point. The downstream AMetric, standing-regime, and same-domain meta-closure sections then ask the narrower question of what architecture remains once that floor is already fixed.

Proposition 2.15 (Mutual closure)**.** *None of the members of K can be coherently asserted while denying the governance role of the others.*

Proof. Admissibility without standing would not govern claim-bearing acts; it would govern at most transitions stripped of licit claim-status. Standing without reference would license empty or equivocal claims rather than determinate claim-content. Governed reference without irreversibility would fail to fix content for inferential use at act-time, because later repair could retroactively alter the force of the act. Governance irreversibility without admissibility would lose the principled distinction between licensed continuation and prohibited repair, and would therefore cease to be governance irreversibility in the relevant sense. So each member depends on the governance role of the others, and the kernel is mutually closed as one necessity set.[2] □

Corollary 2.16 (Minimality)**.** *No proper subset of K suffices for non-degenerate reasoning.*

Proof. Assume some proper subset $K_0 \subsetneq K$ suffices. Let $X \in K \setminus K_0$ be omitted. Then either X is unnecessary or X is generated from the remaining members. The first option contradicts Proposition 2.15, since the governance role of each member depends on the others. The second contradicts Theorem 2.7, since X would then be derivable from below from a strictly smaller basis. Therefore no proper subset suffices. □

2.6 Mechanization boundary

Theorem 2.17 (Mechanization boundary)**.** *A faithful mechanization of the present framework begins at the consequence layer rather than by attempting to generate the kernel from below.*

Proof. Any meaningful proof system is itself a regime in which proof acts are proposition-bearing reasoning acts. By Theorem 2.5, such a regime already has the kernel in force. So no proof assistant, meta-system, or internal formal layer can faithfully generate admissibility, standing, reference, or irreversibility as fresh lower theorems while remaining within the same meaningful regime. Any such attempt is only a restatement, encoding, or circular recapture of what is already presupposed.

Once the kernel is fixed, however, the situation changes. Preservation theorems, transport results, collapse theorems, irrecoverability results, uniqueness theorems, and related consequence families are ordinary derivable mathematics internal to the fixed-kernel regime. Those are proper targets of machine verification. So the principled formalization boundary lies exactly here: the kernel is a governance-level floor; the downstream consequence architecture is mechanizable.[2] □

What the capstone may now treat as fixed. Later sections may use, without reopening, the following facts: the target class is forced rather than protective; the full kernel is already in force wherever the target phenomenon exists; no faithful lower generator exists within the same meaningful regime or by faithful same-domain extension; the kernel is mutually closed and minimal; and faithful mechanization begins only once that floor is fixed.

Any system that appears to violate these results must either fail to instantiate inferentially governed proposition-bearing reasoning or collapse extensionally into a system satisfying the kernel.

3 Modeling necessity and coherent trajectory form

3.1 Why modeling necessity follows the kernel floor

Once the kernel floor is fixed, a new question becomes unavoidable: what structural form must reasoning take if it is to preserve the target phenomenon through lawful redescription, continuation, and commitment? This section does not re-derive the kernel. It shows that, for the already fixed target class, state-transition representation, identity persistence, irreversible commitment, fail-closed bivalence, and the finite fixed-scope path geometry of coherence are forced rather than stylistic.

3.2 History interfaces, stable reference, and the canonical quotient

Definition 3.1 (History interface)**.** Let Act be a set of acts and let $\mathrm{Act}^{<\omega}$ be the set of finite act-sequences, with concatenation written by $\cdot$ and empty history ϵ. A *history interface* is a partial evaluation map

$$\hat{\varepsilon} : \mathrm{Act}^{<\omega} \times \mathrm{Act} \rightharpoonup \{0, 1\},$$

where $\hat{\varepsilon}(h, a)$ is interpreted as the standing value assigned to attempting act a after history h, whenever that value is defined.

Definition 3.2 (Standing-coherence and future-behavior equivalence)**.** A history interface is *standing-coherent* if evaluation is stable under lawful redescription and recomposition: whenever histories h and h' are related by a finite chain of licensed redescriptions and rebracketings, the values $\hat{\varepsilon}(h, a)$ and $\hat{\varepsilon}(h', a)$ agree in definedness and value for every act a.

Given such an interface, define a relation $\sim$ on histories by

$$h \sim h' \iff \forall \sigma \in \mathrm{Act}^{<\omega}\ \forall a \in \mathrm{Act} : \hat{\varepsilon}(h \cdot \sigma, a) \downarrow \Leftrightarrow \hat{\varepsilon}(h' \cdot \sigma, a) \downarrow,$$

and whenever both sides are defined,

$$\hat{\varepsilon}(h \cdot \sigma, a) = \hat{\varepsilon}(h' \cdot \sigma, a).$$

This is the future-behavior quotient, or Myhill–Nerode form, of the history interface.

Definition 3.3 (Stable reference)**.** A history interface has *stable reference* if it is standing-coherent in the sense above. Equivalently, stable reference is the requirement that lawful redescription and recomposition preserve definedness and standing value for every future evaluated context.

Theorem 3.4 (Canonical state-transition representation)**.** *Let $\hat{\varepsilon}$ be a standing-coherent history interface. Let $S := \mathrm{Act}^{<\omega}/\sim$ and write $[h]$ for the $\sim$-class of a history h. Then there is a canonical partial update operator*

$$U : S \times \mathrm{Act} \rightharpoonup S, \qquad U([h], a) := [h \cdot a]$$

whenever $\hat{\varepsilon}(h, a) = 1$, together with a canonical evaluation map

$$\varepsilon : S \times \mathrm{Act} \rightharpoonup \{0, 1\}, \qquad \varepsilon([h], a) := \hat{\varepsilon}(h, a),$$

such that the original interface is recovered exactly. In particular, the quotient by future-behavior equivalence is the coarsest identity-preserving state space for the interface.

Proof. If $h \sim h'$, then the defining condition with continuation $\sigma = \epsilon$ gives agreement of $\hat{\varepsilon}(h, a)$ and $\hat{\varepsilon}(h', a)$ in definedness and value for every a, so $\varepsilon([h], a)$ is well-defined. Likewise, if $\hat{\varepsilon}(h, a) = 1$ and $h \sim h'$, then for every continuation σ and act b,

$$\hat{\varepsilon}((h \cdot a) \cdot \sigma, b) = \hat{\varepsilon}((h' \cdot a) \cdot \sigma, b)$$

in definedness and value, so $h \cdot a \sim h' \cdot a$ and U is well-defined on classes. The original interface is recovered by construction because $\varepsilon([h], a)$ is defined to be $\hat{\varepsilon}(h, a)$.

For the final claim, suppose one tried to identify histories more coarsely than $\sim$ while still preserving identity through all future evaluated contexts. Then some pair of histories that diverge under a future continuation would be treated as the same. That makes future admissibility and standing presentation-dependent. The quotient by $\sim$ is therefore the coarsest identity-preserving state space.[3] □

3.3 Identity persistence and irreversible commitment

Definition 3.5 (Reachability and non-trivial inferential content)**.** Let S and U be as in Theorem 3.4. Write $s \rightsquigarrow t$ if there exists a finite sequence of standing-positive updates taking s to t, and define

$$\mathrm{Reach}(s) := \{t \in S : s \rightsquigarrow t\}.$$

The interface has *non-trivial inferential content* if there exist a history h and an act a with $\hat{\varepsilon}(h, a) = 1$ such that, writing $s = [h]$ and $s' := U(s, a)$,

$$\mathrm{Reach}(s') \subsetneq \mathrm{Reach}(s).$$

Equivalently, some standing-positive commitment has persistent constraint effect on what future evaluated configurations remain reachable.

Theorem 3.6 (Identity persistence)**.** *Stable reference forces identity persistence. More precisely, if the history interface is standing-coherent, then future-behavior equivalence is a congruence for concatenation: if $h \sim h'$, then $h \cdot \sigma \sim h' \cdot \sigma$ for every finite continuation σ. Consequently, admissible evolution preserves equivalence classes.*

Proof. Fix $h \sim h'$ and a continuation σ. For any further continuation τ and act a, the histories $(h \cdot \sigma) \cdot \tau$ and $(h' \cdot \sigma) \cdot \tau$ are just $h \cdot (\sigma \cdot \tau)$ and $h' \cdot (\sigma \cdot \tau)$. Since $h \sim h'$, the corresponding evaluations agree in definedness and value. So $h \cdot \sigma \sim h' \cdot \sigma$. The update operator therefore preserves equivalence classes whenever it is defined.[3] □

Lemma 3.7 (Mutual reachability implies reachability-set equality)**.** *If $s \rightsquigarrow s'$ and $s' \rightsquigarrow s$, then* $\mathrm{Reach}(s) = \mathrm{Reach}(s')$.

Proof. Let $t \in \mathrm{Reach}(s)$. Then $s \rightsquigarrow t$. Since $s' \rightsquigarrow s$, concatenating the path $s' \rightsquigarrow s$ with $s \rightsquigarrow t$ gives $s' \rightsquigarrow t$, so $t \in \mathrm{Reach}(s')$. Thus $\mathrm{Reach}(s) \subseteq \mathrm{Reach}(s')$. The reverse inclusion is symmetric using $s \rightsquigarrow s'$. □

Theorem 3.8 (Irreversible commitment theorem)**.** *Stable reference together with non-trivial inferential content forces at least one irreversible commitment. That is, there exist states $s, s' \in S$ and an act a such that $U(s, a) = s'$ is standing-positive and there is no return path $s' \rightsquigarrow s$.*

Proof. Choose $s \to s'$ witnessing non-trivial inferential content, so $\mathrm{Reach}(s') \subsetneq \mathrm{Reach}(s)$. If there were a return path $s' \rightsquigarrow s$, then Lemma 3.7 would give $\mathrm{Reach}(s') = \mathrm{Reach}(s)$, contradicting strict inclusion. Hence the commitment is irreversible.[3] □

Theorem 3.9 (Coherence necessity theorem)**.** *Let $\hat{\varepsilon}$ be a history interface with stable reference and non-trivial inferential content. Then coherent discourse necessarily instantiates*

1. *identity persistence, in the sense of Theorem 3.6, and*
2. *at least one irreversible commitment, in the sense of Theorem 3.8.*

Proof. Identity persistence is exactly Theorem 3.6. Existence of at least one irreversible commitment is exactly Theorem 3.8.[3] □

Remark 3.10 (What the theorem forbids)*.* The theorem rules out two familiar evasions at once. It rules out reference without persistence, because the canonical quotient is forced by stable reference. And it rules out inference without commitment, because a regime in which every standing-positive step were reversible would fail the non-trivial inferential-content condition that makes commitment matter at all.[3]

3.4 The coherence meta-theorem

Definition 3.11 (Finitary explicitness)**.** A reasoning practice is *finitary-explicit* when whether an update is standing-positive is determined by a finite certificate in the object-language of the regime rather than by an infinitary or semantic oracle external to its admissible action.

Theorem 3.12 (Coherence meta-theorem)**.** *Let a reasoning practice claim coherence in the sense relevant to the capstone. Then coherence forces all of the following:*

1. *stable reference under lawful redescription,*
2. *compositional closure of candidate material under finite recombination,*
3. *canonical state-transition representation together with identity persistence and at least one irreversible commitment, and*
4. *finitary explicitness of the permissibility interface.*

In particular, state-transition form is not an optional modeling preference once coherent reasoning is in view.

Proof. Stable reference is necessary because otherwise the same purported claim can change target under lawful redescription and no claim-bearing identity is preserved. Compositional closure is necessary because if previously evaluated material may be recombined only under hidden restrictions, then the system either launders inadmissibility by presentation or smuggles in an undeclared scope restriction. Given stable reference and non-trivial inferential content, Theorems 3.4, 3.6, and 3.8 force canonical state-transition representation, identity persistence, and at least one irreversible commitment.

For finitary explicitness, note that a standing-bearing interface that depended essentially on an infinitary or semantic oracle would not generate permissibility from the regime's own admissible action. It would import standing from outside the object-language of the same scope. The present theorem family therefore forces a finitely presentable permissibility interface.[3] □

3.5 Bivalence of non-degenerate reasoning

Definition 3.13 (Reference, standing, and irreversibility axioms)**.** A reasoning system satisfies:

1. *Reference* if at least one expression has a determinate denotation and not all expressions collapse to the same trivial object;
2. *Standing* if identity is preserved across licensed inference or update unless an explicit identity-reset is declared; and
3. *Irreversibility* if there exists at least one commitment operation that cannot be globally undone while preserving reference and standing.

Lemma 3.14 (No standing-preserving repair)**.** *In any system satisfying Reference, Standing, and Irreversibility, no violation of admissibility, standing, or reference can be repaired without violating at least one of those conditions.*

Proof. A purported repair can proceed only by reinterpretation, identity drift, or undoing commitment. Reinterpretation changes what the act is about and therefore violates reference. Identity drift changes which act or object is being preserved and therefore violates standing unless an explicit reset is performed. Undoing commitment violates irreversibility. No fourth repair handle remains.[3] □

Theorem 3.15 (Bivalence theorem for non-degenerate reasoning)**.** *If a reasoning system preserves Reference, Standing, and Irreversibility and is non-degenerate, then it must enforce fail-closed admissibility governance on the evaluated fragment. Equivalently, there is no intermediate standing-bearing status between admitted and not admitted.*

Proof. By Lemma 3.14, admissibility violations cannot be repaired post hoc while preserving reference, standing, and irreversibility. So admissibility must be enforced prior to commitment and reuse. Once such prior enforcement is in place, the evaluated fragment is governed by a fail-closed gate: an act is either admitted or it is not. Any purported third standing-relevant status would either refine admissibility by a second independent authorizer or reintroduce a repair/override mechanism. Either move contradicts the theorem family just established.[3] □

Corollary 3.16 (Binary eligibility in state-transition form)**.** *For the canonical update system of this section, admissibility at the governance layer is the bivalent predicate*

$$\mathrm{Adm}(s, a) :\Longleftrightarrow U(s, a) \downarrow .$$

Any richer evaluator contributes no further standing-bearing content unless it changes which updates are actually defined.

Proof. If an act is inadmissible at s, the bivalence theorem forbids any standing-bearing successor state. So definedness of $U(s, a)$ is exactly admissibility. Any multivalued evaluator that never changes the set of defined updates is reachability-inert bookkeeping; if it does change that set, it merely selects a stricter admissible subrelation of the same update graph.[3] □

3.6 Finite fixed-scope trajectory geometry

The path results below are internal to one fixed substrate and concern finite nonempty trajectories only. They do not re-open the later burdens of AMetric collapse, standing emergence, or quotient closure. Their role is narrower and exact: once a fixed-scope substrate with licensed same-scope transitions, lawful tensor-preserving redescription, and standing-bearing states is in place, the finite path geometry of coherence is forced.[5]

Definition 3.17 (Legal trajectory and coherence)**.** Let $(X, A, S, \rightsquigarrow, \sim, \mathrm{Ten})$ be a fixed-scope substrate of the kind recorded in *The Mathematics of Coherent Reasoning: A Bivalent Trajectory Theory*. A finite nonempty sequence $\tau = (x_0, \ldots, x_n)$ is *legal* if each adjacent step $x_i \rightsquigarrow x_{i+1}$ is a licensed same-scope transition. It is *coherent* if it is legal and every state on it has standing:

$$\mathrm{Coherent}(\tau) :\iff \mathrm{Legal}(\tau) \wedge \forall i \leq n,\ S(x_i).$$

Definition 3.18 (Prefix, extension, composition, and pointwise redescription equivalence)**.** Let $\tau = (x_0, \ldots, x_n)$ and $\tau' = (y_0, \ldots, y_k)$.

1. τ' is a *prefix* of τ if $k \leq n$ and $y_i = x_i$ for all $0 \leq i \leq k$.
2. σ is an *extension* of τ if τ is a prefix of σ.
3. If $\tau_1 = (x_0, \ldots, x_m)$ and $\tau_2 = (y_0, \ldots, y_n)$ satisfy $x_m = y_0$, then their *composition* is
$$\tau_1 \circ \tau_2 := (x_0, \ldots, x_m, y_1, \ldots, y_n).$$
4. τ and τ' are *pointwise redescription-equivalent*, written $\tau \approx \tau'$, if $n = k$ and $x_i \sim y_i$ for all $0 \leq i \leq n$.

Definition 3.19 (Failure witness)**.** A *failure witness* for a trajectory τ is an index $i \leq |\tau|$ such that $\neg S(\tau_i)$. A legal trajectory is therefore coherent exactly when it contains no failure witness.

Proposition 3.20 (Coherence characterization)**.** *Let τ be a legal trajectory. Then*

$$\mathrm{Coherent}(\tau) \iff \forall i \leq |\tau|,\ S(\tau_i).$$

Proof. Since τ is assumed legal, the legality clause in the definition of coherence is already satisfied. The definition therefore reduces to the condition that every state on τ has standing at the fixed scope.[5] □

Proposition 3.21 (Failure-witness characterization)**.** *Let τ be a legal trajectory. Then*

$$\neg\mathrm{Coherent}(\tau) \iff \exists i \leq |\tau|,\ \neg S(\tau_i).$$

Proof. By Proposition 3.20,

$$\neg\mathrm{Coherent}(\tau) \iff \neg\forall i \leq |\tau|,\ S(\tau_i).$$

Because the index set $\{0, \ldots, |\tau|\}$ is finite, this is equivalent to

$$\exists i \leq |\tau|,\ \neg S(\tau_i).$$

That index is precisely a failure witness.[5] □

Lemma 3.22 (Prefix legality)**.** *If* $\mathrm{Legal}(\tau)$ *and* τ' *is a prefix of* τ*, then* $\mathrm{Legal}(\tau')$*.*

Proof. Write $\tau = (x_0, \ldots, x_n)$ and $\tau' = (x_0, \ldots, x_k)$ with $k \leq n$. Since τ is legal, each adjacent transition $x_i \rightsquigarrow x_{i+1}$ with $0 \leq i < n$ already occurs in τ as a licensed same-scope step. In particular, for each $i < k$ the same transition holds inside τ'. Hence τ' is legal.[5] □

Theorem 3.23 (Prefix preservation)**.** *If* $\mathrm{Coherent}(\tau)$ *and* τ' *is a prefix of* τ*, then* $\mathrm{Coherent}(\tau')$*.*

Proof. Since $\mathrm{Coherent}(\tau)$, the trajectory τ is legal and every state on τ has standing. By Lemma 3.22, every prefix τ' of τ is legal. Since each state of τ' is also a state of τ, every state of τ' has standing as well. Therefore τ' is coherent.[5] □

Proposition 3.24 (Composition legality)**.** *If* $\mathrm{Legal}(\tau_1)$*,* $\mathrm{Legal}(\tau_2)$*, and* τ_1, τ_2 *are composable, then* $\mathrm{Legal}(\tau_1 \circ \tau_2)$*.*

Proof. Write

$$\tau_1 = (x_0, \ldots, x_m), \qquad \tau_2 = (y_0, \ldots, y_n),$$

and assume $x_m = y_0$. Then

$$\tau_1 \circ \tau_2 = (x_0, \ldots, x_m, y_1, \ldots, y_n).$$

Within the initial block $x_0, \ldots, x_m$, legality follows from $\mathrm{Legal}(\tau_1)$. Within the final block $y_1, \ldots, y_n$, legality follows from $\mathrm{Legal}(\tau_2)$. At the interface, if $n \geq 1$ we need $x_m \rightsquigarrow y_1$. But $x_m = y_0$, and $\mathrm{Legal}(\tau_2)$ gives $y_0 \rightsquigarrow y_1$. Hence the interface transition is licensed as well. Therefore $\tau_1 \circ \tau_2$ is legal.[5] □

Theorem 3.25 (Composition closure)**.** *If* $\mathrm{Coherent}(\tau_1)$*,* $\mathrm{Coherent}(\tau_2)$*, and* τ_1, τ_2 *are composable, then*

$$\mathrm{Coherent}(\tau_1 \circ \tau_2).$$

Proof. Since $\mathrm{Coherent}(\tau_1)$ and $\mathrm{Coherent}(\tau_2)$, both trajectories are legal and every state in each has standing. By Proposition 3.24, the composite trajectory $\tau_1 \circ \tau_2$ is legal. Every state of $\tau_1 \circ \tau_2$ is a state of either τ_1 or τ_2, so every state on the composite has standing. Hence the composite is coherent.[5] □

Theorem 3.26 (Lawful redescription invariance)**.** *Let* τ *and* τ' *be legal trajectories. If* $\tau \approx \tau'$*, then*

$$\mathrm{Coherent}(\tau) \iff \mathrm{Coherent}(\tau').$$

Proof. Write

$$\tau = (x_0, \ldots, x_n), \qquad \tau' = (y_0, \ldots, y_n),$$

so that $x_i \sim y_i$ for each $0 \le i \le n$.

Assume first that Coherent(τ). Since τ is legal, Proposition 3.20 yields $S(x_i)$ for all $0 \le i \le n$. Because lawful redescription preserves standing on the fixed substrate, each relation $x_i \sim y_i$ implies

$$S(x_i) \iff S(y_i).$$

Hence $S(y_i)$ holds for all i. Since τ' is legal by hypothesis, Proposition 3.20 yields Coherent(τ').

The converse implication is symmetric. Therefore Coherent(τ) $\iff$ Coherent(τ').[5] □

Corollary 3.27 (Tensor-determined coherence)**.** *Let $\tau = (x_0, \ldots, x_n)$ and $\tau' = (y_0, \ldots, y_n)$ be legal trajectories. If*

$$\mathrm{Ten}(x_i) = \mathrm{Ten}(y_i) \qquad \textit{for all } 0 \le i \le n,$$

then

$$\mathrm{Coherent}(\tau) \iff \mathrm{Coherent}(\tau').$$

In particular, skin-only variation does not affect coherence.

Proof. By definition of pointwise redescription equivalence on the fixed substrate, equality of tensor content implies $x_i \sim y_i$ for each i, so $\tau \approx \tau'$. The result follows from Theorem 3.26.[5] □

Lemma 3.28 (Failure witnesses persist under extension)**.** *Let τ and σ be trajectories such that σ extends τ. If τ contains a failure witness, then σ contains a failure witness.*

Proof. Write $\tau = (x_0, \ldots, x_n)$. Since σ extends τ, there exists a representation

$$\sigma = (x_0, \ldots, x_n, z_1, \ldots, z_m)$$

for some $m \ge 0$. If $i \le n$ satisfies $\neg S(x_i)$, then the same index i witnesses failure in σ because $\sigma_i = x_i$.[5] □

Theorem 3.29 (Irrecoverability of coherence failure)**.** *Let τ and σ be legal trajectories. If σ extends τ and $\neg$Coherent(τ), then*

$$\neg\mathrm{Coherent}(\sigma).$$

Proof. Because τ is legal and incoherent, Proposition 3.21 yields a failure witness on τ. By Lemma 3.28, that witness persists in every extension σ of τ. Since σ is legal, Proposition 3.21 implies $\neg$Coherent(σ).[5] □

Theorem 3.30 (No deferred coherence)**.** *Let τ be a legal trajectory and let τ' be a prefix of τ. If*

$$\neg\mathrm{Coherent}(\tau'),$$

then

$$\neg\mathrm{Coherent}(\tau).$$

Proof. By Lemma 3.22, every prefix of a legal trajectory is legal; hence τ' is legal. Proposition 3.21 then yields an index $i \leq |\tau'|$ with $\neg S(\tau'_i)$. Since τ' is a prefix of τ, the trajectory τ extends τ'. By Lemma 3.28, the same failure witness occurs in τ. Because τ is legal, Proposition 3.21 now gives $\neg\mathrm{Coherent}(\tau)$.[5] □

Theorem 3.31 (Bivalence of coherence)**.** *For every legal trajectory τ,*

$$\mathrm{Coherent}(\tau) \vee \neg\mathrm{Coherent}(\tau).$$

Proof. Let $\tau = (x_0, \ldots, x_n)$ be legal. By the bivalence theorem of the present section, each standing claim $S(x_i)$ is bivalent on the evaluated fragment. Since the index set $\{0, \ldots, n\}$ is finite, either all of the formulas $S(x_i)$ hold or at least one of them fails. In the first case, Proposition 3.20 yields $\mathrm{Coherent}(\tau)$. In the second case, Proposition 3.21 yields $\neg\mathrm{Coherent}(\tau)$.[5] □

Corollary 3.32 (No partial coherence)**.** *For legal trajectories, coherence is not a primitive graded or intermediate status. Every legal trajectory is either coherent or incoherent.*

Proof. This is an immediate restatement of Theorem 3.31. □

Theorem 3.33 (Exact factorization under composition)**.** *Let τ_1 and τ_2 be legal and composable. Then*

$$\mathrm{Coherent}(\tau_1 \circ \tau_2) \iff \mathrm{Coherent}(\tau_1) \wedge \mathrm{Coherent}(\tau_2).$$

Proof. Assume first that $\mathrm{Coherent}(\tau_1 \circ \tau_2)$. Since τ_1 is a prefix of $\tau_1 \circ \tau_2$, Theorem 3.23 gives $\mathrm{Coherent}(\tau_1)$.

It remains to prove $\mathrm{Coherent}(\tau_2)$. Because τ_2 is legal by hypothesis, it is enough by Proposition 3.20 to show that every state on τ_2 has standing. The interface state of τ_2 is the terminal state of τ_1, which appears in $\tau_1 \circ \tau_2$ and therefore has standing. Every later state of τ_2 also appears in the composite and therefore has standing. Hence every state on τ_2 has standing, so τ_2 is coherent.

Conversely, if both τ_1 and τ_2 are coherent, then Theorem 3.25 yields $\mathrm{Coherent}(\tau_1 \circ \tau_2)$.[5] □

Corollary 3.34 (No coherence by aggregation)**.** *Let τ_1 and τ_2 be legal and composable. If either τ_1 or τ_2 is incoherent, then*

$$\neg\mathrm{Coherent}(\tau_1 \circ \tau_2).$$

Proof. Suppose toward contradiction that $\mathrm{Coherent}(\tau_1 \circ \tau_2)$. Then Theorem 3.33 implies that both τ_1 and τ_2 are coherent, contrary to hypothesis.[5] □

Theorem 3.35 (Trajectory architecture fixed for later use)**.** *Once the fixed-scope substrate is in place, the following finite path geometry is forced.*

1. *Coherent trajectories are closed under prefix extraction and legal composition.*
2. *Lawful tensor-preserving redescription leaves coherence invariant.*
3. *If a legal prefix is incoherent, then every legal extension retaining that prefix is incoherent.*

4. *Every legal trajectory is either coherent or incoherent; no partial coherence remains.*

Proof. Item (1) is exactly Theorems 3.23 and 3.25. Item (2) is Theorem 3.26. Item (3) is Theorem 3.30, with the stronger extension form given by Theorem 3.29. Item (4) is Theorem 3.31 together with Corollary 3.32.[5] □

Remark 3.36 (Boundary of this section). The present section does *not* yet prove the AMetric boundary, no deeper invariant, gate uniqueness, standing emergence, quotient closure, or the unique admissible interior. Those are later burdens. What is fixed here is the modeling architecture that later sections may treat as forced rather than rhetorical: state-transition form, identity persistence, irreversible commitment, fail-closed admissibility bivalence, and the finite fixed-scope trajectory form of coherence.

4 AMetric boundary, singleton admissibility, and no deeper invariant

This section removes every admissibility-relevant pre-fixational structure before the capstone turns to gate uniqueness and the unique admissible interior. Its burden is earlier than standing emergence, conservation, or quotient uniqueness on the admitted side. What must be shown here is exact: before fixation there is no licensed individuator, no admissibility-relevant grading, no selector, no ordering, and no deeper invariant that constrains admissibility without either importing illicit structure or collapsing back to the boundary classification already being determined.

4.1 Pre-fixational non-reference and licensed individuators

Definition 4.1 (Pre-fixational domain)**.** A domain is *pre-fixational* when its elements are candidates prior to any admissible target-fixing act. Before fixation no element is yet licensed as *this* target rather than as one representative among many in a proposal space.

Definition 4.2 (Licensed individuator)**.** A *licensed individuator* on a pre-fixational domain is any admissibility-relevant feature that distinguishes candidates as identity-bearing before fixation: for example a privileged name, index, coordinate, selector, ordering, ranking, score, or representative-specific parameter.

Lemma 4.3 (Non-invariant pre-fixational predicates presuppose fixation)**.** *Let X be a pre-fixational domain and let $P \subseteq X^n$ be a predicate intended to matter to admissibility prior to fixation. If P is not invariant under some bijection $\pi : X \to X$, then P presupposes a licensed individuator on X.*

Proof. Assume that P is not invariant under some bijection $\pi : X \to X$. Then there exists a tuple $\bar{x} \in X^n$ such that either $\bar{x} \in P$ and $\pi(\bar{x}) \notin P$, or conversely. But before fixation the tuples $\bar{x}$ and $\pi(\bar{x})$ differ only by representative assignment: no target has yet been licensed as *this* target rather than another representative of the proposal space. So the difference in P-status cannot be explained by already-fixed standing-bearing content. The only possible explanation is that some representative-sensitive feature is already being treated as authority-bearing. That feature is

exactly a licensed individuator. Hence any admissibility-relevant pre-fixational predicate that is not invariant under bijection already presupposes fixation-level authority.[3, 4] □

Theorem 4.4 (Relabeling indifference is forced by pre-fixational non-reference)**.** *Let X be a pre-fixational domain with no licensed individuators. Then every admissibility-relevant predicate or structure on X must be invariant under the full symmetric group* $\mathrm{Sym}(X)$*. Equivalently, before fixation arbitrary relabelings cannot change admissibility-relevant content.*

Proof. Suppose, toward contradiction, that some admissibility-relevant predicate or structure on X is not invariant under $\mathrm{Sym}(X)$. Then it is not invariant under some bijection $\pi : X \to X$. By the previous lemma, that non-invariance presupposes a licensed individuator on X. But by hypothesis no such individuator exists. Therefore every admissibility-relevant predicate or structure on X must be invariant under every bijection of X, i.e. under the full symmetric group.[3, 4] □

Corollary 4.5 (No admissibility-relevant asymmetry without witness)**.** *Any admissibility-relevant asymmetry prior to fixation requires a licensed individuator, selector, order, metric, or target witness. Absent such witnesses, pre-fixational asymmetry is inadmissible.*

Proof. An admissibility-relevant asymmetry is a non-invariant structure on the pre-fixational domain. By Theorem 4.4, any such non-invariance requires a licensed individuator. A selector, order, metric, ranking, coordinate system, or target witness is precisely the sort of structure by which such authority could be supplied. If none is available, then the asymmetry has no admissible witness and is therefore inadmissible.[3, 4] □

4.2 Equality-pattern completeness and the collapse of pre-boundary structure

Definition 4.6 (Pre-ontological domain and equality pattern)**.** A *pre-ontological domain* is a nonempty set X equipped with no structure beyond equality. For a tuple $\bar{x} = (x_1, \ldots, x_n) \in X^n$, its *equality pattern* is the partition of $\{1, \ldots, n\}$ whose blocks are the index sets of equal coordinates. Two tuples have the same equality pattern exactly when

$$x_i = x_j \iff y_i = y_j$$

for all indices i, j.

Lemma 4.7 (Orbit classification by equality pattern)**.** *Fix $n \geq 1$. Two tuples $\bar{x}, \bar{y} \in X^n$ lie in the same* $\mathrm{Sym}(X)$*-orbit if and only if they have the same equality pattern.*

Proof. First assume that $\bar{y} = \pi(\bar{x})$ for some bijection $\pi \in \mathrm{Sym}(X)$. Because π is injective,

$$x_i = x_j \iff \pi(x_i) = \pi(x_j) \iff y_i = y_j$$

for every pair of indices i, j. So the equality patterns coincide.

Conversely, assume that $\bar{x}$ and $\bar{y}$ have the same equality pattern. List the distinct values appearing among the coordinates of $\bar{x}$ as $a_1, \ldots, a_k$. Because the equality patterns agree, the coordinates of $\bar{y}$

also use exactly k distinct values, say $b_1, \ldots, b_k$, in matching positions. The partial map $a_r \mapsto b_r$ is therefore well-defined and injective. Extend that finite partial bijection arbitrarily to a bijection $\pi : X \to X$. Then $\pi(\bar{x}) = \bar{y}$, so the two tuples lie in the same orbit.[3, 4] □

Theorem 4.8 (Equality-pattern completeness)**.** *Every pre-boundary meaningful relation on a pre-ontological domain is completely determined by equality patterns. Equivalently, every meaningful relation is a union of* $\mathrm{Sym}(X)$*-orbits, and those orbits are exactly the equality-pattern classes.*

Proof. Let $R \subseteq X^n$ be a pre-boundary meaningful relation, i.e. a relation invariant under relabeling. Let $\bar{x}, \bar{y} \in X^n$ have the same equality pattern. By Lemma 4.7, there exists $\pi \in \mathrm{Sym}(X)$ with $\pi(\bar{x}) = \bar{y}$. Since R is meaningful,

$$\bar{x} \in R \iff \pi(\bar{x}) \in R \iff \bar{y} \in R.$$

So membership in R depends only on the equality pattern. Since the orbits are exactly the equality-pattern classes, every meaningful relation is a union of those orbit classes.[3, 4] □

Corollary 4.9 (No selector, no order, no metric beyond equality)**.** *On a pre-ontological domain:*

1. *there is no nonconstant relabeling-invariant grading or indexing of individual proposals,*
2. *there is no relabeling-invariant asymmetric ordering of distinct proposals,*
3. *there is no equivariant selector choosing a canonical representative from a nontrivial set of proposals, and*
4. *every relabeling-invariant metric collapses to the discrete equality metric, so it contributes no graded admissibility content.*

Proof. For item (1), observe that a relabeling-invariant grading or indexing of individual proposals is just a meaningful unary map or predicate. But for unary tuples there is only one equality pattern. By Theorem 4.8, every meaningful unary predicate is therefore constant on all of X, and every meaningful unary map has constant fibers. So there is no nonconstant grading or indexing.

For item (2), suppose $<$ were a relabeling-invariant asymmetric ordering of distinct proposals and choose distinct $x, y \in X$ with $x < y$. Let π be the transposition exchanging x and y. Since $<$ is relabeling-invariant,

$$x < y \iff \pi(x) < \pi(y) \iff y < x,$$

contradicting asymmetry.

For item (3), suppose c were an equivariant selector on nontrivial proposal sets. Choose distinct $x, y \in X$ and set $A = \{x, y\}$. Let π again be the transposition exchanging x and y. Then $\pi(A) = A$, so equivariance gives

$$c(A) = c(\pi(A)) = \pi(c(A)).$$

But $c(A)$ must be either x or y, and the transposition sends each of those to the other, so $c(A) \neq \pi(c(A))$. Contradiction.

For item (4), let $d : X \times X \to \mathbb{R}_{\geq 0}$ be a relabeling-invariant metric. If $x \neq y$ and $u \neq v$, choose a bijection π with $\pi(x) = u$ and $\pi(y) = v$. Invariance gives

$$d(u, v) = d(\pi(x), \pi(y)) = d(x, y).$$

So every distance between distinct points has the same value, say $\lambda > 0$. Together with $d(x, x) = 0$, this is exactly the discrete equality metric up to overall scale.[3, 4] □

Corollary 4.10 (Pre-admissible dimensionlessness)**.** *A pre-admissible proposal regime is dimensionless in the strong sense: there is no relabeling-invariant metric or monotone ordering on the proposal side beyond equality versus disequality.*

Proof. The previous corollary eliminates both nontrivial invariant orderings and nontrivial invariant metrics on the proposal side. So no dimensional, proximity, or graded comparison structure remains available pre-boundarily beyond equality and disequality.[3, 4] □

4.3 The AMetric boundary and the no-deeper-invariant theorem

Definition 4.11 (AMetric boundary)**.** The *AMetric boundary* is the structural interface separating pre-admissible classification and eligibility determination from post-admissible licensed operation. At this boundary objects may be classified and standing may be determined, but no admissible transformation is performed. The boundary is non-transmissive: no admissible structure may be transported backward across it.

Definition 4.12 (Licensed operator)**.** A *licensed operator* is a partial transformation whose domain of application is fixed only after admissibility has been determined. Equivalently, it is an operator of the admitted interior rather than of the raw pre-admissible proposal space.

Theorem 4.13 (Boundary non-action)**.** *No licensed action, execution, or transformation occurs at the AMetric boundary. Boundary work is classificatory rather than operational.*

Proof. Suppose, toward contradiction, that a licensed transformation occurs at the boundary. By the preceding definition, the operator's domain of application is fixed only after admissibility has already been determined. But the AMetric boundary is precisely the interface at which admissibility is being settled. So the alleged boundary action presupposes the very status whose determination the boundary is meant to provide. That is circular. Therefore no licensed action, execution, or transformation occurs at the boundary.[3, 4] □

Theorem 4.14 (Operator undefinedness at the boundary)**.** *No admissibility-conferring operator is defined on the AMetric boundary itself.*

Proof. An admissibility-conferring operator would have to be either licensed or unlicensed. If it is licensed, then by definition it becomes available only after admissibility has already been fixed, so it cannot first determine admissibility. If it is unlicensed, then whatever authority it confers is external to the regime under construction and therefore an illicit import rather than an internal gate mechanism. So no admissibility-conferring operator is defined at the boundary itself.[3, 4] □

Theorem 4.15 (No admissibility-relevant boundary trajectory)**.** *There is no admissibility-relevant process, trajectory, or staged approach by which a proposal becomes "more boundary-positive" before admission.*

Proof. A boundary trajectory would require at least one of the following: a parameter measuring progress toward admission, an ordering of stages from less admissible to more admissible, or an operator that updates the proposal through boundary-adjacent states. But pre-boundary parameters, orderings, selectors, and nontrivial metrics have already been ruled out by Corollary 4.9; and boundary action or operator application is ruled out by Theorems 4.13 and 4.14. So there is no admissibility-relevant boundary trajectory.[3, 4] □

Theorem 4.16 (Boundary non-transmissivity)**.** *No admissible structure may be transported backward across the AMetric boundary. In particular, no post-fixation metric, order, selector, operator, or privileged interior chart may be used to govern pre-admissible classification.*

Proof. Assume that some post-fixation structure S is transported backward to govern pre-admissible classification. Then the pre-admissible side is no longer operating with equality alone; it is now governed by the additional admissibility-relevant structure S. But Theorem 4.8 and Corollary 4.9 show that no such meaningful metric, order, selector, or graded structure survives under full relabeling symmetry on the proposal side. So either S has no pre-boundary authority at all, or it breaks the primitive symmetry assumptions by illicit import. In neither case is same-scope backward transport admissible.[3, 4] □

Definition 4.17 (Purported deeper invariant)**.** A *purported deeper invariant* is a family of pre-boundary meaningful relations claimed to determine or constrain admissibility more fundamentally than the boundary classification itself. The claim of "deeper" means that the invariant is supposed to operate before admissibility is fixed and yet still decide, or partly decide, which proposals qualify.

Lemma 4.18 (Constraint requires pre-boundary availability)**.** *If a purported deeper invariant genuinely constrains boundary classification, then it must be available and meaningful on the pre-admissible side. A purely post-fixation invariant cannot ground admissibility without circularity.*

Proof. If an invariant were available only after admissibility had already been fixed, then using it to determine admissibility would presuppose the very status under determination. So any invariant that genuinely constrains boundary classification must already be available before admissibility is fixed.[4] □

Lemma 4.19 (Invariant data induce invariant unary criteria)**.** *Let Q be any collection of pre-boundary meaningful relations on the proposal side. Any criterion $U(x)$ for individual proposals that is defined solely from Q, without adding non-invariant constants, labels, or privileged representatives, is itself a meaningful unary predicate.*

Proof. The defining rule for U uses only equality, logical operations, quantification over the proposal set, and membership in relations from Q. Because each relation in Q is meaningful, its truth values

are preserved under relabeling, and equality is preserved as well. So for every $\pi \in \mathrm{Sym}(X)$ one has

$$U(x) \iff U(\pi(x)).$$

Thus U is itself a meaningful unary predicate.[4] □

Theorem 4.20 (No deeper pre-boundary invariant with independent constraining force)**.** *Let Q be a purported deeper invariant. Then exactly one of the following holds:*

1. *Q relies on non-invariant labels, coordinates, selectors, or post-fixation structure and is therefore an illicit import;*
2. *every unary admissibility criterion induced from Q is trivial, so Q has no independent constraining force on individual proposals; or*
3. *once admissibility is fixed, the only nontrivial constraining role played by Q is extensionally identical to the boundary classification already being determined.*

There is no fourth option.

Proof. By the previous lemma, any genuinely deeper invariant must be available pre-boundarily. If it is not invariant under relabeling, then by Theorem 4.4 it already assigns admissibility-relevant authority to labels, coordinates, or representatives alone, so it is an illicit import and we are in case (1).

So assume Q is relabeling-invariant. Any way of turning Q into a criterion for whether an individual proposal should pass the boundary produces, by the preceding lemma, a meaningful unary predicate on the proposal side. But by Corollary 4.9, and more fundamentally by the unary-collapse consequence of Theorem 4.8, every such unary criterion is trivial. That yields case (2) unless additional structure is imported.

The only remaining possibility is that, once admissibility is fixed, the nontrivial work done by Q is simply to pick out extensionally the very same proposals already picked out by the admissibility classification. In that case Q adds no independent constraining force and falls under case (3). These three cases exhaust the possibilities.[3, 4] □

Theorem 4.21 (No faithful lower generator at the pre-boundary layer)**.** *No faithful lower generator of admissibility exists at the pre-boundary layer. Any purported lower generator ends in illicit structure import, triviality, violation of the base assumptions, or extensional collapse to the admissibility predicate already being determined.*

Proof. Assume, toward contradiction, that there exists a purported lower generator G together with a rule Φ such that G determines or constrains admissibility prior to fixation while remaining compatible with lawful redescription, non-circular in use, and independently constraining.

If the decisive work of G is genuinely pre-boundary, then by Theorem 4.20 any such object immediately falls into one of three cases: illicit import, triviality, or post-fixational extensional

coincidence with the boundary classification. These are already three of the four outcomes named in the theorem.

It remains only to consider the possibility that the purported lower generator tries to evade Theorem 4.20 by shifting its decisive work outside the pre-boundary proposal regime while still claiming independent constraining force for the same boundary. But then, if that work constrains eligibility at all, it must do so either by transporting post-fixation structure backward across the boundary, or by altering what is preserved under lawful redescription, or by altering the identity conditions under which irreversible commitment is tracked. The first move is illicit structure import. The second and third destroy the base goods already fixed in Sections 2 and 3: stable reference fails, compositional identity drifts, or irreversible commitment is no longer well-defined. So every candidate lower generator ends in illicit import, triviality, base-good violation, or extensional collapse. No fifth outcome remains.[4] □

Theorem 4.22 (Derived boundary axiom: binary, non-eventful, non-parameterized)**.** *On the evaluated fragment, the admissibility boundary is binary and non-eventful. Consequently it admits no admissibility-relevant parameterization, continuous or discrete, including any discrete indexing of multiple positive ways of crossing the boundary.*

Proof. Binary standing at the interface is already fixed by Theorem 3.15 and Corollary 3.16 of Section 3: on the evaluated fragment an act is either admitted or not admitted, and undefinedness is fail-closed rather than a third standing value.

The boundary is non-eventful because Theorem 4.13 rules out licensed action or execution there, Theorem 4.14 rules out admissibility-conferring operators there, and Theorem 4.15 rules out any staged or process-like approach toward admission.

Now suppose, toward contradiction, that there were an admissibility-relevant parameterization of crossing. Then either the parameterization would create more than two standing-relevant statuses on the evaluated fragment — e.g. admissible of type i and admissible of type $j \neq i$ in addition to inadmissible — or else it would require a selector to choose among multiple positive forms of crossing. The first option contradicts binary standing at the interface. The second contradicts the collapse of pre-boundary selectors, rankings, and metrics established above. So no admissibility-relevant parameterization, continuous or discrete, survives at the boundary.[3, 4] □

4.4 Singleton meta-necessity and the binary core of admissibility

Definition 4.23 (Meta-constraint)**.** A *meta-constraint* is any rule that purports to regulate intelligibility by constraining which descriptions count as reference-bearing and composable without collapse.

Theorem 4.24 (Singleton meta-necessity)**.** *There exists exactly one terminal meta-constraint governing intelligibility in non-degenerate irreversible compositional systems: admissibility with standing conservation. Any putative independent meta-constraint either presupposes admissibility-with-standing and is therefore derivative, or else permits collapse of reference under redescription and composition and is therefore inadmissible.*

Proof. Once irreversibility is made well-defined, some admissibility invariant must survive redescription and recomposition strongly enough to stabilize standing. Let C denote admissibility-with-standing. Consider any candidate meta-constraint $C' \neq C$.

If C' is to constrain intelligibility, it must be stable under admissible redescription and composition; otherwise it can be defeated by the system's own recomposition freedoms, and it therefore does not constrain intelligibility at all. But stability under admissible redescription already presupposes admissibility-with-standing, because what counts as stable is defined only relative to admissible redescription and standing conservation. So any stable C' is derivative on C. If instead C' is not stable under admissible redescription and composition, then it permits self-undermining redescriptions and so fails intelligibility. Therefore no independent terminal meta-constraint exists alongside admissibility with standing conservation.[3] □

Corollary 4.25 (Binary standing at the admissibility interface)**.** *On the evaluated fragment, standing has exactly two standing-relevant statuses: eligible and ineligible. Any putative third standing-relevant status would either introduce a second independent authorizer or reintroduce repair, override, or parameterization at the boundary.*

Proof. By Theorem 3.15, admissibility on the evaluated fragment is fail-closed and bivalent. Suppose nevertheless that a third standing-relevant status survives.

If the extra status lies on the positive side of the interface, then admitted acts are split into more than one standing-bearing class. One may then define a new meta-constraint that privileges one positive status over the other. That contradicts Theorem 4.24.

If the extra status lies between admitted and rejected or refines the negative side in a way that changes later eligibility, then it either reintroduces a repair/override route or turns admissibility into a parameterized interface with more than two standing-relevant statuses. That contradicts Theorem 4.22 together with the fail-closed bivalence already fixed in Section 3. So only two standing-relevant statuses survive: eligible and ineligible.[3] □

Proposition 4.26 (Binary core of irreversibility-stabilizing invariants)**.** *Any irreversibility-stabilizing invariant has a binary standing-relevant core. Either it is constant on the fibers of the standing gate and its extra values are bookkeeping, or it refines permissibility only by selecting a stricter subrelation of admissible updates rather than introducing a third standing-bearing status.*

Proof. Let I be an invariant sufficient to stabilize irreversibility. If the extra values of I do not change which updates are admissible or which standing-bearing continuations remain available, then those extra values do no same-scope standing work and are bookkeeping only. In that case I is constant on the fibers of the standing gate for all standing-relevant purposes.

Suppose instead that the extra values of I do change admissible reachability. Then they do so only by selecting a stricter subset of already admissible edges or updates. That changes the admissible update relation, but it still does not introduce a third standing-bearing status at the interface. The standing-relevant core remains binary: an update is either licensed or it is not. So any irreversibility-stabilizing invariant either factors through the binary gate or acts only by selecting a

stricter admissible subrelation. In neither case does it generate an additional same-scope standing value.[3] □

5 Standing regime, quotient closure, and the unique admissible interior

This section completes the first full interior-closure layer of the capstone. Section 4 already removed every admissibility-relevant selector, order, grading, metric, and deeper invariant from the proposal side. What remains is the positive side itself: once the boundary is already binary, non-eventful, and non-parameterized, what same-scope structure is left on the admitted side? The answer is exact. Gate existence is forced; competing same-scope gates collapse; standing is reuse-stable admissibility rather than an extra classifier; no repair, generator, or carrier route survives; every admissibility-preserving standing presentation factors through one standing quotient; no stricter same-scope standing-bearing refinement survives beneath that quotient; and the admissible interior is unique up to lawful equivalence.[4, 20, 21, 5, 6, 7]

5.1 Gate existence and unique same-scope classification

Once the proposal side is already stripped of selectors and parameterizations, the admitted side cannot remain indefinite. Some gate must exist, and it cannot bifurcate into rival same-scope classifications.

Definition 5.1 (Evaluated fragment and gate)**.** Fix a typed same-scope domain D with evaluated fragment $X_D^\bullet$. A *gate predicate* on D is a binary admissibility predicate

$$\mathrm{Adm}_D : X_D^\bullet \to \{0, 1\}$$

intended to classify acts or states into eligible and ineligible members of the evaluated fragment.

Theorem 5.2 (Gate existence on the fixed scope)**.** *If a fixed same-scope regime is intelligible, supports non-degenerate reasoning, and contains at least one irreversible commitment, then some gate predicate* Adm_D *is forced on the evaluated fragment.*

Proof. Assume toward contradiction that no gate predicate is instantiated on the evaluated fragment. Then there is no principled eligible/ineligible partition on which the regime can state that some acts are licit while others are irreversibly excluded. But an irreversible commitment regime requires exactly that: it must be able to state which acts are excluded and preserve that status under admissible continuation. If no gate exists, then either every evaluated act is treated as licit, in which case irreversible exclusion is void; or no evaluated act is eligible, in which case non-degenerate reasoning collapses; or eligibility varies silently by representation, in which case claim identity and evidential reference are not stable. Each branch contradicts the fixed-domain intelligibility and irreversibility package. Therefore some gate predicate is forced.[4, 6] □

Theorem 5.3 (Unique same-scope admissibility classification)**.** *Let* Adm_D *and* Adm'_D *be two candidate same-scope gate predicates for the same typed domain* D*. Then they agree extensionally*

on $X_D^\bullet$.

Proof. Assume toward contradiction that they disagree on some evaluated act a. Then one gate treats a as admitted while the other treats it as not admitted on the same fixed scope. If both verdicts are allowed to stand, then standing on the fixed domain becomes indeterminate. If instead one tries to decide which gate governs a, that decision requires an admissibility-relevant differentiator selecting between the two gates. But Section 4 already proved that the boundary is binary, non-eventful, and non-parameterized. So there is no admissibility-relevant selector by which two disagreeing same-scope gates can coexist. The disagreement is therefore impossible, and the two predicates must coincide extensionally.[4, 6] □

Corollary 5.4 (No competing same-scope gate)**.** *There is no second same-scope admissibility predicate that changes eligibility or standing on the evaluated fragment while preserving the fixed scope.*

Proof. Any such predicate would either disagree extensionally with the unique gate, which Theorem 5.3 rules out, or else merely rename it. □

5.2 Standing emergence as reuse-stable admissibility

The next theorem does not derive standing from a lower layer. It identifies what standing is once the gate and the same-scope continuation discipline are already fixed.

Definition 5.5 (Reuse-stable admissibility)**.** Let D be fixed and let $a \in X_D^\bullet$. We say that a is *reuse-stably admissible* if

1. $\mathrm{Adm}_D(a) = 1$, and
2. every licensed same-scope continuation of a preserves the standing-relevant invariant bundle — equivalently, the standing-relevant tensor content — that a is supposed to carry on that scope.

Definition 5.6 (Standing predicate at fixed scope)**.** A *standing predicate* on D is any predicate

$$\mathrm{Stand}_D : X_D^\bullet \to \{0, 1\}$$

claimed to mark exactly those evaluated acts that are eligible for standing-bearing continuation, commitment, or licensed reuse on the fixed domain.

Proposition 5.7 (Admitted same-scope continuation preserves the standing-relevant invariant bundle)**.** *If an evaluated act is admitted on a fixed domain and later reused under a licensed same-scope continuation, then that continuation preserves the standing-relevant members of the invariant bundle the act is supposed to carry.*

Proof. Suppose admitted same-scope continuation failed to preserve the relevant invariant bundle. Because the continuation is same-scope, no explicit reset or domain change has been declared. So either the continuation silently retargets the act while presenting it as continuation of the same standing-bearing content, violating stable reference, or it consults some hidden selector or criterion

that changes what counts as relevant midstream. The latter would reintroduce hidden gate work forbidden by the AMetric boundary and no-parameterization theorems of Section 4. Hence admitted same-scope continuation preserves the standing-relevant invariant bundle.[4] □

Theorem 5.8 (Standing emergence)**.** *For every evaluated act $a \in X_D^\bullet$,*

$$\mathrm{Stand}_D(a) = 1 \iff a \text{ is reuse-stably admissible on } D.$$

Equivalently, standing is not an additional interface-level classifier beyond admissibility on a fixed scope. It is admissibility as preserved through licensed same-scope continuation.

Proof. Suppose first that $\mathrm{Stand}_D(a) = 1$. Because standing is claimed to govern eligibility for commitment and licensed reuse on the fixed scope, it is already an admissibility-relevant status. By Theorem 5.3, no second gate remains on the same scope, so a must be admitted. If a were not reuse-stable, some licensed same-scope continuation would alter the invariant bundle while still counting as continuation of the same standing-bearing content, contradicting Proposition 5.7. Hence every standing-positive act is reuse-stably admissible.

Conversely, suppose that a is reuse-stably admissible. Then $\mathrm{Adm}_D(a) = 1$ by definition. Suppose nevertheless that $\mathrm{Stand}_D(a) = 0$. Since standing is supposed to govern commitment or later licensed continuation, this would split admitted acts into two substantive subclasses: those merely admitted and those admitted-with-standing. If the split affects what may later be done on the fixed scope, it is an admissibility-relevant same-side split on the positive side, forbidden by the binary interface theorems. If the split affects nothing, then it is inert bookkeeping and does not define a substantive standing predicate at all. So denying standing to a reuse-stably admissible act is impossible as a same-scope standing-bearing classification.[4] □

Theorem 5.9 (Standing–admissibility equivalence)**.** *For every evaluated act $a \in X_D^\bullet$,*

$$\mathrm{Stand}_D(a) = 1 \iff \mathrm{Adm}_D(a) = 1.$$

Equivalently, there is no coherent same-scope regime in which admissibility is standing-neutral, standing is deferred past admissibility, or standing exists independently of admissibility.

Proof. The forward implication is immediate from Theorem 5.8: if $\mathrm{Stand}_D(a) = 1$, then a is reuse-stably admissible, and the first clause of reuse-stable admissibility is $\mathrm{Adm}_D(a) = 1$.

For the reverse implication, assume $\mathrm{Adm}_D(a) = 1$ but $\mathrm{Stand}_D(a) = 0$. Then admissibility and standing are being separated on the same fixed domain. By *Standing–Admissibility Equivalence*, every non-equivalence regime must instantiate at least one of the following structural features: standing-neutral admissibility, deferred admissibility, admissibility reset, factorized governance, or illicit standing creation, suspension, or partition. But standing-neutral admissibility is already ruled out by Theorem 5.8: once an act is admitted and reuse-stably admissible, no extra standing-positive layer remains to be conferred. Deferred admissibility and admissibility reset are ruled out by the irreversibility and no-repair package proved below. Factorized governance would require a second same-scope authorizer beyond the unique gate, which Theorem 5.3 forbids. Illicit standing creation or suspension is ruled out by the no-generators and no-carriers theorems proved below. So

every non-equivalence route is blocked. Therefore $\mathrm{Adm}_D(a) = 1$ implies $\mathrm{Stand}_D(a) = 1$, and the equivalence follows.[4, 20] □

Corollary 5.10 (Rejection of dual-track governance)**.** *Any same-scope distinction between "formal admissibility" and "substantive standing" is either inert bookkeeping or a prohibited second authorizer. In particular, admissibility without standing, standing without admissibility, and parallel governance bookkeeping are inadmissible on the fixed domain.*

Proof. Immediate from Theorem 5.9.[20] □

Corollary 5.11 (Standing is lawful-redescription invariant)**.** *If $a \sim_D b$, then*

$$\mathrm{Stand}_D(a) \iff \mathrm{Stand}_D(b).$$

Proof. Lawful redescription preserves admissibility and the standing-relevant invariant bundle. So it preserves reuse-stable admissibility, and Theorem 5.8 gives the result.[4] □

5.3 No repairs, no generators, and no carriers

Once standing is identified with reuse-stable admissibility, every attempted positive-side "repair" must either change the act, add a new gate, or collapse to bookkeeping. The next three theorems isolate those impossibilities in the exact forms later sections will need.

Theorem 5.12 (No same-act repair)**.** *Let $a \in X_D^\bullet$. If $\mathrm{Stand}_D(a) = 0$, then no internal same-scope post-processing that still treats the output as the same evaluated act can yield $\mathrm{Stand}_D(a) = 1$.*

Proof. Assume an internal same-scope step repairs a while still treating the result as the same evaluated act. If the repair changes admissibility, then one has a same-scope promotion from ineligible to eligible for the same act, contradicting irreversibility. If the repair leaves admissibility unchanged but changes standing, then by Theorem 5.8 it changes reuse-stable admissibility. That can happen only by silently changing which invariant bundle the act is supposed to preserve or by declaring an undeclared reset. The first move violates same-scope identity and Proposition 5.7; the second is not same-act repair. So no internal same-scope post-processing can repair a standing-negative act as the same evaluated act.[4, 6] □

Theorem 5.13 (No generators)**.** *There is no internal operation T and no evaluated act $a \in X_D^\bullet$ with $\mathrm{Stand}_D(a) = 0$ such that $\mathrm{Stand}_D(T(a)) = 1$ as a consequence of a alone and without fresh evaluation on D.*

Proof. If $T(a)$ is still treated as the same evaluated act as a, then the claim is exactly same-act repair and is impossible by Theorem 5.12. If $T(a)$ is treated as a distinct act, then its positive standing depends on fresh evaluation of that new act rather than on the prior failed standing of a alone. So no generator exists.[4, 6] □

Definition 5.14 (Carrier attempt)**.** A *carrier attempt* on the fixed domain D is any derived predicate $Q_D(a)$ proposed to authorize standing independently of the unique admissibility gate. Typical examples are scores, confidences, probabilities, losses, utilities, rankings, or structural invariants offered as primitive authorizers.

Lemma 5.15 (Any putatively independent authorizer induces gate work)**.** *If a predicate Q_D is used as an independent authorizer to decide whether an evaluated act may enter standing-bearing continuation on D, then Q_D performs gate work. It either changes which acts are admitted for licensed reuse or refines the admitted side into a further substantive eligibility partition.*

Proof. If Q_D changes whether an act may enter standing-bearing continuation, then it directly changes eligibility. If instead every act admitted by Adm_D remains admitted but only some of them are counted by Q_D as genuinely standing-positive, then Q_D introduces a substantive same-side split among admitted acts. No third possibility exists.[4] □

Theorem 5.16 (No independent carriers)**.** *No carrier attempt Q_D can serve as an independent authorizer of standing on the fixed domain except by coinciding extensionally with reuse-stable admissibility. Any admissible use of Q_D is therefore downstream bookkeeping inside an already gated interior.*

Proof. Let Q_D be a carrier attempt. By Lemma 5.15, if it is used as an independent authorizer then it performs gate work. If it changes who is admitted, it competes with the unique gate, contradicting Theorem 5.3. If it leaves admission unchanged but refines the admitted side substantively, it creates an admissibility-relevant same-side split on the positive side, contradicting Theorem 5.9 together with the binary fixed-domain interface. The only remaining possibility is extensional coincidence with reuse-stable admissibility, in which case Q_D is merely a renaming or bookkeeping layer.[4, 6] □

Corollary 5.17 (No primitive score, probability, or structural authorizer)**.** *No numeric threshold, confidence score, probability assignment, utility ranking, explanation score, or structural invariant may function as a primitive same-scope authorizer of standing.*

Proof. Each such proposal is a carrier attempt in the sense of the previous definition. Theorem 5.16 therefore excludes it unless it collapses extensionally to the unique gate, in which case it ceases to be an additional authorizer.[4, 6] □

Inline proof anchor III: standing emergence, equivalence, and no carriers

The local hinge proved in *Necessity of Admissibility in Non-Degenerate Compositional Systems* and then closed positively in *Standing–Admissibility Equivalence* is that standing is not a second gate layered over admission. The forcing chain is finite.

1. If $\mathrm{Stand}_D(a) = 1$, then a is already marked as eligible for standing-bearing continuation on the fixed scope. That status is admissibility-relevant. By Theorem 5.3, no second gate remains available on the same scope, so a must already satisfy $\mathrm{Adm}_D(a) = 1$.

2. If a standing-positive act were not reuse-stable, some licensed same-scope continuation would alter the invariant bundle it is supposed to preserve while the act was still treated as the same standing-bearing content. That is silent retargeting or hidden selector work, and so is impossible by Proposition 5.7.

3. Conversely, if a is reuse-stably admissible but $\mathrm{Stand}_D(a) = 0$, then admitted acts split into two subclasses: merely admitted and admitted-with-standing. If that split affects later licensed use, it is a second positive-side classifier on the same scope; if it affects nothing, it is inert bookkeeping. Neither option yields a coherent same-scope standing predicate.

4. Once standing emergence is in place, every putative non-equivalence route between admissibility and standing must instantiate one of the residual dualism forms listed in *Standing–Admissibility Equivalence*: standing-neutral admissibility, deferred admissibility, admissibility reset, factorized governance, or illicit standing creation/suspension. Each route is blocked by the fixed-domain closure package.

5. The carrier theorem is then immediate. A derived predicate Q_D either changes who is admitted to standing-bearing continuation, in which case it competes with the unique gate, or it leaves admission unchanged but refines the admitted side into a further substantive partition, in which case it recreates the same-side split just eliminated. If it does neither, it is only bookkeeping. No fourth role remains.

So the fixed-domain positive side is rigid: standing emerges as reuse-stable admissibility, admissibility and standing are equivalent, and every primitive score-, probability-, confidence-, or structure-based authorizer collapses either into a forbidden second gate or into bookkeeping inside an already gated interior.

5.4 Standing conservation and cross-domain non-portability

Theorem 5.18 (Standing conservation)**.** *For each fixed domain D, no internal same-scope operation creates, amplifies, transfers, or recovers standing. Standing can only be preserved or destroyed.*

Proof. Recovery is excluded by Theorem 5.12. Creation from a failed act without fresh gating is excluded by Theorem 5.13. Amplification would require standing to admit same-scope degrees beyond the binary gate/standing theorem family already fixed; that is impossible once standing is identified with reuse-stable admissibility. Transfer would require some predicate or structure to carry standing from one act or context to another independently of fresh gating, which is exactly an independent carrier and is therefore impossible by Theorem 5.16. These four cases exhaust the standard ways of increasing standing.[4, 6] □

Corollary 5.19 (No cross-domain portability without fresh gating)**.** *Let D_1 and D_2 be distinct domains. Standing in D_1 does not by itself confer standing in D_2. Independent target gating on D_2 is required.*

Proof. Assume standing in D_1 automatically conferred standing in D_2 without fresh target gating. Then either the gate of D_1 would be reused to evaluate acts on D_2, which is an illicit scope transport,

or some derived property of "having standing in D_1" would serve as an independent carrier on D_2. The first move violates fixed-scope discipline; the second is excluded by Theorem 5.16. Hence independent target gating on D_2 is required.[4, 6] □

5.5 Standing quotient uniqueness and no stricter same-scope refinement

The admitted side is now rigid enough that every standing-bearing presentation must factor through a unique same-scope quotient. This is the earliest point at which the capstone may safely speak of quotient closure and no second same-scope standing-bearing invariant.

Definition 5.20 (Standing quotient)**.** Fix a same-scope substrate with presentation carrier P_D and standing-equivalence relation $\sim_D$, where

$$p \sim_D p' \iff \mathrm{Ten}_D(p) = \mathrm{Ten}_D(p').$$

The *standing quotient* of the domain is

$$\mathcal{Q}_D := P_D/\sim_D .$$

Its classes represent exactly the same-scope standing-bearing tensor content of the domain.

Definition 5.21 (Standing presentation)**.** A *standing presentation* for D is a triple (Y, r, Pred_Y) where $r : P_D \twoheadrightarrow Y$ is a surjection and every same-scope standing-bearing value factors through r. The presentation is *complete* if $r(p) = r(p')$ implies $p \sim_D p'$, and *sound* if $p \sim_D p'$ implies $r(p) = r(p')$.

Lemma 5.22 (Completeness of standing presentations)**.** *Every standing presentation is complete.*

Proof. If $r(p) = r(p')$, then every standing-bearing value factors through the same point of Y, so for every admitted observable or same-scope test one has

$$\mathrm{Pred}_D(h; p) = \mathrm{Pred}_Y(h; r(p)) = \mathrm{Pred}_Y(h; r(p')) = \mathrm{Pred}_D(h; p').$$

Hence p and p' agree on every same-scope standing-bearing consequence, i.e. $p \sim_D p'$.[7] □

Lemma 5.23 (Soundness forced by stable reference and exhaustion)**.** *Assume minimal coherence at the fixed scope. Then every standing presentation is sound.*

Proof. Suppose $p \sim_D p'$ but $r(p) \neq r(p')$. Then Y distinguishes two putative standing states that already agree on tensor content and on every same-scope standing-bearing consequence. To keep that distinction as standing-bearing, one would need some additional admissible same-scope discriminator separating $r(p)$ from $r(p')$. But by the fixed-domain exhaustion and factorization results, no such same-scope discriminator exists beneath the quotient. So the extra distinction is unwitnessed representative privilege, i.e. skin treated as tensor load. That is inadmissible. Therefore $r(p) = r(p')$, and the presentation is sound.[7, 21] □

Theorem 5.24 (Standing-quotient uniqueness)**.** *Let (Y, r, Pred_Y) be any same-scope standing presentation for D. Then r is automatically sound and complete and induces a unique bijection*

$$\phi_D : \mathcal{Q}_D \longrightarrow Y, \qquad \phi_D([p]) = r(p).$$

In particular, the standing quotient $\mathcal{Q}_D$ is the unique same-scope standing-bearing content space for D, up to unique isomorphism.

Proof. Completeness is Lemma 5.22. Soundness is Lemma 5.23. Define $\phi_D([p]) := r(p)$. Soundness implies that ϕ_D is well-defined. Completeness implies injectivity. Surjectivity holds because r is surjective. Finally, uniqueness follows from the quotient universal property: any map constant on standing classes factors uniquely through the quotient map $q_D : P_D \to \mathcal{Q}_D$. Hence $r = \phi_D \circ q_D$, and the standing quotient is unique up to unique isomorphism.[7] □

Corollary 5.25 (Skin non-authority)**.** *No admissible same-scope standing-bearing consequence can depend on a choice of representative inside a standing class. Equivalently, every admissible same-scope standing-bearing consequence factors uniquely through the standing quotient.*

Proof. Let $F : P_D \to Y$ be any admissible same-scope standing-bearing consequence functional. Lawful redescription neutrality implies that F is constant on standing classes. So by the quotient universal property there exists a unique $\widetilde{F} : \mathcal{Q}_D \to Y$ with $F = \widetilde{F} \circ q_D$. Thus F depends only on the quotient class $[p]$, not on the representative p itself. So representative-only variation is skin and carries no same-scope authority.[7] □

Theorem 5.26 (Same-scope factorization discipline)**.** *Let F be any same-scope standing-relevant discriminator, compatibility rule, consequence functional, or auxiliary presentation on the fixed domain D. Then F is constant on standing-equivalence classes and factors uniquely through the standing quotient $\mathcal{Q}_D$. If F fails to factor, then there exist $p, p' \in P_D$ with $p \sim_D p'$ but $F(p) \neq F(p')$; that witness is exactly a representative-only, metadata-level, or path-dependent discriminator and therefore cannot be admitted as same-scope standing-bearing authority.*

Proof. If F is same-scope and standing-relevant, then by hypothesis it is supposed to affect admissible same-scope standing-bearing consequences. Corollary 5.25 already states that every such consequence factors uniquely through the standing quotient. So F must be constant on standing classes and hence factor through $\mathcal{Q}_D$.

Suppose instead that F fails to factor. Then there exist $p \sim_D p'$ with $F(p) \neq F(p')$. Because p and p' already determine the same standing class, the difference detected by F is not carried by tensor content. The only remaining source of the distinction is representative-only, metadata-level, or path-dependent skin. But the fixed-domain typed-boundary discipline excludes any same-scope standing-relevant discriminator beyond the quotient-determined content. So non-factorization is not a legitimate same-scope refinement; it is illicit same-scope authority import.[7, 21] □

Theorem 5.27 (No stricter admissibility-preserving standing refinement)**.** *Within the fixed substrate of the present capstone, there exists no stricter admissibility-preserving standing generator beneath*

tensor content. Equivalently, any admissibility-preserving auxiliary representation (Y, I, Φ) *of standing satisfies*

$$\mathrm{Ten}_D(x) = \mathrm{Ten}_D(y) \implies \big(\Phi(I(x)) \iff \Phi(I(y))\big).$$

Consequently, any admissibility-preserving attempt to derive standing from an auxiliary invariant collapses to tensor-determined bookkeeping and yields no deeper standing-bearing layer within this substrate.

Proof. Let (Y, I, Φ) be an admissibility-preserving auxiliary representation of standing on the fixed substrate, and suppose that $\mathrm{Ten}_D(x) = \mathrm{Ten}_D(y)$. By definition of standing equivalence, this means $x \sim_D y$. Lawful redescription invariance therefore gives

$$\mathrm{Stand}_D(x) \iff \mathrm{Stand}_D(y).$$

On the other hand, admissibility-preservation of the auxiliary representation gives

$$\mathrm{Stand}_D(x) \iff \Phi(I(x)), \qquad \mathrm{Stand}_D(y) \iff \Phi(I(y)).$$

Combining these equivalences yields

$$\Phi(I(x)) \iff \Phi(I(y)).$$

So no admissibility-preserving auxiliary representation can refine standing more finely than tensor content. Any such representation factors through tensor-determined standing and yields no deeper standing-bearing layer.[5] □

Corollary 5.28 (No second same-scope standing-bearing equivalence)**.** *There is no admissibility-preserving same-scope equivalence relation on the fixed presentation carrier that preserves standing-bearing content while remaining genuinely distinct from* $\sim_D$*. Any such purported second equivalence either collapses to* $\sim_D$ *or else changes scope.*

Proof. If an equivalence relation is coarser than $\sim_D$, it identifies acts with different standing-bearing consequences and therefore fails to preserve same-scope standing-bearing content. If it is finer than $\sim_D$, then it distinguishes acts that already agree on every standing-bearing consequence; by Theorem 5.27, that extra distinction cannot be standing-bearing. So the only admissibility-preserving same-scope equivalence is $\sim_D$ itself, up to bookkeeping.[5, 7] □

Corollary 5.29 (Exhaustion of same-domain discriminator routes)**.** *Within the fixed domain, every purported refinement of admissibility or standing is exactly one of the following:*

1. *quotient-factorizing bookkeeping with no independent authority;*
2. *extensional renaming of the unique gate / standing predicate;*
3. *a representative-only, metadata-level, or path-dependent discriminator barred by same-scope factorization; or*
4. *explicit scope change.*

No fifth same-domain route remains.

Proof. Theorem 5.26 eliminates every purported same-scope discriminator that fails quotient factorization. Theorem 5.9 removes residual dual-track governance between admissibility and standing, so a same-scope refinement cannot survive by splitting the two notions apart. Theorem 5.3 eliminates genuinely distinct same-scope gates, leaving only extensional renaming when no new authority is introduced. Anything left over is either bookkeeping or explicit scope change.[21, 7, 20] □

Inline proof anchor IV: standing quotient uniqueness and factorization discipline

The quotient closure used later is the formal content of *Anchor / Tensor / Skin: A Role Decomposition for Admissible Description* together with the typed boundary discipline of *The Exhaustion Theorem*. The forcing chain is short.

1. Any standing presentation (Y, r, Pred_Y) is complete: if $r(p) = r(p')$, then every same-scope standing-bearing value agrees on p and p', so they lie in the same standing class.
2. Soundness is then forced by stable reference and exhaustion: if $p \sim_D p'$ but $r(p) \neq r(p')$, the presentation is distinguishing two acts that already agree on tensor content and on every same-scope standing-bearing consequence. To maintain that distinction as standing-bearing, one would need a further admissible same-scope discriminator. But the fixed domain has no such discriminator beneath the quotient.
3. Soundness and completeness therefore yield a unique bijection from the standing quotient to any same-scope standing presentation. The quotient is not merely one convenient presentation among many; it is the unique same-scope standing-bearing content space up to unique isomorphism.
4. The factorization theorem is then immediate. If a same-scope standing-relevant functional F fails to factor through the quotient, then there exist $p \sim_D p'$ with $F(p) \neq F(p')$. But p and p' already carry the same tensor content. So the difference detected by F is representative-only, metadata-level, or path-dependent skin, and cannot be admitted as same-scope standing-bearing authority.

This is why non-factorization itself serves as the witness of illicit same-scope authority import.

5.6 The unique admissible interior

The unique-interior theorem is now a short consequence of the gate, standing, conservation, and quotient theorems already fixed.

Definition 5.30 (Admissible interior)**.** For a fixed domain D, the *admissible interior* is the class

$$\mathrm{Int}_D := \{a \in X_D^\bullet : \mathrm{Stand}_D(a) = 1\}.$$

By Theorem 5.8, this is exactly the class of reuse-stably admissible acts on D.

Definition 5.31 (Lawful equivalence of interiors)**.** Two subsets $I_1, I_2 \subseteq X_D^\bullet$ are *lawfully equivalent* if they determine the same subset of the standing quotient $\mathcal{Q}_D$. Equivalently, after identifying

lawfully equivalent presentations, they contain exactly the same standing-positive acts.

Lemma 5.32 (Difference of interiors yields a standing split)**.** *Let $I_1, I_2 \subseteq X_D^\bullet$ be closed under lawful redescription. If I_1 and I_2 are not lawfully equivalent, then there exists an evaluated act $a \in X_D^\bullet$ such that one interior treats a as standing-positive and the other does not.*

Proof. If the two interiors are not lawfully equivalent, then their images in the standing quotient differ. So some lawful-redescription class belongs to one image but not the other. Choose any representative a of that class. Because both interiors are closed under lawful redescription, membership of the class is equivalent to membership of a itself. Hence one interior treats a as standing-positive and the other does not.[4] □

Theorem 5.33 (Unique admissible interior)**.** *For each fixed domain D, the admissible interior is unique up to lawful equivalence. More precisely, if I_1 and I_2 are two claimed admissible interiors for the same domain, then exactly one of the following holds:*

1. *I_1 and I_2 are lawfully equivalent; or*
2. *one of I_1 and I_2 contains inadmissible or reuse-unstable surplus.*

In particular, there is no genuine same-scope plurality of admissible interiors.

Proof. Assume that neither interior contains inadmissible or reuse-unstable surplus. Then every member of either interior is standing-positive by Theorem 5.8. Suppose nevertheless that I_1 and I_2 are not lawfully equivalent. By Lemma 5.32, there exists an act a such that one interior contains a and the other does not. Because neither interior contains surplus, this means that a receives two different standing verdicts on the same fixed domain. But standing is not an independent additional classifier beyond reuse-stable admissibility on the fixed interface. So differing standing verdicts on the same fixed domain would amount either to a second gate or to a substantive same-side split among admitted acts. Both are impossible by Theorems 5.3 and 5.9. Hence the interiors must be lawfully equivalent.[4] □

Corollary 5.34 (No discrete or continuous plurality of admissible interiors)**.** *For a fixed domain there is no genuine plurality of admissible interiors, either discrete or continuously parameterized.*

Proof. A discrete plurality gives two non-equivalent interiors and contradicts Theorem 5.33. A continuously parameterized plurality makes interior membership depend on an admissibility-relevant parameter, which the AMetric boundary and no-parameterization theorems of Section 4 already forbid. The discrete side is the explicit multiplicity elimination proved in *Unus Solus Possibilis Est*; the continuous side is blocked by the AMetric theorem ladder already fixed in Section 4.[6, 3, 4] □

Theorem 5.35 (Non-representability of admissibility by weaker same-domain structure)**.** *Fix one same-scope domain D. Then no weaker same-domain structure can underwrite admissibility or standing as an independent lower authorizer on D. In particular, no metric, probability, score, ranking, selector, privileged representative, structural invariant, categorical surrogate, or same-domain extension may serve as an admissibility-conferring structure except by one of the following four degenerate outcomes:*

1. *it is inert bookkeeping with no admissibility-relevant force;*
2. *it is an illicit import of additional same-scope authority;*
3. *it violates lawful redescription, stable reference, irreversibility, or fixed-domain determinacy; or*
4. *it collapses extensionally to the unique admissibility gate / reuse-stable admissibility already fixed on D.*

Equivalently: admissibility and standing are not representable, on the same fixed domain, by any independent weaker surrogate structure.

Proof. On the pre-admissible side, Section 4 excludes admissibility-relevant metric, ranking, selector, and parameter structure; only equality-pattern content survives there. So no pre-boundary score, metric, ranking, or selector can confer admissibility. On the evaluated side, standing is exactly reuse-stable admissibility on the fixed domain, and any putatively independent authorizer performs gate work. If it changes who is admitted, it competes with the unique gate; if it leaves admission unchanged but refines the positive side substantively, it creates a forbidden same-side split; and if it does neither, it is bookkeeping only. Hence no independent carrier survives except by extensional collapse to the gate itself.

Finally, the same-domain non-derivability closure fixed earlier in the capstone shows that no faithful lower generator or regime-preserving same-domain extension can derive admissibility, standing, reference, or irreversibility for the same original acts without illicit structure import, triviality, violation of the base assumptions, or extensional collapse back to the gate. Therefore no weaker same-domain representer of admissibility exists.[3, 4, 6, 2] □

5.7 Structural necessity and closure of the admissibility architecture

The preceding theorems are now strong enough to be compressed into one fixed-domain closure statement before the capstone turns to operators. The point of the next theorem is not to add a new burden. It makes explicit that the kernel floor, the AMetric boundary, the same-scope gate, the standing quotient, the unique interior, and the conservation law are not a menu of interface options but one forced architecture on the fixed domain.

Theorem 5.36 (Structural necessity and closure of the admissibility architecture)**.** *Fix one same-scope domain D inside the target class already secured in Sections 2–4. Then, up to lawful equivalence, exactly one admissibility architecture is available on D:*

1. *a prior binary admissibility gate on the evaluated fragment;*
2. *standing and admissibility equivalent on that gate;*
3. *one standing quotient carrying all same-scope standing-bearing content, with every same-scope standing-relevant discriminator forced to factor through it;*
4. *one admissible interior, unique up to lawful equivalence; and*
5. *a conservation law under which standing can only be preserved or destroyed on the same scope.*

Any purported alternative same-domain architecture must instantiate at least one of the following defeated forms:

1. *pre-boundary selector, metric, order, ranking, parameter, or deeper invariant;*
2. *residual dual-track governance separating admissibility from standing;*
3. *a second same-scope gate or an admissibility-relevant same-side split;*
4. *same-act repair, generator, or independent carrier;*
5. *a representative-only, metadata-level, or path-dependent discriminator that fails factorization through the standing quotient;*
6. *illicit scope transport or fresh same-domain authority supplied by weaker surrogate structure; or*
7. *extensional collapse to the unique gate / standing predicate already fixed on D.*

Hence no alternative same-domain admissibility architecture exists.

Proof. Section 4 already closes the proposal side. The AMetric boundary theorem, the no-deeper-invariant theorem, and the derived non-parameterization theorem eliminate every pre-boundary selector, metric, order, ranking, parameter, and purported deeper invariant as same-domain admissibility authority. On the evaluated side, Theorem 5.2 forces a prior admissibility partition, and Theorem 5.3 rules out any competing same-scope gate. Theorem 5.9 removes residual dual-track governance between admissibility and standing, so no same-scope split remains in which admissibility is neutral while standing is separately conferred. Theorems 5.12, 5.13, and 5.16 remove every same-scope route by which positive-side authority could be recreated beneath or beside the gate. Theorem 5.24, Corollary 5.25, and Theorem 5.26 force all same-scope standing-bearing content and every same-scope standing-relevant discriminator through one quotient, while Theorem 5.33 and its plurality corollary force one admissible interior up to lawful equivalence. Theorem 5.18 and its cross-domain corollary close the remaining routes of creation, amplification, transfer, or recovery. Finally, Theorem 5.35, read together with the kernel-floor non-derivability theorems of Section 2, excludes weaker same-domain surrogate structures and faithful same-domain extensions except by illicit authority import, violation of the base goods, inert bookkeeping, or extensional collapse back to the gate. These are exactly the defeated alternative forms listed above, so no seventh same-domain admissibility architecture remains. □

Corollary 5.37 (No design-choice reading of the admissibility architecture)**.** *Within the relevant regime, the kernel floor together with the admissibility architecture of the present section is a theorem-level necessity rather than an optional axiomatic or interface-design choice. To reject it, one must deny either the target-domain theorems of Section 2, the AMetric theorems of Section 4, or one of the fixed-domain closure theorems collected above.*

Proof. The kernel floor is fixed as derived rather than axiomatic in Section 2. The present theorem then shows that, once that floor and the AMetric boundary are in place, the same-domain admissibility architecture has no rival surviving form. So what remains is theorem-level necessity, not a menu of undischarged design alternatives. □

Remark 5.38 (Counterexample trichotomy and primitive-floor clarification). Within the fixed-domain meaningful regime defined in this manuscript, no alternative admissibility architecture exists. Any purported counterexample either (i) changes scope, (ii) violates the base conditions required for non-degenerate reasoning, or (iii) collapses extensionally to the unique admissibility / standing gate. This is not a merely presentational way of stopping the derivation. The admissibility–standing–reference–irreversibility kernel is fixed in the capstone as a theorem-level floor: its non-derivability, minimality, and uniqueness are proved within the regime, the residual separation between admissibility and standing is eliminated by equivalence closure, and same-scope standing-relevant discriminators are exhausted by quotient factorization rather than by an open list of unchecked alternatives. Primitive status therefore means structural non-derivability within the same fixed-domain meaningful regime, not temporary axiomatization or interface-design choice.

Remark 5.39 (Why this compression matters). The capstone does not gain new mathematics by stating Theorem 5.36. It gains proof visibility. The theorem compresses a distributed chain into the exact hostile-reader claim that matters later: mathematics, physics, and same-domain meta-level escape are all downstream of an interface whose same-domain architecture is already closed.

Remark 5.40 (What is now fixed for later sections). The capstone may now treat the following as theorem-local fixed points: some gate exists on every non-degenerate fixed domain; no competing same-scope admissibility classifier survives; admissibility and standing are equivalent on that fixed domain; no repair, generator, or carrier route survives on the same scope; standing can only be preserved or destroyed; every same-scope standing-bearing presentation and standing-relevant discriminator factors through the standing quotient; no stricter admissibility-preserving standing refinement remains beneath tensor content; and the admissible interior is unique up to lawful equivalence. This is the exact closure layer required before operator exhaustion begins.

6 Operator exhaustion and mathematics closure

6.1 Why the capstone can now turn to operators

Section 5 fixed the proof-bearing center on which later closure depends: a unique same-scope gate, standing emergence as reuse-stable admissibility, standing conservation, standing-quotient uniqueness, the no-second-closure-relevant-equivalence theorem, the unique admissible interior, and the compressed closure claim of Theorem 5.36 that no alternative same-domain admissibility architecture survives once that interface package is in place. Once that closure layer is in place, the next and unavoidable question is whether mathematics can reopen the interior by operator means. The point of the present section is to show that it cannot. The burden is not to deny the legitimacy of completion, forcing, compactness, saturation, universal constructions, or readback arguments. The burden is narrower and more exact: to classify their fixed-scope status once the admissible interior is already fixed, and to show that no same-scope operator family yields new standing-bearing power there.

The local proof surface for this section is carried primarily by *Closure by Exhaustion for Same-Scope Operators under Admissibility* together with the conservative-extension and post-closure collapse results of *Clausura Ex Necessitate Structurae*; the older fixed-base presentation is retained only as a predecessor and cross-check, and no later synthesis paper is used here as an independent theorem

engine.[8, 10, 9, 3, 11]

6.2 Standing-bearing carrier and same-scope operator setting

Definition 6.1 (Fixed admissibility base)**.** A *fixed admissibility base* is a triple

$$B = (X, \Omega, \mathcal{A}),$$

where X is a carrier, Ω is a family of partial operations on X, and $\mathcal{A}$ is the admissibility regime specifying which primitive applications and compositions count as licensed on that carrier. In the capstone, this language is the operator-side translation of the same-scope standing regime already fixed in Section 5: "old-base-relevant content" means the previously admitted same-scope content of the fixed interior.

Definition 6.2 (same-scope admissible extension)**.** Let $B = (X, \Omega, \mathcal{A})$ be a fixed admissibility base. An admissible extension

$$B^* = (X^*, \Omega^*, \mathcal{A}^*)$$

is *same-scope* iff it is presented as preserving X as the standing-bearing carrier under discussion rather than openly replacing X by a quotient, completion, extension, new ambient structure, or new admissibility-bearing regime.

Definition 6.3 (Carrier-preserving operator)**.** A purported operator F on admissible inputs from X is *carrier-preserving* if it is presented as acting on that same base carrier X rather than by openly replacing X with a quotient, completion, extension, new ambient structure, or new same-scope gate.

Definition 6.4 (Conservative action on old-base-relevant content)**.** A purported operator or redescription on $B = (X, \Omega, \mathcal{A})$ acts *conservatively on old-base-relevant content* if it adds no new values for old operations on old tuples and no new same-scope consequences about the old base beyond those already fixed by the original regime.

Definition 6.5 (Internally fixed auxiliary datum)**.** Let $x \in X$ be an admissible input. An auxiliary datum $a(x)$ is *internally fixed over* x if every admissibility-preserving redescription of the base that leaves the standing content of x unchanged also leaves $a(x)$ unchanged, up to the normalizations already licensed by the regime. If the relevant datum is not internally fixed in that sense, it is *external*: it must be supplied by a selector, witness, evaluator, comparison map, completion rule, generic object, transfer criterion, or other ambient structure not already fixed by the base.

Lemma 6.6 (Old-graph change is non-conservative)**.** *Let $A = (X, \Omega)$ and let $A^* = (X^*, \Omega^*)$ be an admissible extension. Suppose there exist $\omega \in \Omega$ of arity n and $\bar{a} \in X^n$ such that*

$$\bar{a} \notin \mathrm{dom}(\omega) \quad \textit{but} \quad \bar{a} \in \mathrm{dom}(\omega^*).$$

Then the induced partial-algebra structure on the old carrier X has changed. In particular, the maneuver is not conservative on the standing-bearing carrier.

Proof. The graph of a primitive operator on X is part of the partial-algebra structure carried by X. If $\bar{a}$ was previously outside $\mathrm{dom}(\omega)$ but becomes defined under ω^*, then the graph of the old

primitive has been enlarged on old data. That is a change in the induced partial-algebra structure on X, not a mere notational enrichment. □

Lemma 6.7 (Domain fixity on the standing-bearing carrier)**.** *Let $A = (X, \Omega)$ be a nondegenerate partial algebra, and let $A^* = (X^*, \Omega^*)$ be a same-scope admissible extension. Then for every $\omega \in \Omega$ of arity n,*

$$\mathrm{dom}(\omega^*) \cap X^n = \mathrm{dom}(\omega).$$

Proof. Because ω is the restriction of ω^*, every old defined tuple remains defined, so

$$\mathrm{dom}(\omega) \subseteq \mathrm{dom}(\omega^*) \cap X^n.$$

Assume the inclusion is strict. Then some tuple $\bar{a} \in X^n$ is newly defined in the extension. By Lemma 6.6, the induced partial-algebra structure on the old carrier X has changed. But a same-scope admissible extension is presented precisely as preserving X as the same standing-bearing carrier. Changing the primitive graph on X is therefore not lawful redescription but an actual change of scope. Hence strict enlargement is impossible, and equality follows. □

Theorem 6.8 (No internal totalization)**.** *Let $B = (X, \Omega, \mathcal{A})$ be a nondegenerate fixed admissibility base. No same-scope carrier-preserving extension of B totalizes a genuinely partial old operation on old tuples.*

Proof. Suppose some old primitive $\omega \in \Omega$ is partial on X but becomes total on old tuples in a same-scope carrier-preserving extension. Then

$$\mathrm{dom}(\omega^*) \cap X^n = X^n$$

while

$$\mathrm{dom}(\omega) \subsetneq X^n.$$

This contradicts Lemma 6.7. So no same-scope carrier-preserving extension totalizes a genuinely partial old operation on old tuples. □

Corollary 6.9 (No post hoc repair and no deferred admissibility on old tuples)**.** *If an old tuple is outside the domain of an old operation at the stage of formation, no later same-scope carrier-preserving extension makes that old application defined on the same old data. In particular, an undefined operator application on the standing-bearing carrier cannot be rendered admissible after the fact by same-scope extension.*

Proof. Any such repair or deferral would have to turn a previously undefined old tuple into a newly defined one for an old primitive on the old carrier. That is exactly the graph change forbidden by Lemma 6.7 and Theorem 6.8. □

Theorem 6.10 (Auxiliary-data dichotomy)**.** *Let F be a carrier-preserving operator on admissible input $x \in X$ for a fixed base $B = (X, \Omega, \mathcal{A})$. Suppose the old-base-relevant effect of $F(x)$ is determined, up to conservative action on old-base-relevant content, by x together with an auxiliary datum $a(x)$. Then exactly one of the following holds.*

1. *If $a(x)$ is internally fixed over x, then F acts conservatively on the old base.*
2. *If $a(x)$ is external, then F is not internal to the base structure.*

Proof. If $a(x)$ is internally fixed, then admissibility-preserving redescription leaves it unchanged wherever the standing content of x is unchanged. So the effect of F contributes no genuinely new distinction beyond what was already fixed by the base together with x. In that case F only reorganizes or reads off previously fixed content and is therefore conservative on the old base.

If $a(x)$ is external, then the effect of F depends essentially on a datum not determined by the old base together with x. The operator's effect is then not fixed by the base alone, so it is not internal to that base. No third option remains. □

6.3 Finite exhaustion of totalization and standing-extension moves

Definition 6.11 (Covered totalization mechanisms)**.** A purported strategy for rendering a partial operator total on the standing-bearing carrier falls under the following covered mechanisms whenever it uses the corresponding move.

1. **Domain enlargement.** Old tuples on the standing-bearing carrier become newly defined.
2. **Carrier collapse.** The effective carrier is changed by quotient, identification, completion, or replacement.
3. **Evaluative completion.** A new global evaluator, comparison rule, semantic predicate, limit device, metric, measure, probability rule, or analogous decision procedure settles cases previously undefined on the old carrier.
4. **Conservative definitional extension.** Only notation or abbreviatory language changes.
5. **Non-conservative family change.** The operator family or admissibility regime is altered in a way not reducible to the first four mechanisms.

Lemma 6.12 (Finite exhaustion of totalization moves)**.** *Let τ be a purported same-scope strategy that renders a partial operator total on the standing-bearing carrier. Then τ falls under at least one of the five covered mechanisms above.*

Proof. A successful totalization must make some previously undefined input evaluable. Inspect the first point at which the strategy differs from the original regime. If an old tuple on the standing-bearing carrier becomes newly defined, the move is domain enlargement. If old undefinedness disappears only because the object under discussion has been identified, collapsed, or replaced, the move is carrier collapse. If the old input is decided by adjoining an evaluator, semantic predicate, comparison totality, limit device, metric, measure, or probability rule, the move is evaluative completion. If none of those changes occurs and the only modification is notational, the move is conservative definitional extension. The only remaining possibility is that the strategy changes the governing operator family or admissibility regime in a way not captured by pure notation, which is non-conservative family change. Therefore every same-scope totalization attempt is covered. □

Proposition 6.13 (Disposition of the covered mechanisms)**.** *Within the standing-relative framework fixed by Sections 4 and 5, each covered mechanism is blocked or vacuous.*

1. *Domain enlargement is forbidden by Theorem 6.8.*
2. *Carrier collapse changes the carrier and so is not a same-scope extension of the original algebra.*
3. *Evaluative completion hits the AMetric boundary and therefore presupposes an imported evaluator or other same-scope boundary-side authority.*
4. *Conservative definitional extension is bookkeeping and leaves undefinedness unchanged.*
5. *Non-conservative family change is an open regime change rather than an internal same-scope strengthening of the original base.*

Proof. Clause (1) is exactly the old-graph obstruction already proved. Clause (2) is immediate from the same-scope convention fixed above. Clause (3) decides old undefined cases by adjoining a standing-relevant evaluator not generated by the standing-bearing carrier; that is boundary selector work or imported authority and is blocked by the AMetric boundary and the no-carriers theorem of Section 5. Clause (4) alters only language, so by definition it cannot change domains or make an old undefined tuple defined. Clause (5) changes the operator family or admissibility policy itself and is therefore an explicit change of regime rather than a same-scope extension. □

Theorem 6.14 (Finite exhaustion schema for same-scope standing-extension)**.** *Every same-scope standing-extending operator attempt falls under at least one of the five covered mechanisms stated just above. Each covered mechanism is blocked or vacuous. Consequently, no same-scope operator can totalize old undefined applications, repair standing on the old carrier, or otherwise enlarge standing-bearing coverage on the original scope.*

Proof. Any same-scope standing-extending operator attempt must do at least one of the following: make an old undefined tuple newly defined, change the carrier on which admissibility is measured, add a new evaluator or completion device to settle old undefinedness, claim only notational enrichment, or alter the governing operator family or admissibility regime. By Lemma 6.12, those possibilities are exhaustive. By Proposition 6.13, each is blocked or vacuous. Therefore no same-scope standing-extending operator survives. □

Corollary 6.15 (No terminal same-scope totalization)**.** *There is no universal, terminal, reflective, or best same-scope admissible completion that renders partial operators total on the standing-bearing carrier.*

Proof. Any such object would itself realize a surviving same-scope totalization strategy, contradicting Theorem 6.14. □

6.4 Conservative extension collapse

Theorem 6.16 (Conservative extension collapse)**.** *Relative to the fixed admissible interior, every purported same-scope operator enlargement is of exactly one of the following two kinds.*

1. *It is conservative on already admitted content: it adds no new old-tuple values, no new same-scope consequences, and no new standing-bearing content beyond definitional or bookkeeping redescription.*

2. *It depends on external witnesses, evaluators, selectors, generic objects, carrier change, regime enlargement, or fresh same-scope authority, and is therefore not a genuinely internal enlargement of the old scope.*

In particular, there is no same-scope operator enlargement that is both nonconservative and genuinely internal.

Proof. The direct route in which an old primitive acquires new values on old tuples is blocked by Theorem 6.8. The indirect route in which the apparent effect of an operator is mediated by an auxiliary datum is handled by Theorem 6.10: if the datum is internally fixed, the move is conservative; if it is external, the move is not internal to the base.

It remains to consider admissibility-preserving theory extensions. Here *Clausura Ex Necessitate Structurae* supplies the decisive collapse result: once the old operator behavior on the old carrier is fixed, any conservative standing-bearing extension is definitional/bookkeeping on the old scope. A new symbol, evaluator, or operator-role that genuinely participates in standing-bearing claims must either be explicitly definable from the old language on the old carrier or else rely on operational, semantic, witness-based, or meta-foundational authority not already fixed there. In the first case the extension is definitional bookkeeping; in the second it is not an internal same-scope enlargement at all. No third status remains. □

Corollary 6.17 (Interpretability collapse)**.** *If an operator symbol, enrichment, or theory extension is uniquely determined by the already fixed operator core on the evaluated fragment, then its same-scope standing-bearing content is definitional bookkeeping rather than new power.*

Proof. Uniqueness of interpretation relative to the fixed operator core means that the enrichment adds no new same-scope distinctions about what the old base can do. In the standing vocabulary of the capstone, that is exactly the conservative case of Theorem 6.16. So same-scope internal enrichments determined entirely by the existing core are bookkeeping rather than new operator power. □

Inline proof anchor V: operator exhaustion on the standing-bearing carrier

The operator-closure claim is the conjunction of the graph-rigidity theorem in *Closure by Exhaustion for Same-Scope Operators under Admissibility*, the conservative-extension and interpretability collapse theorems in *Clausura Ex Necessitate Structurae*, and the cleaner fixed-base predecessor in *Fixed-Base Exhaustion for Internal Operators* [8, 10, 9]. The forcing chain has five steps.

1. *Old-graph rigidity.* If some old tuple $\bar{a} \in X^n$ with $\bar{a} \notin \mathrm{dom}(\omega)$ becomes newly defined in a same-scope carrier-preserving extension, then the graph of an old primitive has changed on old data. That is already a change in the induced old structure on X, not a harmless redescription.

2. *No same-scope totalization.* A same-scope carrier-preserving extension therefore cannot make a genuinely partial old primitive total on old tuples. The same argument yields no post hoc repair and no deferred admissibility for old undefined applications: changing the timing still changes the old graph.

3. *Auxiliary-data dichotomy.* If the apparent effect of an operator F is mediated by an auxiliary datum $a(x)$, then exactly two cases remain. If $a(x)$ is internally fixed over x, the operator only reads off structure already fixed by the base and is conservative on old-base-relevant content. If $a(x)$ is not internally fixed, then F depends essentially on an external selector, witness, evaluator, generic object, comparison map, completion rule, or other imported datum and is therefore not internal to the base.

4. *Finite exhaustion of totalization and standing-extension moves.* Any same-scope attempt to totalize, complete, or enlarge standing-bearing coverage must instantiate at least one of five covered mechanisms: domain enlargement, carrier collapse, evaluative completion, conservative definitional extension, or non-conservative family change. Each is already blocked or vacuous.

5. *Conservative extension collapse.* Once old operator behavior on the old carrier is fixed, every remaining admissibility-preserving same-scope enrichment is either definitional/bookkeeping on the old scope or else depends on fresh same-scope authority, imported witness load, or open regime change. No third same-scope status remains.

This is the exact operator-side hinge compressed in Theorems 6.8, 6.10, 6.14, and 6.16: same-scope internal strengthening is exhausted by conservative bookkeeping on the old base or explicit dependence on external data, witnesses, evaluators, or carrier change.

6.5 Closure of the major mathematical operator families

The point of operator exhaustion is not to deny the mathematical usefulness of ambient methods. It is to classify their fixed-scope status once the admissible interior is already fixed. The families below are not disconnected mini-arguments. They are recurring sites at which the same conservative/external dichotomy reappears.

Family	**Load-bearing datum or move**	**Fixed-scope status after closure**
Choice and selection	selector rule, ranking, or representative criterion	Conservative only if the relevant selector is already internally fixed; otherwise imported and inadmissible.
Genericity and forcing	generic witness, surrogate evaluator, or transfer criterion	Generic output is external; readback or absoluteness is conservative only when no new same-scope standing-bearing content is added.
Completion and evaluative closure	completion domain, boundary, embedding, or evaluator	Either external by carrier change or conservative on old-base-relevant content if the evaluative rule was already fixed.
Compactness, saturation, and model-theoretic existence	test-family totality, local-to-global rule, larger model, or certifying witness structure	Conservative only in the vacuous/definable case; otherwise external ambient enlargement or witness import.
Universal-property existence	completed comparison totality or universal witness object	Conservative only if the witness is already fixed strongly enough that no new same-scope content is added; otherwise external and typically carrier-changing.
Infinity escalation	completed or exhaustively idealized totality	Explicit regime enlargement rather than internal strengthening of the original base.
Typicality, measure, and probabilistic authorization	comparison class, weighting rule, evaluative carrier, or selector principle	Carrier laundering or boundary-side selection; not a primitive same-scope operator source of standing.
Canonical representatives and quotients	normalization or identification rule	Conservative only if the normalization rule is internally fixed; otherwise selector import or carrier change.
Meta-foundational reanchoring	standing-conferring meta-criterion or authority	Inadmissible as an operator on the old scope.
Repair and rescue schemes	repair rule, rescue authority, or post hoc standing override	No admissible same-scope instance exists.

Theorem 6.18 (Flagship family closure)**.** *Every major mathematical operator family classified above is exhausted by the same fixed-scope dichotomy: the move is either conservative on already admitted content because its load-bearing datum is already internally fixed, or it depends on external*

data, witness import, carrier change, evaluative completion, or regime enlargement and is therefore not a new same-scope source of standing-bearing power.

Proof. Choice and selection reduce to the status of the selector or representative rule: if that rule is already standing-fixed, the move is conservative; if not, it is imported and therefore inadmissible. Genericity and forcing reduce to the status of the generic witness or transfer criterion: direct generic output is external, while readback is conservative only when no new same-scope standing-bearing content is added. Completion reduces to whether the completion domain, boundary, embedding, or evaluator is already fixed; if not, the move is external or carrier-changing. Compactness, saturation, and model-theoretic existence proceed by larger models, realized types, certifying witness structures, or local-to-global rules; unless the witness or comparison totality was already fixed, the move is external. Universal constructions are posed with respect to ambient comparison data and witness objects, so they are conservative only when that witness data was already fixed strongly enough that no new same-scope content is added. Infinity escalation openly changes the regime being measured. Typicality, measure, and probabilistic authorization attempt to turn an evaluative carrier into a same-scope authorizer and are therefore blocked by the no-carriers theorem and by boundary non-parameterization. Canonical representatives and quotients depend on a normalization or identification rule and therefore either collapse to bookkeeping or change carrier. Meta-foundational re-anchoring imports standing-conferring authority rather than generating same-scope standing. Repair schemes are ruled out by the no-repair / no-generator / no-carrier closure package already fixed in Section 5. In every case the result is conservative readback, bookkeeping, external dependence, or regime change — never a third same-scope source of standing-bearing power. □

Corollary 6.19 (Why the standard detours do not reopen mathematics)**.** *Forcing, compactness, saturation, completion, choice, typicality, genericity, universal-property existence, model-theoretic existence, canonicalization, meta-foundational re-anchoring, and repair schemes do not reopen mathematics inside the fixed admissible interior. What they yield, when read back to the old base, is either conservative content already fixed there or externally dependent content that is not internal to that base.*

Proof. This is the direct capstone reading of Theorem 6.18. The ambient methods remain mathematically legitimate, but their fixed-scope status is exhausted by conservative readback, bookkeeping, and external dependence. Since no third same-scope status remains, they do not reopen the interior. □

6.6 Interaction closure: no emergent same-scope family

Definition 6.20 (Audited kernel)**.** An *audited kernel* is one of the imported datum types already represented in the family-wise layer: a selector or representative rule, a generic or forcing witness, a completion or total evaluator, a global family of constraints, a global comparison totality, a certifying witness, an infinitary completion surrogate, a weighting class, a privileged identification rule, a repair rule, or a meta-foundational authority.

Lemma 6.21 (Kernel provenance)**.** *Let F be a same-scope operator on admissible input $x \in X$. If F is not bookkeeping-equivalent on x, then the action of F depends on at least one audited kernel.*

Proof. If F is not bookkeeping-equivalent, then by contraposition of Theorem 6.10 some datum needed by F is not internally fixed over x. Such a non-standing-fixed datum is imported. The family audit above records the possible imported datum types that can do standing-bearing work on the old carrier. Those are exactly the audited kernels listed here. Hence every non-bookkeeping same-scope operator exhibits at least one audited kernel. □

Lemma 6.22 (Bookkeeping is stable under finite composition)**.** *A finite composite of bookkeeping-equivalent same-scope stages is bookkeeping-equivalent.*

Proof. Each bookkeeping-equivalent stage changes presentation only and adds no new operator power on the carrier it receives. Composing finitely many such stages still changes presentation only. Hence the composite is bookkeeping-equivalent. □

Theorem 6.23 (Composition and alternation closure)**.** *No finite composition, alternation, or same-scope pipeline assembled from same-scope operators creates a new admissible operator family. If a composite is not bookkeeping-equivalent, then some stage already instantiates an audited kernel, and the composite is therefore either a rebranded registry family or inadmissible.*

Proof. Let

$$F = F_n \circ F_{n-1} \circ \cdots \circ F_1$$

be a finite same-scope pipeline on admissible input x. If every stage is bookkeeping-equivalent on the input it receives, then F is bookkeeping-equivalent by Lemma 6.22. Assume instead that F is not bookkeeping-equivalent. Choose the least index j such that the prefix

$$F_j \circ \cdots \circ F_1$$

is not bookkeeping-equivalent. Then the shorter prefix $F_{j-1} \circ \cdots \circ F_1$ is bookkeeping-equivalent, so the first standing-relevant departure occurs at stage F_j. By Lemma 6.21, stage F_j already depends on an audited kernel. Later stages can only propagate, rename, or mask that kernel; they cannot turn it into a genuinely new same-scope family. Hence the composite is either a rebranded registry family or inadmissible. □

Theorem 6.24 (Iteration, self-reference, and transfer closure)**.** *No staged or iterative procedure, no same-scope diagonal or self-referential construction, and no representation-change, foundation-swap, absoluteness, or transfer maneuver creates a new admissible same-scope operator family.*

Proof. If alleged novelty appears at some finite stage of an iterative procedure, then a finite prefix is already non-bookkeeping, and Theorem 6.23 reduces the novelty to an audited kernel already present at that stage. If every finite stage is bookkeeping-equivalent and novelty is claimed only in an eventual output, then some extra datum is needed to turn the stagewise trace into a standing-bearing result. That datum can only be a completion or total evaluator, an infinitary or idealized passage, a forcing or genericity move, a weighting or stabilization criterion, or a repair of nonstanding intermediate stages — all already closed by Theorem 6.18 and Theorem 6.14.

A nontrivial diagonal or self-referential maneuver must choose a privileged code or representative, totalize a self-applied evaluator, certify the result by a witness structure or meta-authority, or pass

to an infinite or completed object not locally generated at the point of action. Those possibilities are exactly the already closed families of choice/canonicalization, completion, model-theoretic existence, infinity escalation, and meta-foundational re-anchoring. If none of those occur, the construction is only a self-map inside the fixed regime and therefore collapses, after normalization, to bookkeeping identity capture rather than a new operator source.

Finally, if a purported operator moves from one representational regime to another and back again on a shared same-scope domain, then either the round trip is admissible on a normalization-fixed overlap, in which case it changes presentation only and is bookkeeping, or else it depends on a privileged interpretation, selector, witness, or authority not fixed by the overlap. That is imported standing-bearing load and is inadmissible. Hence representation change, foundation swap, absoluteness, and transfer never create a new same-scope admissible family. □

Theorem 6.25 (No admissible emergent family)**.** *No emergent operator family is admissible. Equivalently, no same-scope family not already audited in the family-wise registry can acquire standing-bearing force merely by hidden-kernel disguise, composition, iteration, diagonalization, representation change, foundation swap, absoluteness, or transfer among already closed families.*

Proof. Let F be a purported emergent same-scope operator family. If a genuinely non-bookkeeping member of F exists, then by Lemma 6.21 it exhibits an audited kernel. If that kernel is already one of the named registry families, then F is merely that family under redescription. If novelty is instead claimed to arise from assembly rather than from a primitive kernel, then Theorems 6.23 and 6.24 show that the assembly is bookkeeping or inadmissible. Thus no same-scope admissible family remains outside the registry. □

6.7 Mathematics closure

Theorem 6.26 (Mathematics closure)**.** *Mathematics, treated as an operator system under admissibility, is closed by exhaustion. Equivalently, once the admissible interior is fixed, no same-scope operator family can generate new standing-bearing mathematical content beyond conservative readback, bookkeeping-equivalent refinement, or explicit scope change.*

Proof. Theorem 6.8 blocks the direct route in which old partiality is totalized on the old standing-bearing carrier. Theorem 6.14 then shows that every same-scope totalization or standing-extension attempt falls under one of the five covered mechanisms and is blocked or vacuous. Theorem 6.16 classifies every remaining apparent operator enlargement as conservative bookkeeping or external dependence. Theorem 6.18 shows that the major mathematical operator families instantiate no further same-scope case beyond that dichotomy. Theorem 6.25 closes the last loophole by ruling out interaction-generated novelty. Therefore mathematics, as an admissible operator system, is closed by exhaustion. □

Corollary 6.27 (What survives mathematics after closure)**.** *After closure, mathematics is not empty and not dispensable. What remains is internally fixed, standing-preserving structural coordination within the already closed admissible interior: refinement, lawful redescription, definitional enrichment,*

invariant readback, proof, mapping, and disciplined transformation that do not claim new same-scope standing-bearing power.

Proof. The closure theorem blocks same-scope standing-generative enlargement, not standing-preserving work inside the already admitted interior. Conservative readback, definitional redescription, invariant comparison, and proof remain legitimate because they do not add new old-base-relevant standing-bearing content. This is exactly the post-closure classification of mathematics drawn from *Clausura Ex Necessitate Structurae* and restated in capstone form by *Eadem Natura, Duo Spectra*: mathematics survives as internally fixed standing-preserving structural coordination, not as a standing-generative foundation. □

7 Physics closure and descriptive role discipline

7.1 Why physics appears only after mathematics closure

Section 6 closed the same-scope operator question for mathematics. The present section asks the parallel physical question: once the unique admissible interior has already been fixed and no same-scope operator family can reopen it, what becomes of physical description itself? The answer must be narrower than an empirical theory of nature and stronger than a purely rhetorical claim that everything is bookkeeping. The burden here is classificatory and role-theoretic: to show, under the already fixed closure spine, what descriptive status physical structure may still carry without reopening standing. This section should therefore not be read as a second proof engine parallel to Sections 2–6. Its local task is to derive the descriptive-role discipline forced once the earlier theorem chain and the imported physical closure sources are already in place. The burden is to show that physical description may be rich, metric, analytic, and dynamically elaborate while still remaining non-generative with respect to standing.

This burden is classificatory rather than reconstructive.

This section classifies standing-bearing roles of physical descriptions under the fixed admissibility architecture; it does not introduce an independent generative theory of physical law.

What this section does not claim is equally important. It does not replace successful physical formalisms, derive new empirical predictions, or deny the mathematical and experimental richness of physical theory. Its burden is narrower: to classify which physical structures can still carry standing-bearing force once the earlier closure spine is fixed.

Unlike the fully local forcing carried in Sections 2–6 and Section 8, the present section does not introduce a second independent theorem engine beneath the physical applications. It states, under explicit theorem support, what physical description may and may not do once the prior closure spine is already fixed.

For this classificatory layer the live theorem support is limited and one-directional: *Clausura Ex Necessitate Structurae* carries the metric-licensing and physics-closure hinge, *Anchor / Tensor / Skin: A Role Decomposition for Admissible Description* contributes only the derived role discipline, and the Standard Model and measurement papers enter only as compact downstream applications

rather than as independent proof engines.[10, 7, 12, 13]

7.2 Licensed metric description

Definition 7.1 (Licensed physical description)**.** A *licensed physical description* is a same-scope physical presentation whose metric, analytic, probabilistic, state-descriptive, or representational content is admitted only inside an already fixed interior and only insofar as it preserves standing under lawful redescription. Such a description may be dynamically rich, but it does not thereby become a same-scope source of standing.

Definition 7.2 (L-metric articulation)**.** An *L-metric articulation* is the admissible interior use of metric or analytic descriptors once the AMetric boundary has already been crossed and the relevant scope has been fixed. It is licensed exactly as interior bookkeeping and comparison structure, not as a boundary selector or primitive same-scope authorizer.

Theorem 7.3 (Licensed metric description theorem)**.** *Within the fixed admissibility-governed interior, metric, probabilistic, analytic, and state-descriptive structure may appear only as licensed interior bookkeeping. In particular, no such structure is primitive at the boundary and none can function as a same-scope authorizer of standing.*

Proof. At the boundary, the capstone has already fixed a binary, non-eventful, and non-transmissive AMetric interface: there is no admissibility-relevant selector, ranking, grading, metric, or operator there. So any metric or analytic object that is asked to determine standing before admissibility is fixed would reintroduce exactly the kind of parameterization or boundary action already blocked in Section 4.

Suppose, then, that metric or analytic structure is introduced only after admission. If it is now asked to decide which acts stand, then it functions as a carrier attempt in the sense of Section 5: it becomes a primitive authorizer distinct from the unique same-scope gate. But the no-carriers theorem already rules that out. So the only admissible remaining role for such structure is licensed interior bookkeeping: it can describe, compare, coordinate, track, or calculate within the admitted interior, but it does not thereby become an independent same-scope source of standing-bearing force.

This is precisely the physical-side content imported from *Clausura Ex Necessitate Structurae* in capstone language: the L-metric boundary licenses metric articulation without making it generative. Hence metric, probabilistic, analytic, and state-descriptive structure are licensed only as interior bookkeeping and may not function as primitive same-scope authorizers of standing. □

Corollary 7.4 (No cross-fixation comparability from metricity alone)**.** *Metric articulation is fixation-local. If a fixed physical scope admits metric description, that does not by itself license cross-fixation comparison, cross-domain identification, or global ontic transport.*

Proof. A cross-fixation comparison would require an additional same-scope identification aligning relata, comparison standards, or domain structure across more than one fixation. But metricity alone is only licensed interior description within a fixation. Without an independently licensed

cross-scope gate, metricity therefore cannot by itself authorize cross-fixation comparison or transport. So the admissible role of metric articulation remains fixation-local. □

7.3 Physical description is non-generative

Definition 7.5 (Physical non-generativity)**.** A fixed-scope physical description is *non-generative* iff it does not create, amplify, repair, transfer, or primitively authorize standing. It may coordinate already admitted content, but it cannot produce new same-scope standing-bearing force.

Theorem 7.6 (Physical non-generativity theorem for admissible physical description)**.** *No admissible physical description on the fixed scope becomes an independent same-scope source of standing-bearing force by metric articulation, typicality, boundary narrative, ontic carrier laundering, or descriptive enrichment alone. All admissible physical descriptions are therefore standing-preserving bookkeeping within the unique admissible interior unless they openly change scope by adding new witness load, new selector structure, or new admissibility-bearing data.*

Proof. Any alleged same-scope elevation of physical description into a standing-bearing authorizer would have to proceed by one of the routes already excluded elsewhere in the capstone. If it works by a derivative score, typicality, probability, fit, or threshold, then it is carrier laundering and is blocked by the no-carriers theorem. If it works by a story of metric emergence, boundary productivity, or pre-admissible dynamic crossing, then it violates the AMetric and non-eventful boundary theorem. If it works by transporting standing from another scope, another observational frame, or another descriptive envelope, then it is illicit scope transport and is blocked by the fixed-scope discipline. If it works by local redescription or by representational choice alone, then it violates skin non-authority and redescription neutrality.

So a physical description may be rich, but any same-scope standing-bearing force it is claimed to add must either collapse to already admitted bookkeeping or enter by explicit scope change. The latter case is not same-scope generation but a new regime. Hence admissible physical description is non-generative on the fixed scope: the theorem classifies what physical theories may do as admissible descriptions, not what physical reality must be independently of that classificatory setting. □

Theorem 7.7 (Conditional locality eliminator)**.** *Assume locality is retained as a structural admissibility constraint on the fixed physical scope. Then any purported locality-preserving theory extension falls into exactly one of the following classes:*

1. *it is redundant, adding no genuine new constraint content beyond locality itself;*
2. *it requires hidden global coordination to remain coherent and therefore ceases to be locality-preserving; or*
3. *it parasitically offloads its enforcement onto already fixed structure and therefore contributes no independent same-scope standing-bearing content.*

In no case does such an extension generate new same-scope standing-bearing physical content.

Proof. If an extension adds no genuine new constraint content beyond locality, then it is redundant

by definition. If it does add such content, then that content must be coordinated somewhere. If the coordination is genuinely new and same-scope, the extension requires a global structure or external synchronization not already contained in purely local articulation; that breaks the locality hypothesis. If the extension instead claims to get its force "for free" from the already fixed locality discipline, then nothing new is enforced and the added structure is parasitic on the existing regime. These cases are exhaustive once locality is treated as a structural admissibility constraint. Therefore locality alone eliminates most purported same-scope physical extensions before empirical adequacy is even reached. □

Remark 7.8. The locality eliminator is conditional and secondary. It sharpens physics closure when locality is explicitly retained, but the general physical non-generativity theorem does not depend on it. The capstone therefore records it as an optional strengthening, not as the whole of physics closure.

Theorem 7.9 (Exhaustion of same-domain physical closure escapes)**.** *Let Φ be any purported same-scope physical enrichment on the fixed domain that claims standing-bearing authority beyond the closure results of Sections 4–6. Then exactly one of the following holds:*

1. *Φ preserves anchors, tensor content, and standing and therefore collapses to licensed interior bookkeeping;*
2. *Φ functions as a carrier attempt or other derived same-scope authorizer and is blocked by the no-carriers theorem;*
3. *Φ reintroduces a boundary selector, metric primitive, ranking, or probabilistic authorizer and is blocked by the AMetric boundary;*
4. *Φ depends on a representative-sensitive or non-factorizing discriminator and is blocked by standing-quotient uniqueness, skin non-authority, and same-scope factorization discipline;*
5. *Φ adds a witness, selector, ontic privileging device, or other same-domain authority not already licensed on the fixed scope, and so is not an enrichment of the same scope but an illicit authority import or explicit scope change.*

No sixth same-domain physical escape route remains. In particular, every same-scope physical closure escape is either bookkeeping, already ruled out by earlier theorems, or not a same-scope counterexample at all.

Proof. If Φ preserves anchors, tensor content, and standing, then by the licensed-metric-description theorem and the derived role discipline it contributes no new standing-bearing force and is licensed interior bookkeeping. If instead Φ tries to decide which acts stand by means of a derived score, weight, confidence, probability, or comparable criterion, then it is a carrier attempt and is ruled out by Theorem 5.16. If it shifts the work to a boundary ranking, metric, selector, or other admissibility-relevant parameter, then it reopens precisely the kind of pre-admissible structure excluded by Section 4. If it preserves present eligibility while distinguishing representative-sensitive or non-factorizing structure as standing-bearing, then it is blocked by Theorem 5.24, Corollary 5.25, and Theorem 5.26. The only remaining way to produce additional same-scope authority is to add

witness load, ontic privileging, or some further admissibility-bearing structure not already licensed on the fixed scope. That is not a same-scope physical enrichment but either illicit same-domain authority import or explicit scope change. These cases are exhaustive because any purported physical closure escape must either preserve the fixed standing architecture, attempt to refine or replace it, or leave the fixed scope altogether. □

7.4 Derived role discipline for physical description

Definition 7.10 (Anchor / tensor / skin in physical description)**.** Within a fixed physical scope, anchors are the non-violable preconditions of admissible description, tensors are the quotient-level invariant content and compatibility relations that carry standing-relevant load, and skin is representational surplus whose alteration changes no same-scope standing-bearing consequence.

Theorem 7.11 (Derived role-discipline theorem for physical description)**.** *Every admissibility-preserving same-scope physical description factors through the anchor / tensor / skin decomposition, and no fourth admissibility-bearing role exists there. In particular:*

1. *anchors are preconditions of admissibility rather than objects of same-scope explanation or optimization;*
2. *tensors are quotient-level invariant content rather than mechanisms or operators; and*
3. *skin is expressive surplus with zero same-scope standing-bearing authority.*

This role discipline is derived and non-generative: it introduces no new admissibility question and no new admissibility level.

Proof. The role-decomposition theorem from *Anchor / Tensor / Skin: A Role Decomposition for Admissible Description* already shows, within fixed-scope admissibility-governed regimes, that every admissible description decomposes into anchors, tensor content, and skin, uniquely up to lawful redescription. Its minimality theorem shows that no fourth role remains, because any purported additional role must either change definedness (hence anchor), change same-scope admissible consequences (hence tensor), or change neither (hence skin).

Applying that result to physical description adds no new generative power. On the contrary, it forbids the two most common confusions at precisely the point where the descriptive role discipline matters. First, anchors are not new foundational objects to be explained from within the same scope; attempting to move them would require boundary action or illicit transport. Second, tensors are not operators or mechanisms; treating them as such would either introduce a new operator envelope or mis-type invariant content as generative. Third, skin cannot be promoted to standing-bearing authority without violating quotient factorization and same-scope uniqueness. So the role discipline is derived and non-generative: it classifies physical description after closure and does not create a fourth foundational level. □

Corollary 7.12 (Tensors are not operators; mechanism language is skin unless typed otherwise)**.** *A tensor may constrain admissible physical description by quotient-level invariance and compatibility,*

but it is not itself an operator envelope. Mechanism language may therefore appear only as explicitly non-load-bearing skin unless an actual licensed operator envelope has already been fixed independently.

Proof. If a tensor were treated as an operator, it would either act without licence, which is inadmissible, or else require a new licensed operator envelope. In the latter case the purported mechanism is either just an already fixed operator under another name or a new same-scope extension, which is blocked by exhaustion. So tensors are not operators. Correlatively, mechanism language may be used only as skin unless a genuine operator envelope has already been independently licensed. □

7.5 Standard Model bookkeeping as a worked transport-closure example

Proposition 7.13 (Witness-normal-form collapse in the fixed Standard Model load)**.** *Within the fixed renormalizable chiral gauge load audited in* Structura Ex Necessitate Standardis Modelis*, apparent same-scope Standard Model parameter freedoms collapse to a witness-normal form under admissible redescription and explicit transport. In particular, the low-energy and fixed-load witness results may be read in the following capstone form.*

Sector	***Witness-normal form***	***Capstone reading***
Charged current	G_F	*The pair* (g_2, v) *overcounts same-scope load in the contact-interaction regime; the standing-bearing content collapses to the Fermi witness.*
Electromagnetic sector	e	*The pair* (g_1, g_2) *overcounts same-scope electromagnetic load once the pure-photon scope is fixed.*
Yukawa / mass presentation	*transport-equivalent mass and Yukawa witnesses*	*The two presentations are same-scope bookkeeping variants rather than independent standing-bearing structures.*
QCD CP-odd sector	$\bar{\theta}$	*The physically relevant CP-odd QCD content collapses to one invariant witness.*
Flavor and oscillation sector	*oscillation invariants*	*Same-scope flavour descriptions collapse to the invariant oscillation family rather than to presentation-dependent coordinates.*
Broken-gauge high-energy sector	*longitudinal–Goldstone amplitude-class equivalence*	*The two presentations remain transport-equivalent amplitude bookkeeping within the fixed admitted scope.*

Moreover, anomaly cancellation and Yukawa compatibility force the hypercharge class in the minimal chiral skeleton, hypercharge normalization is redundant, and tree-level FCNC are excluded at the fixed load.

Proof. This is the capstone reading of the transport-certificate ledger in *Structura Ex Necessitate Standardis Modelis*. Each collapse is exhibited there by the same pattern: the input skin is named, an admissible transport map is written, the invariant witness set is computed, and witness rank is compared to apparent parameter rank. The charged-current and electromagnetic examples show directly that pairs such as (g_2, v) and (g_1, g_2) overcount standing-bearing content on restricted scopes because only one witness survives transport on the fixed scope. The Yukawa/mass, $\bar{\theta}$, oscillation, and longitudinal–Goldstone subsectors repeat the same pattern in more structured settings. The forced discrete skeleton then shows that anomaly cancellation and Yukawa compatibility fix the hypercharge class and leave no room for a further same-scope equivalence completion within the declared load. So the Standard Model application does not create new physics-closure burden; it exhibits the already proved non-generativity theorem in a worked transport ledger. □

Corollary 7.14 (No further same-scope equivalence completion in the fixed Standard Model load)**.** *Within the declared renormalizable Standard Model load, no further same-scope equivalence completion remains without adding new witness load, leaving the scope, or invoking a standing-failure mode.*

Proof. Once witness-normal form has been reached, any further same-scope equivalence completion would have to preserve standing while collapsing distinctions not already removed by the transport certificates. But by the same ledger, any such further collapse either adds no new witness information and is redundant, or alters the witness set and is therefore a scope change, or else relies on privileged representation or hidden load and is a standing failure. So no further same-scope equivalence completion remains. □

7.6 Measurement as a problem of reference

Definition 7.15 (Physically non-selective theory)**.** A theory is *physically non-selective* with respect to a measurement episode if it realizes multiple stable post-measurement record-bearing histories while supplying no physical law, state variable, or dynamically privileged trajectory that singles out one such history as the unique referent of the pre-measurement diachronic discourse.

Theorem 7.16 (Measurement-reference application theorem)**.** *If a physically non-selective theory is used to underwrite truth-apt diachronic discourse in which measurement records function as theory-level evidential anchors, then admissible reference assignments for those records must collapse into a single standing-bearing quotient class on the fixed scope. If they do not, then confirmation, memory, anticipation, and deliberation become assignment-relative rather than theory-relative.*

Proof. The measurement source proves that admissible reference assignments form a fixed-domain space whose standing-bearing content factors through a unique measurement-reference quotient. It also proves that confirmational-role divergence is quotient-relevant rather than semantic skin: if admissible assignments induce different comparative evidential roles, then they belong to distinct standing-bearing quotient classes. In a physically non-selective theory, the bare dynamics may stabilize multiple branch-local records without supplying a selector that fixes which one is the diachronic referent of the pre-measurement discourse. Bare Everett is the paradigmatic example.

If the admissible assignments nevertheless remain multiple in a quotient-relevant way, then the same physical episode supports different standing-bearing evidential uses, so theory-level confirmation is no longer fixed by the theory as such. By the same fixed-domain theorem family, memory, anticipation, and agency inherit that same lineage dependence. Therefore either admissible assignments are restricted strongly enough that fixation commutes with lawful redescription and the quotient collapses to one standing-bearing class, or else the discourse becomes assignment-relative rather than theory-relative. □

Corollary 7.17 (Bare physical multiplicity does not yet fix evidential reference)**.** *Branch-local determinacy, decoherence-based record stability, centered descriptions, self-location, and decision-theoretic branch-weighting do not by themselves fix diachronic evidential reference. They presuppose successor typing and record reference rather than creating them.*

Proof. The measurement source shows that branch-local determinacy and diachronic referential determinacy are distinct. Bare Everettian dynamics and decoherence yield stable local records, but they remain physically non-selective with respect to which record is the referent of pre-measurement evidential discourse. Centered descriptions, self-location, and branch-weighting presuppose a domain of already typed successors and records over which they distribute centers, credences, or utilities. So these devices are downstream of referential fixation rather than solutions to it. □

7.7 Physics closure

Theorem 7.18 (Physics closure theorem)**.** *Physical description is closed under the same invariant that already closed mathematics: no same-scope physical description becomes a new standing-bearing authorizer. More precisely, within the unique admissible interior every purported same-scope physical enrichment is exactly one of the following:*

1. *licensed interior bookkeeping or redescription that preserves anchors, tensor content, and standing;*
2. *a carrier attempt, boundary selector, or metric/probabilistic authorizer blocked by the earlier sections;*
3. *an added witness, selector, ontic privileging device, or admissibility-bearing structure that therefore changes scope rather than enriching the same scope.*

No fourth same-scope physical status remains.

Proof. The first case is exactly the licensed metric description theorem together with the derived role discipline: physical description may be rich, but if it preserves anchors, tensor content, and standing, it is licensed interior bookkeeping. The second case is excluded by the AMetric boundary theorem, the no-carriers theorem, the no-repair and no-generator theorems, and the measurement source's no-selector consequences for physically non-selective theories. The third case is not same-scope enrichment at all; it is explicit scope change because it adds new witness content or new admissibility-bearing structure.

These cases are exhaustive. A purported same-scope physical enrichment must either preserve standing while adding no new standing-bearing content, or else attempt to authorize standing by a new same-scope source, or else enlarge the witness-bearing content of the scope. Since the second class is blocked and the third is scope change, only licensed non-generative description remains. Therefore admissible physical description is closed under the same invariant already governing mathematics: no same-scope physical description becomes an independent same-scope source of standing-bearing force. This is a theorem about the classificatory role of physical description inside the admissible regime, not an ontological theorem asserting that physical reality is exhausted by bookkeeping. □

Remark 7.19 (Claim posture of this section)*.* Theorems in this section are local only in the following sense: they classify what descriptive and metric structure may still do once the earlier closure layers are fixed. The section does not claim to reproduce, inside the present manuscript alone, every physics-facing source derivation that motivated those classifications. Its force is therefore controlled

and one-directional: the core spine constrains the physical layer, while the physical layer is not allowed to back-support the spine.

8 Meta-closure against same-domain incompleteness

Section 7 closed the physical side of the capstone. The next and last pre-synthesis burden is to show that the fixed domain cannot be reopened from above by code, truth, infinitary rule-power, or richer same-scope extension. The claim is exact. This section does not deny classical Gödel incompleteness in its ordinary arithmetized setting. It proves only that a same-domain Gödel architecture threatening governance-level closure is not instantiable inside the unique admissible interior already fixed in Section 5.[1]

The local theorem support for this section is now straightforward: the dedicated same-domain non-instantiability manuscript supplies the meta-closure package, while Sections 3 and 5 provide the already-fixed trajectory and standing architecture on which the present route-exhaustion argument depends.[14]

8.1 Inferential life and closure relevance

Definition 8.1 (Inferential life of the fixed domain)**.** The *inferential life* of the fixed domain is the totality of what may be admitted, rejected, committed, reused, or licensed as same-scope continuation on that domain.

Theorem 8.2 (Inferential-life exhaustion)**.** *On the fixed scope already determined by Sections 3 and 5, inferential life is exhausted by exactly two structures:*

1. *the standing classification of evaluated acts on the fixed domain; and*
2. *the terminal standing behavior of legal same-scope trajectories.*

More precisely, if two same-domain descriptions agree on the standing classification of every evaluated act and on the terminal standing behavior of every legal same-scope trajectory, then they agree on what may be admitted, rejected, committed, reused, or licensed as same-scope continuation.

Proof. At the act level, Section 5 already fixed a unique same-scope gate, standing emergence as reuse-stable admissibility, standing-quotient uniqueness, and the no-second-same-scope-equivalence theorem. So once standing classification of evaluated acts is fixed, no additional same-domain act-level standing-bearing distinction remains.

At the path level, Section 3 fixed coherence as standing preservation along legal same-scope trajectories and proved that incoherent prefixes remain incoherent under extension. Standing failure is therefore irrecoverable within the retained lineage. So once terminal standing behavior of legal same-scope trajectories is fixed, what may be licensed as same-scope continuation or commitment is fixed as well.

[1] This is the precise capstone use of *Non-Instantiability of Gödel Incompleteness in the Unique Admissible Interior* [14]: same-domain non-instantiability, not denial of classical metamathematics.

Since inferential life consists exactly of what may be admitted, rejected, committed, reused, or licensed as same-scope continuation, agreement on those two structures exhausts it. □

Theorem 8.3 (Closure-relevance transmission)**.** *Let Θ be any same-domain distinction on the fixed scope. If Θ changes inferential life, then Θ changes at least one of the following:*

1. *the standing classification of some evaluated act; or*
2. *the terminal standing behavior, and hence the licensed continuation structure, of some legal same-scope trajectory.*

Consequently, every same-domain incompleteness objection can threaten closure only if it survives into the standing quotient or the coherent trajectory structure of the fixed domain.

Proof. Assume that Θ changes neither the standing classification of any evaluated act nor the terminal standing behavior of any legal same-scope trajectory. Then, by Theorem 8.2, two descriptions differing only by Θ agree on the entire inferential life of the fixed domain. That contradicts the hypothesis that Θ changes inferential life. So at least one of the two listed kinds of change must occur.

If the change is act-level, then by standing-quotient uniqueness it survives into the standing quotient: there is no second same-scope equivalence relation that can absorb it while leaving admissible standing unchanged. If the change is trajectory-level, then by the finite fixed-scope trajectory results it survives into coherent trajectory structure itself, because no later same-scope continuation can repair an incoherent prefix. Hence every same-domain closure threat must survive into one of those two structures. □

Corollary 8.4 (External undecidability inertness)**.** *An undecidable or unprovable sentence that never re-enters the fixed scope so as to change standing classification or coherent same-scope trajectory structure is closure-inert on that scope.*

Proof. If neither kind of change occurs, then Theorem 8.3 shows that the distinction changes no inferential-life fact and therefore threatens no same-domain closure. □

Theorem 8.5 (Inferential necessity of closure distinctions)**.** *Let F be any formal system associated with the fixed domain, and let Θ be a sentence, code-construction, or metamathematical distinction formulated in F. Suppose the claim is that the undecidability, unprovability, or fixed-point behavior of Θ undercuts closure of reasoning on the fixed domain in the ordinary inferential sense, namely that it changes what may be admitted, rejected, committed, or licensed as same-domain continuation when Θ is used in reasoning on that domain. Then Θ must induce at least one of the following:*

1. *a difference in the standing classification of some evaluated act; or*
2. *a difference in the terminal standing of at least one legal same-domain trajectory, equivalently a divergence in what may be licensed as standing-bearing continuation or commitment on the fixed domain.*

Consequently, every ordinary Gödel-style phenomenon can threaten closure on the fixed domain only if it survives into the standing quotient or coherent trajectory structure of that domain.

Proof. By hypothesis, the presence or absence of Θ changes the inferential life of the fixed domain. Therefore Theorem 8.3 yields either a standing-classification difference or a trajectory-level terminal-standing difference.

If the difference is act-level, then by standing-quotient uniqueness and the collapse of second same-scope standing-bearing equivalence relations it survives into the standing quotient itself. If the difference is trajectory-level, then by the finite fixed-scope trajectory theorems it survives into coherent trajectory structure, because terminal standing certifies coherence, incoherent prefixes remain incoherent under extension, and standing failure cannot be repaired downstream within the same lineage. Therefore every ordinary Gödel-style phenomenon that genuinely threatens closure must survive into the standing quotient or coherent trajectory structure of the fixed domain. □

Corollary 8.6 (Trajectory-neutral closure relevance is act-level)**.** *If an alleged distinction on the fixed domain alters no legal same-domain trajectory—equivalently, changes neither any terminal standing outcome nor any licensed same-domain continuation—then any closure relevance it has must appear as a difference in standing classification of some evaluated act.*

Proof. Apply Theorem 8.3. If the trajectory branch is absent by hypothesis, only the act-level branch remains. □

Remark 8.7. Why Gödel must become governance-relevant to matter. Classical incompleteness by itself is not enough to threaten closure in the present sense. To matter here, an undecidable or unprovable sentence must change the inferential life of the fixed domain; and by Theorems 8.3 and 8.5, any such change must survive into standing classification or coherent trajectory structure. The bridge from standard metamathematical talk to governance-level closure is therefore theorematic rather than interpretive.

8.2 Governance-relevant code predicates are hidden gate work

Definition 8.8 (Same-domain faithful coding layer)**.** A *same-domain faithful coding layer* for a class A of syntax-bearing acts on the fixed scope consists of a coding map

$$\ulcorner . \urcorner : A \to C$$

into a code-object space C such that predicates on C may be read back, without scope reset or domain change, as standing-bearing constraints on the original acts or on their licensed same-scope continuations.

Definition 8.9 (Readback and governance-relevant code predicate)**.** If $P \subseteq C$ is a predicate on code-objects, its same-domain readback is the induced predicate

$$\widehat{P}(a) :\Longleftrightarrow P(\ulcorner a \urcorner) \qquad (a \in A).$$

The code predicate P is *governance-relevant* when this readback changes inferential life on the fixed domain.

Lemma 8.10 (Hidden gate-work lemma)**.** *If a same-domain code predicate is governance-relevant, then its readback performs hidden gate work on the original fixed domain: it either changes which acts are admitted or standing-positive, or it refines the positive side or negative side of the gate into a further substantive same-scope classification affecting licensed continuation.*

Proof. If the readback is governance-relevant, then by definition it changes inferential life. Theorem 8.3 therefore forces the change to appear either in standing classification or in coherent trajectory structure. If it changes standing classification directly, it changes which acts are admitted or standing-positive. If it leaves admission unchanged but changes what same-scope continuation is licensed, then it refines one side of the gate into a further substantive same-scope classification. No third same-domain option remains. □

Theorem 8.11 (Code-predicate role exhaustion)**.** *Let $P \subseteq C$ be a same-domain code predicate with readback $\widehat{P}$. Then exactly one of the following roles is instantiated on the fixed domain:*

1. *inert bookkeeping: $\widehat{P}$ changes no standing classification and no licensed same-scope continuation;*
2. *extensional gate renaming: $\widehat{P}$ coincides extensionally with the unique same-scope admissibility gate and therefore adds no independent governance content;*
3. *same-side refinement: $\widehat{P}$ leaves gate verdicts unchanged but purports to refine the admitted side or rejected side into a further substantive same-scope classification affecting licensed continuation;*
4. *independent standing-bearing authorizer: $\widehat{P}$ purports to authorize admissibility or standing-bearing continuation independently of the unique same-scope gate and is not extensionally identical with it.*

No fifth same-domain governance role exists.

Proof. If $\widehat{P}$ is closure-inert, then role (1) holds by definition. Assume instead that $\widehat{P}$ is governance-relevant. By Lemma 8.10, it either changes who is admitted or standing-positive, or it leaves admission fixed while changing licensed same-scope continuation. In the first case, either the induced classification agrees extensionally with the unique gate, giving role (2), or it does not, giving role (4). In the second case it refines one side of the gate without changing admission, giving role (3). These cases are mutually exclusive and exhaustive. □

Theorem 8.12 (No independent governance-relevant code predicate)**.** *No same-domain code predicate can serve as an independent governance-relevant authorizer on the original fixed scope. Every same-domain code predicate is therefore either inert bookkeeping or extensional renaming of the unique admissibility gate.*

Proof. By Theorem 8.11, every same-domain code predicate must instantiate exactly one of the four roles listed there. Role (3) is impossible because Section 5 already eliminated substantive same-side refinements: a positive-side refinement would amount to a second same-scope gate, while

a negative-side refinement would open a same-scope promotion channel, both of which are excluded by the standing theorem family. Role (4) is impossible because the no-carriers theorem of Section 5 already excludes any independent same-scope authorizer not extensionally identical with the unique gate. So only roles (1) and (2) remain. □

Corollary 8.13 (One numbering is not enough)**.** *A single fixed coding map does not evade Theorem 8.12. One numbering is harmless unless predicates on its codes re-enter the fixed scope as governance-relevant constraints; and once they do, code-role exhaustion applies.*

Proof. The danger is governance-relevant readback, not multiplicity of coding as such. A single numbering that never changes standing or trajectories is inert bookkeeping; a single numbering that does change them is already covered by Theorems 8.11 and 8.12. □

8.3 Same-domain diagonalization and the gate-collapsed fixed point

Definition 8.14 (Same-domain negative fixed-point construction)**.** A *same-domain negative fixed-point construction* on the fixed scope consists of:

1. a same-domain faithful coding layer on a class A of syntax-bearing acts;
2. a governance-relevant code predicate $P \subseteq C$ with readback $\widehat{P}$;
3. an admissible same-domain substitution/readback operation producing an act $g \in A$ whose standing-bearing content is lawfully equivalent to a negative fixed-point formula about its own code, typically of the form $\neg P(\ulcorner g \urcorner)$.

Theorem 8.15 (Necessary package for a same-domain Gödel threat)**.** *If a Gödel-style undecidability, unprovability, or fixed-point phenomenon is claimed to threaten governance-level closure on the fixed scope from within the same domain, then any such threat must realize all of the following simultaneously:*

1. *a same-domain faithful coding layer;*
2. *governance-relevant readback of at least one code-based distinction to the original fixed domain;*
3. *a standing-bearing code-predicate role that survives the code-role exhaustion of this section;*
4. *an admissible same-domain fixed-point production; and*
5. *survival of the resulting distinction into the standing quotient or coherent trajectory structure of the fixed domain.*

Proof. Without same-domain faithful coding, the phenomenon remains external bookkeeping. Without governance-relevant readback, it changes no standing fact and no same-scope continuation fact, so it is closure-inert by the previous subsection. Without a standing-bearing code-predicate role, there is no way for the code-level distinction to matter to the fixed domain at all. Without an admissible same-domain fixed-point production, there is no Gödel-style self-reference on the original scope. And if the resulting distinction does not survive into the standing quotient or coherent

trajectory structure, then by Theorem 8.3 it does not threaten closure. So every same-domain Gödel threat requires all five items. □

Theorem 8.16 (Non-existence of admissible same-domain diagonalization)**.** *No same-domain diagonal apparatus producing a governance-relevant negative fixed point is instantiable on the fixed scope.*

Proof. Assume, for contradiction, that a same-domain diagonal apparatus exists. By Theorem 8.15, it must realize a same-domain faithful coding layer, governance-relevant readback, a standing-bearing code-predicate role, an admissible negative fixed-point production, and survival of the resulting distinction into the standing quotient or coherent trajectory structure. By Theorem 8.12, the only remaining governance-relevant code-level case after role exhaustion is extensional gate renaming. So the standing-bearing content of the fixed point g is lawfully equivalent to "g is not admitted" or "g is not standing-positive."

If g is admitted or standing-positive, then the same evaluated act is both admitted and, by its own standing-bearing content, not admitted. That contradicts the unique same-scope gate and the bivalent standing theorem family of Section 5.

If g is not admitted or not standing-positive, then it carries no standing-bearing force on the fixed scope. For it nevertheless to threaten closure, its failed status would have to be reversed or made governance-relevant again either by same-act repair / generation or by later same-scope continuation within the same lineage. The first route is blocked by the no-repair and no-generator theorems of Section 5; the second is blocked by the no-deferred-coherence and irrecoverability results of Section 3. In either case the purported fixed point becomes closure-inert. That contradicts item (5) of Theorem 8.15. Therefore no admissible same-domain diagonalization exists. □

Corollary 8.17 (No governance-relevant negative fixed point)**.** *There is no governance-relevant fixed-domain act g whose standing-bearing content is lawfully equivalent to* $\neg\mathrm{Prov}(\ulcorner g \urcorner)$ *for any same-domain provability predicate* Prov*.*

Proof. Such an act would be a same-domain negative fixed point and therefore part of an admissible same-domain diagonal apparatus. Theorem 8.16 eliminates that possibility. □

Inline proof anchor VI: why same-domain diagonalization fails

The full same-domain obstruction is proved in *Non-Instantiability of Gödel Incompleteness in the Unique Admissible Interior* [14]. Once the fixed-domain standing package is already in place, the diagonal failure argument is finite.

1. Any same-domain Gödel threat must realize the full prerequisite package: a faithful coding layer, governance-relevant readback, a standing-bearing code-predicate role, an admissible fixed-point production, and survival of the resulting distinction into standing classification or coherent trajectory structure.

2. Code-role exhaustion leaves only four roles for the governing code-predicate: inert bookkeeping, extensional gate renaming, same-side refinement, and independent authorizer. The last two are

impossible on the fixed domain, so the only governance-relevant surviving case is extensional gate renaming.

3. The putative negative fixed point is therefore gate-collapsed. Its standing-bearing content is lawfully equivalent not to an independent provability split but simply to "not admitted" or "not standing-positive."

4. If the fixed-point act g is admitted, then the same evaluated act is both admitted and, by its own standing-bearing content, not admitted. That contradicts bivalence together with unique gate classification.

5. If g is not admitted, then the construction threatens closure only if its failed status can later regain same-domain governance force. Same-act repair and same-source generation are already blocked on the fixed domain, and same-lineage downstream restoration is blocked by irrecoverability of standing failure. So the non-admitted branch is closure-inert.

Every same-domain negative fixed point therefore collapses into one of two dead ends: contradiction on the admitted branch or inertness on the non-admitted branch. That is why Theorem 8.16 is stronger than the slogan "Gödel does not apply here": it shows that the relevant same-domain apparatus never reaches a stable governance-relevant fixed point inside the unique admissible interior.

8.4 Semantic, infinitary, and extension rescue routes

Theorem 8.18 (No semantic or meta-foundational standing import)**.** *Truth predicates, satisfaction predicates, intended-model semantics, global closure predicates, reflection-on-truth, and similar same-domain semantic or meta-foundational resources cannot function as independent same-domain standing-bearing authorizers on the fixed scope. Any such appeal is therefore either closure-inert bookkeeping, extensional gate renaming, explicit scope change, or inadmissible same-domain authority import.*

Proof. If such a semantic or meta-foundational appeal changes no standing classification and no coherent same-scope trajectory fact, then by Theorem 8.3 it is closure-inert bookkeeping. If it agrees extensionally with the unique same-scope gate, it is only gate renaming. If it purports to alter admissibility, standing, or licensed continuation while remaining on the old scope, then it either functions as an independent same-scope authorizer or attempts to preserve a second closure-relevant equivalence relation alongside admissible standing classification. Section 5 excludes the first; the same section's no-second-equivalence theorem excludes the second. So the only remaining non-inert use is explicit scope change or inadmissible same-domain authority import. □

Definition 8.19 (Finite same-domain witness)**.** A *finite same-domain witness* for a purported same-domain distinction Θ is either an evaluated act whose standing classification differs when Θ is imposed, or a finite legal same-scope trajectory whose coherence, terminal standing, or licensed continuation status differs when Θ is imposed.

Theorem 8.20 (Infinitary admissibility collapse)**.** *Let Θ be a same-domain standing-bearing distinction attributed to reflection, infinitary rule-power, or transfinite closure. If Θ is governance-relevant on the fixed scope, then Θ has a finite same-domain witness. Equivalently, every admissible*

same-domain standing-bearing use of infinitary rule-power collapses to finite witness at the level of evaluated acts or finite legal trajectories; otherwise the appeal is closure-inert or an inadmissible same-domain authorizer.

Proof. If Θ is governance-relevant, then by Theorem 8.3 it must survive either into standing classification or into coherent trajectory structure. In the first case, some evaluated act changes standing status and is therefore a finite same-domain witness. In the second case, some legal same-scope trajectory changes coherence, terminal standing, or licensed continuation. But Section 3 fixes the trajectory theory only for finite nonempty same-scope trajectories, so this is again a finite same-domain witness. Conversely, if no such finite witness exists, then no standing classification changes and no finite legal same-scope trajectory changes; the alleged infinitary distinction is therefore closure-inert or else an attempt to smuggle in a new same-domain authorizer. □

Corollary 8.21 (No primitive infinitary same-domain standing acts)**.** *There is no primitive same-domain act-token whose alleged standing-bearing force depends on an infinitary verification burden not reduced to a finite same-domain witness already visible on the fixed scope.*

Proof. If the alleged infinitary burden changes no standing or trajectory fact beyond the ordinary gate verdict on the act-token, then it is bookkeeping only. If it is claimed to matter to governance-level closure, then by Theorem 8.20 there must already be a finite same-domain witness. If one denies such a witness while still claiming same-domain standing-bearing force, then the token can do that work only by semantic or meta-foundational import, which Theorem 8.18 excludes. □

Theorem 8.22 (Same-domain extension trichotomy)**.** *Let E be any attempt to rescue a same-domain incompleteness objection by adjoining a richer theory, additional primitives, stronger rules, or auxiliary structure. Then exactly one of the following holds:*

1. *E changes no standing classification and no coherent trajectory fact on the old scope, in which case it is definitional or bookkeeping only;*
2. *E operates only after explicit reset, fresh gating, or domain change, in which case it is an explicit scope change rather than a same-domain rescue; or*
3. *E purports to change standing classification or coherent trajectory structure on the old scope, in which case it is inadmissible same-domain authority import.*

In particular, there is no fourth same-domain extension route to a governance-relevant incompleteness threat.

Proof. If the extension changes no standing classification and no coherent same-scope trajectory fact on the old scope, then by Theorem 8.2 it changes no inferential-life fact at all and is therefore bookkeeping only. If it works only after explicit reset, fresh gating, or domain change, then by definition it is a scope change rather than a same-domain rescue. Assume finally that it remains on the old scope while changing standing classification or coherent trajectory structure there. Then Theorem 8.3 forces the difference to survive into the standing quotient or coherent trajectory structure. But Section 5 already fixed the standing quotient as unique and eliminated any second

same-scope standing-bearing equivalence. So such an extension cannot be a benign enrichment of the old scope; it is independent same-domain authority import. These three cases are exhaustive and mutually exclusive. □

Theorem 8.23 (Rescue-route exhaustion)**.** *Every same-domain attempt to rescue a Gödel-style incompleteness objection by semantic truth, satisfaction, intended-model semantics, reflection, infinitary rule-power, transfinite closure, meta-foundational re-anchoring, or richer same-domain extension falls into exactly one of the following classes:*

1. *closure-inert bookkeeping;*
2. *extensional gate renaming;*
3. *finite same-domain witness already captured by standing classification or coherent trajectory structure;*
4. *explicit scope change; or*
5. *inadmissible same-domain authority import.*

None of these classes yields a new same-domain governance-level closure threat on the fixed scope.

Proof. Theorem 8.18 classifies semantic and meta-foundational routes as inert, gate renaming, scope-changing, or inadmissible. Theorem 8.20 classifies infinitary and reflection-based routes: any governance-relevant same-domain force collapses to a finite witness already captured by standing or trajectory structure, and any residual claim without such a witness is inert, scope-changing, or inadmissible. Theorem 8.22 classifies richer same-domain extensions as bookkeeping, explicit scope change, or inadmissible authority import. These families exhaust the rescue space. None of the resulting classes yields a new same-domain threat: inert bookkeeping changes no inferential fact, gate renaming adds no new authority, finite witness collapse adds nothing beyond the standing/trajectory structure already fixed, scope change is not same-domain, and inadmissible authority import is excluded by the regime. □

Theorem 8.24 (Gödel threat alignment)**.** *Let F be any formal system or coding architecture associated with the fixed domain. If a Gödel-style undecidability or unprovability phenomenon in F is claimed to threaten governance-level closure on that domain, then that phenomenon must be realized as a same-domain governance-relevant distinction. In particular, it must survive into the standing quotient or coherent trajectory structure of the fixed domain.*

Proof. This is an immediate application of Theorem 8.5. If the alleged phenomenon does not survive into standing classification or coherent trajectory structure, then it is closure-inert and does not threaten governance-level closure on the fixed domain. Therefore any genuine same-domain Gödel threat must survive there. □

Theorem 8.25 (Translation filter for ordinary Gödel objections)**.** *Let F be an ordinary recursively axiomatized formal system with an undecidable or unprovable sentence G_F. To threaten governance-level closure on the fixed domain, the phenomenon associated with G_F must be translated into a same-domain distinction affecting admissibility, standing, or licensed continuation on that domain.*

If no such translation exists, then the phenomenon is closure-inert relative to the fixed domain. If such a translation does exist, then it requires the same-domain prerequisite package of Theorem 8.15; any code-based negative fixed-point translation is blocked by Theorems 8.12 and 8.16, and any semantic, infinitary, or extension rescue translation is blocked by Theorem 8.23. Hence ordinary Gödel phenomena threaten closure only by a same-domain translation that the present regime excludes.

Proof. If the phenomenon associated with G_F is never translated into a same-domain distinction affecting admissibility, standing, or licensed continuation on the fixed domain, then by Theorem 8.24 it is closure-inert relative to that domain.

Assume instead that such a same-domain translation exists. Then by Theorem 8.24 it must survive into standing classification or coherent trajectory structure, and by Theorem 8.15 it must realize the same-domain Gödel prerequisite package. If the translation is code-based and negative-fixed-point in form, then Theorems 8.12 and 8.16 eliminate the only remaining governance-relevant self-reference candidate after code-role exhaustion. If the translation instead attempts to rescue the objection by semantic truth, satisfaction, reflection, infinitary rule-power, or richer same-domain extension, then Theorem 8.23 blocks those same-domain rescue routes. Therefore no ordinary Gödel phenomenon threatens closure on the fixed domain except by a same-domain translation that the present regime forbids. □

Remark 8.26. Ordinary PA-style objections are now subsumed by Theorem 8.25. An undecidable sentence such as G_F threatens closure on the fixed domain only if it is translated into a same-domain distinction affecting admissibility, standing, or licensed continuation; if no such translation occurs, the phenomenon is inert, and if it does occur, it requires the same-domain coding and self-reference package already eliminated above.

8.5 Terminal route exhaustion and non-instantiability

Theorem 8.27 (Exhaustion of same-domain incompleteness-objection routes)**.** *Let Ω be any alleged incompleteness objection to governance-level closure on the fixed scope, offered as a same-domain objection. Then exactly one of the following routes must be in play:*

1. *code-level readback of code judgments as governance-relevant constraints on the original acts or trajectories;*
2. *semantic or meta-foundational import;*
3. *reflection, infinitary rule-power, or transfinite closure;*
4. *a second same-scope equivalence, quotient, or standing-bearing refinement beside admissible standing classification;*
5. *a richer same-domain extension or new gate; or*
6. *explicit scope change.*

The first five routes are impossible or closure-inert on the fixed scope; the sixth is not same-domain. Hence no same-domain incompleteness-objection route remains.

Proof. Any genuine threat must survive into standing classification or coherent trajectory structure by Theorem 8.3. To acquire such same-domain force, the objection must proceed by code-level readback, semantic or meta-foundational import, reflection or infinitary rule-power, a second equivalence or parallel quotient, a richer same-domain extension or new gate, or explicit scope change. No seventh same-domain route remains.

Route (1) is exhausted by Theorems 8.11, 8.12, and 8.16. Route (2) is eliminated by Theorem 8.18. Route (3) is eliminated by Theorem 8.20. Route (4) is eliminated by the no-second-same-scope-equivalence theorem of Section 5. Route (5) is eliminated by Theorem 8.22 together with the unique same-scope gate. Route (6) is, by definition, not same-domain. Therefore the first five routes are impossible or closure-inert and the sixth falls outside the claim. □

Theorem 8.28 (Non-instantiability of Gödel incompleteness in the unique admissible interior)**.** *Under the fixed-domain theorem package already proved in Sections 3–7, no same-domain Gödel threat is instantiable in the unique admissible interior. Equivalently, classical incompleteness remains valid in its standard setting, but a same-domain Gödel architecture relevant to governance-level closure cannot be instantiated inside the admissibility-governed regime of this capstone.*

Proof. Assume, for contradiction, that a same-domain Gödel threat is instantiable on the fixed scope. By Theorems 8.24 and 8.15, the alleged threat must be realized as a same-domain governance-relevant distinction surviving into the standing quotient or coherent trajectory structure. But Theorem 8.27 already exhausts every same-domain incompleteness-objection route and shows that the only remaining route is explicit scope change, which is not same-domain. That contradicts the assumption that a same-domain Gödel threat is instantiable on the fixed scope. Therefore no such threat exists inside the unique admissible interior. □

Corollary 8.29 (No governance-relevant "true but unprovable" split on the fixed scope)**.** *There is no same-domain admissibility-relevant positive-side split between "admitted and provable" and "admitted but merely true or unprovable" on the fixed scope.*

Proof. Such a split would be a standing-bearing distinction among admitted acts. By Theorem 8.3, it would have to survive into standing classification or coherent trajectory structure. That makes it either a same-side refinement, a second same-scope equivalence relation, or a new same-domain gate. Each possibility is excluded by the route-exhaustion theorem and the standing theorem family already fixed earlier. So no such split exists. □

Corollary 8.30 (Governance-certified results are not vulnerable to same-domain incompleteness objections)**.** *Any result whose certification remains inside the fixed admissibility-governed regime of this capstone is not vulnerable to a same-domain Gödel incompleteness objection.*

Proof. Immediate from Theorem 8.28. The point is structural: once same-domain route exhaustion is complete, no internal incompleteness route remains by which the fixed domain could be reopened from above. □

This completes the exhaustion at the level of same-domain counterexamples: no same-domain construction and no regime-exiting construction yields a counterexample to closure.

Kernel-level exhaustion pressure. The route-exhaustion result of this section should be read together with Theorem efthm:kernel-exhaustion-apparent-counterexamples. An apparent incompleteness-based escape either collapses back into the same kernel-governed regime, fails as a lower-generator construction, or exits the target phenomenon altogether. Accordingly the meta-closure layer does not merely answer one objection among others; it closes the remaining same-domain objection space after kernel-level exhaustion has already ruled out rival realizations of the same target.

9 Capstone synthesis: two spectra, parity, and descriptive closure

9.1 Why the final synthesis is now forced

By the end of Section 8 no same-scope escape route remains open. The kernel floor is fixed, state-transition form and irreversible commitment are forced, the AMetric boundary is non-parameterized, the admissible interior is unique, mathematics is closed by operator exhaustion, physics is closed by the same standing-preserving invariant, and same-domain incompleteness objections are non-instantiable inside that interior. What remains is to classify the relation between mathematics and physics after closure and then state what that classification implies for the totality of admissible description.

The synthesis proved here is downstream and classificatory: it draws only on the already-fixed closure results for mathematics, physics, and same-domain meta-escape, together with the synthesis theorems of *Eadem Natura, Duo Spectra*, and does not reopen any earlier foundational burden.[11, 10, 2, 3]

9.2 Minimal background for the synthesis

Definition 9.1 (Grounding)**.** Within the capstone, *grounding* is standing-generative authority: a domain grounds another only if it can confer standing-bearing force, foundational privilege, or admissibility-bearing status on that other domain rather than merely organize, translate, or redescribe it.

Definition 9.2 (Fixation mode)**.** A post-closure domain is *internally fixed* when its admissible objects and transformations are fixed symbolically within the admitted interior itself; it is *externally fixed* when its admissible use remains tied to externally fixed reports, records, or evidential anchors. These are fixation modes, not different standing values.

Definition 9.3 (Representational spectrum)**.** The *representational spectrum* of a post-closure domain is the way one and the same non-generative admissible order is articulated under a fixation mode. Mathematics occupies the internally fixed symbolic spectrum; physics occupies the externally fixed empirical spectrum.

Definition 9.4 (Shared invariant)**.** The *shared invariant* of the capstone is the theorem-local fact

already fixed by Sections 6, 7, and 8: standing cannot be generated by formal operators, licensed metric description, external coding, semantic import, infinitary rule-power, or same-scope extension. What survives in either domain is standing-preserving coordination inside an already fixed admissible interior.

9.3 The fixation split and two-spectrum exhaustion

Theorem 9.5 (Fixation dichotomy / two-spectrum exhaustion)**.** *After the closure theorems of Sections 6, 7, and 8 are fixed, every standing-preserving post-closure descriptive discipline falls into exactly one of two spectra:*

1. *the* internally fixed symbolic spectrum*, or*
2. *the* externally fixed empirical spectrum*.*

No third same-level admissible spectrum remains.

Proof. A post-closure discipline must still belong to the target class fixed in Sections 2–4: it must support determinate reference, admissible redescription, and standing-preserving use. If its objects and constraints are fixed symbolically from within the admitted interior itself, then it belongs to the internally fixed spectrum. If instead its evidential anchors remain externally fixed — as reports, records, measurements, or observation-bearing carriers — while its descriptive work remains licensed interior bookkeeping, then it belongs to the externally fixed spectrum.

No third same-level spectrum remains. A purported third spectrum would have to differ from both existing spectra by a standing-bearing invariant other than fixation mode. But Section 4 already eliminated same-scope boundary selectors and admissibility-relevant parameterization, Section 5 fixed the unique standing quotient and excluded any second same-scope standing-bearing equivalence, and Section 8 excluded same-domain code, semantic, infinitary, and extension escape routes. So any purported third spectrum is either bookkeeping-only, a notational redescription of one of the two spectra, or an actual scope change rather than a third post-closure spectrum. Therefore the split is exhaustive. □

Corollary 9.6 (Fixation is classificatory rather than hierarchical)**.** *The distinction between the two spectra classifies how post-closure work is fixed and articulated. It does not rank one spectrum above the other in standing-bearing force.*

Proof. The proof of Theorem 9.5 introduced only a difference in fixation mode and representational articulation. No new standing-bearing source, gate, or admissibility-bearing asymmetry was added. So the split classifies the surviving difference between the two domains; it does not create a hierarchy of standing. □

9.4 Closure parity

Theorem 9.7 (Closure parity theorem)**.** *After closure, mathematics and physics are equal in foundational status, equal in grounding incapacity, and equal in legitimate post-closure role.*

Proof. Section 6 proved that mathematics is closed by exhaustion: no same-scope operator family generates new standing-bearing mathematical content, and what survives is internally fixed standing-preserving structural coordination. Section 7 proved the parallel result for physics: no same-scope physical description generates standing, and what survives is externally fixed standing-preserving empirical coordination. Section 8 then removed the only same-domain meta-level routes by which one domain might still claim standing-bearing privilege from above.

So the two domains are already equal in foundational status: neither remains foundational in a standing-generative sense. They are equal in grounding incapacity because the same shared invariant blocks mathematics from generating standing formally and blocks physics from generating standing descriptively. Finally, they are equal in legitimate post-closure role, because in each case what survives is disciplined coordination, refinement, mapping, and engineering within an already fixed admissible envelope. What differs is fixation mode and spectrum, not the standing-bearing type of role each domain is permitted to perform. □

9.5 From parity to mirror classification

Proposition 9.8 (No residual distinguishing invariant)**.** *Once foundational status, grounding incapacity, and legitimate post-closure role are fixed as in Theorem 9.7, no admissible invariant distinguishes mathematics and physics beyond fixation mode and the corresponding representational spectrum.*

Proof. By Theorem 9.7, mathematics and physics already coincide in every standing-relevant classificatory dimension carried by the capstone: foundational status, grounding incapacity, and legitimate post-closure function. By Theorem 9.5, the only same-level admissible distinction left available is fixation mode and the resulting spectrum. Therefore any purported further invariant must be one of three things: bookkeeping that leaves the standing profile unchanged, a notational redescription of one of the already admitted spectra, or a scope change that alters the standing-bearing conditions under which the domain is classified. None of these is a residual same-level invariant beyond fixation mode. □

Theorem 9.9 (Mirror classification theorem)**.** *Mathematics and physics are equivalent under the equivalence relation induced by admissible standing-preserving structure and differ only by fixation mode and representational spectrum.*

Proof. Theorem 9.7 fixes equality of foundational status, grounding incapacity, and legitimate post-closure role. Theorem 9.5 fixes the only surviving admissible distinction between the two domains: mathematics occupies the internally fixed symbolic spectrum and physics occupies the externally fixed empirical spectrum. Proposition 9.8 removes any further admissible invariant beyond that split. So the two domains are equivalent under admissible standing-preserving structure and differ only by fixation mode and the representational spectrum through which the same non-generative admissible order is articulated. □

Corollary 9.10 (*Eadem natura, duo spectra*)**.** *After closure, mathematics and physics are two spectra of one constrained admissible order.*

Proof. Immediate from Theorem 9.9. The corollary states in compressed form the same mirror-classification result: sameness in post-closure nature, difference only in spectrum. □

9.6 The prohibition package

Theorem 9.11 (No grounding asymmetry)**.** *After closure, mathematics does not ground physics and physics does not ground mathematics.*

Proof. Grounding in the capstone means standing-generative authority. But Theorem 9.7 states that mathematics and physics are equal in grounding incapacity. If mathematics grounded physics, then mathematics would possess a standing-generative role denied to physics; if physics grounded mathematics, the reverse would hold. Either claim contradicts parity. Corollary 9.6 confirms that fixation mode introduces no standing-bearing hierarchy that could rescue either asymmetry. □

Theorem 9.12 (No cross-domain rescue)**.** *The non-grounding status of mathematics cannot be repaired by appeal to physics and the non-grounding status of physics cannot be repaired by appeal to mathematics.*

Proof. Suppose first that physics could rescue mathematics. Then mathematics would acquire standing-bearing force from relation to an externally fixed domain. But Section 5 already excluded same-scope transfer and carrier laundering, and Theorem 9.9 states that the two domains differ only by fixation mode rather than by standing-bearing kind. So physics cannot rescue mathematics without smuggling in transferred standing. The converse case is symmetric: if mathematics could rescue physics by formal exactness, it would reinstall the grounding asymmetry already ruled out by Theorem 9.11. Therefore neither domain can repair the other's non-grounding status. □

Theorem 9.13 (No meta-foundational re-escalation)**.** *No higher mixed standpoint, meta-foundational layer, or combined mathematics–physics platform restores foundational privilege to either domain after closure.*

Proof. A purported meta-foundational re-escalation must either leave the standing profile of both domains unchanged or alter it. If it leaves the standing profile unchanged, then it is a bookkeeping redescription of the already closed relation between the two domains. If it alters that profile, then at least one domain must acquire standing-bearing authority not already available under Theorems 6.26, 7.18, and 8.28. But every such route would have to proceed by generation, transfer, repair, selector import, or scope change, and those routes are already exhausted by earlier sections. So no meta-foundational re-escalation remains. □

Corollary 9.14 (No surviving reduction thesis)**.** *After closure, neither reduction of physics to mathematics nor reduction of mathematics to physics survives as a standing-relevant thesis.*

Proof. A standing-relevant reduction thesis would have to do more than provide translation or representational economy; it would have to make one domain derivative in standing-bearing force with respect to the other. Theorems 9.11 and 9.12 exclude exactly that. What may remain are

translations, embeddings, or reformulations, but these are bookkeeping relations within the mirror structure, not reductions that reinstall privilege. □

Corollary 9.15 (Prohibition package)**.** *The following post-closure moves are inadmissible:*

1. *re-grounding physics via mathematics;*
2. *re-grounding mathematics via physics;*
3. *cross-domain rescue attempts;*
4. *meta-foundational re-escalation; and*
5. *any standing-relevant reduction thesis between the two domains.*

Proof. Immediate from Theorems 9.11, 9.12, and 9.13 together with Corollary 9.14. □

9.7 Post-closure practice

Theorem 9.16 (Post-closure practice rule)**.** *After closure, legitimate work in mathematics and physics is restricted to standing-preserving internal transformation, representational refinement, constraint mapping, and engineering within already established admissible envelopes.*

Proof. Theorem 9.9 identifies mathematics and physics as two spectra of one constrained admissible order. Corollary 9.15 removes every route by which either domain could recover standing-generative privilege over the other or over the admissible descriptive totality. So any legitimate post-closure work must remain within the standing profile already fixed for each domain. For mathematics, Section 6 identified the surviving role as internally fixed structural coordination. For physics, Section 7 identified the surviving role as externally fixed empirical coordination. In each case the legitimate residue is therefore transformation, refinement, mapping, and engineering within the admitted interior, not new grounding. □

Corollary 9.17 (Closure is not nihilism)**.** *Closure eliminates standing-generative ambition, not substantive practice. Mathematics and physics remain fully legitimate as disciplined domains of work after closure.*

Proof. Immediate from Theorem 9.16. The theorem restricts the type of legitimate work that remains; it does not eliminate that work. □

Remark 9.18 (Seriousness without privilege)*.* The capstone therefore ends positively rather than nihilistically. Mathematics remains serious because exact symbolic articulation, proof, mapping, and engineering remain legitimate inside the internally fixed spectrum. Physics remains serious because measurement, modeling, empirically anchored refinement, and engineering remain legitimate inside the externally fixed spectrum. What closure removes is only the claim that either domain stands beneath the other, or beneath reality, as ground.

9.8 Descriptive closure of admissible description

Definition 9.19 (Admissible descriptive totality)**.** For the purposes of the capstone, the relevant totality is the totality of admissible descriptions under standing preservation. This is a classificatory notion, not a free-standing ontological posit over and above the fixed admissible order. The capstone therefore states closure only for admissible description, not for ontology as such.

Theorem 9.20 (Descriptive-closure theorem)**.** *Reality, in the sense of Definition 9.19, is closed by exhaustion. Equivalently, once the theorem chain of this capstone is in place, no admissible description lies outside the two spectra fixed by Theorem 9.5, and neither spectrum contains any standing-generative route by which foundational privilege could be reintroduced.*

Proof. Sections 6, 7, and 8 already closed the three substantive re-entry routes: mathematics cannot reopen the interior by operator means, physics cannot reopen it by descriptive or measurement-side means, and same-domain meta-level coding, semantic import, infinitary escalation, or extension cannot reopen it from above. Theorem 9.5 then exhausts the remaining admissible descriptive universe into exactly two post-closure spectra. Theorem 9.9 shows that these are not two rival foundations but two spectra of one constrained admissible order, and Corollary 9.15 rules out every remaining route by which either spectrum could recover standing-generative privilege.

Therefore the totality of admissible description is closed: there is no third spectrum, no residual grounding asymmetry, no cross-domain rescue, and no same-scope meta-level reopening route. This is exactly descriptive closure in the classificatory sense fixed above. The theorem therefore closes the space of admissible description, not ontology independent of admissible description. □

10 Downstream consequences: placement, roadmaps, and profiles

Downstream consequence-facing only. From this point onward, the manuscript is packaging the already-closed theorem chain as downstream consequence material. Section 10, Section 11, and the appendices record consequence placement, audit posture, source-role discipline, and atlas typing. None of this material reopens Sections 1–9 or adds a new foundational burden.

10.1 Why application placement comes only after descriptive closure

By the end of Section 9 the foundational theorem arc is already complete. The kernel floor is fixed, state-transition form and irreversible commitment are forced, the AMetric boundary is non-parameterized, the standing quotient and unique interior are fixed, mathematics and physics are closed under the same non-generative invariant, same-domain incompleteness objections are exhausted, and the manuscript has already reached the two-spectrum synthesis and the descriptive-closure theorem. What remains is therefore not a new foundational burden but a placement burden: how downstream manuscripts should appear without silently becoming proof sources for what is already closed.

Import note 10.1 (Downstream sources used here)**.** The application manuscripts are used here

only in the exact roles their own front matter, theorem maps, and denial maps assign to them. *Carrier Universality and Global Smoothness for the Periodic 3D Navier–Stokes Equations* is used for official-problem alignment, claim-class discipline, theorem-map hygiene, same-object conventions, one-route dependency audit, and theorem-local denial targets. *Vacuum-sector Mass Gap for the Local Gauge-Invariant Sharp-Local Yang–Mills Net* and its three companions are used for endpoint hygiene, proof-home crosschecks, the theorem-level object of the local sharp-local net, the qualitative scope of faithful-Wilson universality, and the separation between local-net endpoints and nonlocal sector claims. *The Measurement Problem as a Problem of Reference* is used for the fixed-domain quotient theorem, the standing conditional, the confirmation criterion, the Everettian non-selection theorem, and the selector-exhaustion taxonomy. *Structura Ex Necessitate Standardis Modelis* is used for witness-normal-form and transport-closure bookkeeping in the Standard Model case. None of these sources reopens Sections 1–9.

10.2 Claim-type discipline

Definition 10.2 (Foundational closure theorem)**.** A *foundational closure theorem* fixes same-scope admissibility, standing, non-generativity, quotient structure, or closure of admissible description at the level of the governing theorem architecture. Sections 1–9 belong to this class.

Definition 10.3 (Downstream internal problem solution)**.** A *downstream internal problem solution* begins only after a closure theorem is already fixed and then uses that theorem to eliminate, transport, or classify the remaining licit endpoint statuses on a specific carrier. It may be theorem-heavy, but it does not function as the proof source of the foundational closure theorem from which it starts.

Definition 10.4 (Theorem roadmap and consequence profile)**.** A *downstream consequence roadmap* is a compact main-body presentation of a downstream manuscript that records its exact claim type, theorem-level endpoint, live dependency spine, denial map, and non-claims. A *downstream consequence profile* is a shorter placement statement used when the application clarifies the force of the closure theorem without needing its own extended theorem map inside the capstone. In this section, roadmap and profile are therefore both downstream consequence formats.

Theorem 10.5 (No downstream backflow)**.** *Let M be a downstream application manuscript introduced after the foundational arc of Sections 1–9 is already complete. Under the declared source set, M cannot function as a proof source for Sections 1–9 unless the theorem arc is explicitly rebuilt with M promoted to a declared premise.*

Proof. The foundational arc is already closed before any application manuscript enters. If a later application were allowed to backfill a premise of Sections 1–9 without explicit promotion, the manuscript would cease to have one strict dependency order. Therefore any such backflow must be blocked unless the theorem arc is rebuilt with that manuscript declared as a live premise. □

Proposition 10.6 (Method-neutral reading of theorem roadmaps)**.** *A downstream manuscript may appear in the capstone as a theorem roadmap even when its proof class is not constructive, analytic, or estimate-driven in the conventional style of the target field. What matters for roadmap status is that the source manuscript itself fixes the official theorem object, states its exact claim class, identifies its live dependency spine, and localizes its own non-claims and denial map.*

Proof. The capstone does not classify downstream manuscripts by conformity to a preferred proof style. It classifies them by theorem-role discipline. If a source manuscript states the official theorem object, fixes the method class actually claimed, exposes the live dependency spine, and marks what the manuscript does and does not prove, then the capstone can preserve that structure without importing the manuscript as a hidden premise. A theorem roadmap is therefore method-neutral in this exact sense: it records a theorem object and its audit architecture, not a stylistic preference for how such a theorem must be proved. □

10.3 Navier–Stokes as a theorem roadmap

Proposition 10.7 (Why Navier–Stokes belongs here as a theorem roadmap)**.** *Under the declared source set, the periodic Navier–Stokes manuscript should appear in the capstone as a theorem roadmap rather than as a proof-home appendix and rather than as a hidden foundational source. This remains true even though its proof class is explicitly a closure-induced impossibility theorem rather than a constructive continuation argument.*

Proof. The manuscript *Carrier Universality and Global Smoothness for the Periodic 3D Navier–Stokes Equations* is already claim-facing. Its opening states that the endpoint is Fefferman's periodic statement (B), that the proof class is a *closure-induced impossibility theorem* rather than a constructive continuation argument, that the theorem map is explicit, and that the official result is reached through one live route rather than through ambient corpus pressure. It also localizes the standard objections by theorem number and makes explicit what the result does *not* claim: it does not assert that admissibility manufactures the missing analytic bridge inequality or a new continuation estimate; it asserts only that, once the official periodic classical scope and same-scope continuation question are fixed, any purported finite-horizon endpoint has no licit same-scope completion. Those are theorem-roadmap features rather than a mere application note, and they should be preserved in exactly that form here. In particular, the capstone should not rephrase the result as though its legitimacy depended on looking like a standard PDE continuation proof; the source manuscript itself already fixes the official theorem object and method-neutral claim class.[15] □

Imported hardening from the Navier–Stokes source. The capstone should preserve six features from *Carrier Universality and Global Smoothness for the Periodic 3D Navier–Stokes Equations* whenever it mentions the result: (i) the endpoint is the official periodic classical theorem, (ii) the proof class is closure-induced impossibility rather than constructive PDE continuation, (iii) the realized maximal classical solution is treated as one same-scope determination rather than as a sequence of independently reclassified time slices, (iv) no hidden endgame premises are allowed onto the live route, (v) the theorem-local denial map is part of the argument rather than a rhetorical add-on, and (vi) the same-scope status trichotomy is closed before the official theorem is compressed.[15]

Proposition 10.8 (Method-neutral placement of the periodic Navier–Stokes manuscript)**.** *The periodic Navier–Stokes manuscript retains theorem-roadmap status inside the capstone precisely*

because its source-level hardening already fixes three things theorem-locally: the official theorem object, the exact method class, and the one-route dependency audit. The capstone therefore preserves those features rather than translating the source into a conventional PDE-proof template.

Proof. The source manuscript states on its first pages that the endpoint is Fefferman's periodic statement (B), that the proof class is a closure-induced impossibility theorem, and that the live route to the official theorem is unique and theorem-local. It also states what the manuscript does not claim: no new continuation estimate, no manufactured analytic bridge inequality, and no hidden endgame premise off the live route. Those declarations are not rhetorical. They are part of the theorem-local hardening. Preserving them is therefore the faithful capstone treatment.[15] □

Theorem 10.9 (Closure–classical equivalence for the official periodic endpoint)**.** *Fix the official periodic scope: smooth divergence-free data on $\mathbb{T}^3$, fixed viscosity $\nu > 0$, zero forcing, the fixed mean-zero pressure gauge, the lawful-redescription class generated by the standard periodic carrier skins, and the same-scope continuation question fixed by the source manuscript. Within that fixed scope, the following are equivalent.*

1. *The official periodic classical endpoint holds: for every official datum $u_0 \in C^\infty(\mathbb{T}^3;\mathbb{R}^3)$ with $\nabla\cdot u_0 = 0$, there exists a unique smooth periodic Navier–Stokes solution pair (u,p) on $\mathbb{T}^3 \times [0,\infty)$ with the fixed mean-zero pressure gauge.*

2. *No licit same-scope finite-horizon endpoint status survives for a purported periodic breakdown. Equivalently: on the fixed periodic carrier, every purported finite-horizon endpoint proposal is either (i) represented by a standing-positive admissible in-scope witness, (ii) terminal standing failure, or (iii) an illicit same-scope gate / scope move; only clause (i) is a licit same-scope endpoint status, and that remaining witness branch is then exhausted and eliminated.*

In particular, for the official periodic problem, the closure-induced impossibility result is not an adjacent interpretive claim but a theorem-local restatement of the official classical endpoint.

Proof. (2) ⇒ (1) is exactly the manuscript-local live route compressed in the source text:

$$\text{19C–19F} \;\to\; \text{20B} \;\to\; \text{ST4/ST5} \;\to\; \text{20C} \;\to\; 21.x.$$

On that route, the fixed periodic problem first admits the unique admissible interior; the standard periodic carrier is identified with that interior; official data enter that carrier; the realized local class is shown to be exactly the full official local classical class; finite-horizon failure is already carrier-visible; the same-scope realized-lineage status trichotomy is proved and collapsed locally; and the remaining standing-positive witness branch is exhausted and eliminated. The source manuscript then states this official conclusion first as the transfer endpoint corollary and immediately afterward as the official classical restatement. (1) ⇒ (2) is the theorem-local negative form of the same official endpoint: if the official periodic classical statement holds for the full official datum class, then no same-scope finite-horizon breakdown exists in that class, so no licit same-scope endpoint status survives for such a purported breakdown. Thus the two formulations are equivalent on the fixed official periodic scope.[15] □

Theorem 10.10 (Exhaustiveness of the same-scope realized-lineage endpoint trichotomy)**.** *Fix an admitted carrier state x for the official periodic problem and let $R_{\mathrm{NS}}(x) = (u_x, p_x)$ be its realized maximal classical lineage on $[0, T^*(x))$. Assume $T^*(x) < \infty$. Then every purported same-scope finite-horizon endpoint proposal for this realized periodic lineage has exactly one of the following theorem-local statuses:*

(i) it is represented by a standing-positive admissible in-scope witness on the fixed torus carrier;

(ii) it is terminal standing failure; or

(iii) it is an illicit same-scope gate / scope move, namely a path-dependent reclassification, representative-only or metric-only gate, new selector / tolerance / discriminator, or undeclared scope change.

No fourth same-scope endpoint status exists.

Proof. The realized maximal classical periodic object is treated as one same-scope determination with one fixed continuation question, not as a sequence of independently reclassified time slices. On that fixed same-scope lineage, admissibility and standing are global rather than time-indexed, and every standing-relevant same-scope discriminator factors through the future-behavior quotient. Therefore an endpoint proposal cannot survive as a fresh stage-indexed status, a path-dependent status, or a representative-only / skin-level status.

If the endpoint is presented through the carrier-side standing-relevant horizon witness, it lies in clause (i). If instead the proposal says that the same realized lineage can be completed only by leaving admissible representability while still claiming the same torus carrier and lineage identity, that is exactly terminal standing failure, giving clause (ii). If it tries to avoid both routes while keeping the endpoint alive, it must smuggle in a new same-scope discriminator, representative-only or metric-only gate authority, selector, tolerance, or undeclared scope change, giving clause (iii). These cases are exhaustive because the same-object convention, quotient / history-factorization packet, no-admissibility-dynamics theorem, and ATS non-authority result leave no additional same-scope way to distinguish or preserve an endpoint claim. Hence no fourth same-scope endpoint status exists.[15] □

10.4 Yang–Mills as a theorem roadmap

Proposition 10.11 (Why Yang–Mills belongs here as a theorem roadmap)**.** *Under the declared source set, the Yang–Mills cluster should appear in the capstone as a theorem roadmap rather than as a proof-home appendix and rather than as a hidden foundational source.*

Proof. The Yang–Mills endpoint manuscript *Vacuum-sector Mass Gap for the Local Gauge-Invariant Sharp-Local Yang–Mills Net* already states the theorem-level object, the exact claim boundaries, the theorem-map reduction, the seam roles of the three companions, and the local-net endpoint. It fixes from the outset that the theorem-level object is the bounded-region local gauge-invariant sharp-local net generated by flowed curvature composites together with its OS/Wightman vacuum representation and vacuum-sector mass gap; that faithful-Wilson universality is qualitative only within the admissible gradient-flow renormalization class and after coherent normalization; that

the explicit quantitative Route 1 ledger remains ρ-indexed; and that extended-support line/surface sectors remain outside the theorem-level claim. The three companions then preserve fixed role splits rather than adding theorem-level ambiguity: Companion I isolates the ultraviolet gate and the live Route 1 mass-gap chain, Companion II isolates the Lane A sharp-local construction and the finite-cap / inductive-union seam discipline, and Companion III isolates reconstruction, non-triviality, faithful-Wilson universality, and the exact local-net endpoint. That is theorem-roadmap architecture, and the capstone should preserve it in exactly that form rather than compressing it into a generic existence narrative.[16, 17, 18, 19] □

Imported hardening from the Yang–Mills sources. The capstone should preserve five items from the Yang–Mills cluster: (i) the theorem-level object is local and gauge-invariant, (ii) faithful-Wilson universality is qualitative and normalization-bound rather than an unqualified quantitative uniformity claim, (iii) the Route 1 ledger is explicitly ρ-indexed, (iv) the companions carry fixed proof-home role splits, and (v) nonlocal sectors are excluded from the theorem-level endpoint rather than silently absorbed into it.[16, 17, 18, 19]

Proposition 10.12 (Endpoint-hygiene placement of the Yang–Mills roadmap)**.** *The Yang–Mills materials retain theorem-roadmap status inside the capstone because the source cluster already fixes the theorem-level object, the exact endpoint boundaries, the role of qualitative faithful-Wilson universality, the ρ-indexed status of the explicit quantitative ledger, and the exclusion of nonlocal sectors from the theorem claim.*

Proof. The endpoint manuscript and the three companions divide these burdens explicitly. The main endpoint text fixes the theorem-level local sharp-local net and the vacuum-sector mass gap, while the companions freeze the ultraviolet gate, Route 1 chain, Lane A seam discipline, reconstruction, non-triviality, and faithful-Wilson universality. Because these role splits are already source-level and theorem-facing, the capstone should preserve them as a roadmap rather than repackage them as a single undifferentiated application note.[16, 17, 18, 19] □

10.5 Measurement as a compact consequence profile

Proposition 10.13 (Why the measurement manuscript belongs here as a consequence profile)**.** *Under the declared source set, the measurement-reference manuscript should appear in the capstone as a compact consequence profile rather than as a theorem roadmap and not as a foundational proof source.*

Proof. The manuscript *The Measurement Problem as a Problem of Reference* already carries a complete fixed-domain theorem ladder: admissible reference assignments, the measurement-reference quotient theorem, global equivalence collapse, the Everettian non-selection theorem, the fixation-commutation criterion for theory-level confirmation, diachronic inheritance for memory, anticipation, deliberation, and confirmation, and selector exhaustion. It also states the exact target discourse and its denial-condition matrix. Those hardenings are real, but the application is narrower than the foundational arc and is explicitly conditional on the downstream closure package already fixed

earlier. Its role in the capstone is therefore to show what the closure theorem does to a physically non-selective measurement theory, not to introduce a second theorem spine in full. The correct placement is a compact consequence profile stating the claim type, the standing conditional, and the exact denial targets without reproducing the entire measurement manuscript.[13] □

Imported hardening from the measurement source. The capstone preserves six source-level features whenever it cites *The Measurement Problem as a Problem of Reference*: (i) the standing conditional that fixes the target explanatory practice, (ii) the measurement-reference quotient theorem, (iii) global equivalence collapse on the same scope, (iv) the Everettian non-selection theorem for the bare physical package, (v) the fixation-commutation criterion for theory-level confirmation, and (vi) the selector-exhaustion taxonomy together with the denial-condition matrix.[13]

10.6 Standard Model as a compact consequence profile

Proposition 10.14 (Why the Standard Model manuscript belongs here as a consequence profile)**.** *Under the declared source set, the Standard Model bookkeeping manuscript should appear in the capstone as a compact consequence profile rather than as a theorem roadmap.*

Proof. The role of *Structura Ex Necessitate Standardis Modelis* in the capstone is explanatory rather than proof-bearing. It gives a witness-normal-form and transport-closure ledger for a fixed renormalizable chiral gauge load, but it does not reopen the theorem class of Sections 1–9. Its contribution is to show, in a concrete physical domain, how standing-preserving redescription and transport closure collapse apparent parameter freedoms and certify which inputs remain irreducible. That is exactly the work of a compact consequence profile.[12] □

10.7 Exact placement theorem

Theorem 10.15 (Main-body placement theorem)**.** *Under the declared source set, the main body of the capstone should present downstream applications in the following exact form:*

1. *the periodic Navier–Stokes and Yang–Mills manuscripts appear as theorem roadmaps;*
2. *the measurement-reference and Standard Model manuscripts appear as compact consequence profiles;*
3. *a Zenodo-only appendix atlas may extend the downstream layer by adding further theorem roadmaps or compact consequence profiles, provided each appendix is consequence-facing only, explicitly conditional on Sections 1–9, and removable without altering the foundational arc; and*
4. *none of these downstream manuscripts is a live dependency of the foundational closure arc already completed in Sections 1–9.*

Proof. The first clause follows from the theorem-map, claim-type, denial-map, and endpoint-hygiene hardening already present in *Carrier Universality and Global Smoothness for the Periodic 3D*

Navier–Stokes Equations and in the Yang–Mills cluster. The second follows from the narrower, downstream, and explicitly conditional role of *The Measurement Problem as a Problem of Reference* and *Structura Ex Necessitate Standardis Modelis*. The third clause is the no-backflow theorem applied to a controlled appendix atlas: once additional materials are typed as theorem roadmaps or compact consequence profiles and are declared removable without changing Sections 1–9, they extend explanatory coverage without becoming proof sources. The fourth clause is Theorem 10.5 itself. □

10.8 Zenodo appendix-atlas extension

Proposition 10.16 (Zenodo appendix-atlas rule)**.** *A Zenodo consequence atlas may be appended after the audit architecture if and only if every appendix satisfies all of the following conditions:*

1. *it is typed explicitly as a theorem roadmap or a compact consequence profile;*
2. *it declares its fixed-domain object and the exact sections of the foundational arc on which it depends;*
3. *it states an exact no-backflow clause; and*
4. *deleting the appendix leaves Sections 1–9 unchanged.*

Proof. Each condition is forced by the claim-type discipline already fixed in this section. Without explicit typing, a downstream appendix drifts toward hidden-proof-source status. Without a fixed-domain declaration and dependency note, scope drift is no longer controlled. Without a no-backflow clause, the appendix can be misread as silently rebuilding an earlier theorem burden. And if deleting the appendix changes Sections 1–9, then the appendix was never merely consequence-facing. Therefore the four conditions are necessary and jointly sufficient for a Zenodo-only appendix atlas compatible with the capstone architecture. □

Exact wording for the 42 version. The capstone proves the foundational closure theorem chain in Sections 1–9. Navier–Stokes and Yang–Mills appear in the main body as theorem roadmaps, and measurement and Standard Model materials appear there as compact consequence profiles. The Zenodo-only appendix atlas extends the downstream layer across further controlled theorem roadmaps and consequence profiles, each consequence-facing only and each removable without altering the foundational arc. None of the appendix manuscripts functions as a premise of Sections 1–9.

Downstream consequence-facing only. This section is archival packaging downstream of Sections 1–9 and of the placement rules in Section 10. It records formalization posture, dependency routing, source-role discipline, and audit consequences of the already closed theorem chain. Nothing in this section adds a new proof burden to the foundational arc.

11 Downstream audit consequences: formalization posture and archival architecture

The substantive theorem chain is already closed before this section begins. What remains is downstream audit architecture: what depends on what, what is proved locally, what is imported, what Lean support is actually documented, what a hostile reader must deny to reject the paper, and which sources are permanently outside the live theorem arc.

Import note 11.1 (Audit sources used here)**.** This section uses only the documentation actually supplied by the declared source set: *Foundational Closure of Admissibility and Standing* for the fixed-kernel formalization boundary and documented package facts; *The Mathematics of Coherent Reasoning: A Bivalent Trajectory Theory* for the public theorem surface, theorem-family correspondences, and dependency spine of the trajectory-side Lean development; *Non-Instantiability of Gödel Incompleteness in the Unique Admissible Interior* for the manuscript-specific axiom-free theorem-ladder audit; *Standing–Admissibility Equivalence* for the audit-grade elimination of residual dual-track governance between admissibility and standing; *The Exhaustion Theorem* for typed boundary discipline, fixed-signature factorization against same-scope discriminators, and the capstone-facing necessity-layer witnesses; and the completed capstone itself as the theorem object whose dependencies are now being frozen. No broader Lean claim is taken from the latter two sources than those audit-facing closure roles.

11.1 Downstream formalization posture

Definition 11.2 (Capstone formalization boundary)**.** The *capstone formalization boundary* is the exact point at which the manuscript stops making formalization claims stronger than the supplied sources support. In the present manuscript that boundary is as follows:

1. the kernel floor is *not* claimed as a theorem generated inside a lower internal Lean layer;
2. documented Lean surfaces are cited only where a source manuscript names public theorem wrappers, theorem families, package facts, or a manuscript-specific audit statement; and
3. no capstone-wide merged Lean development is claimed unless a dedicated capstone audit exists in the declared source set.

Theorem 11.3 (No-overclaim formalization rule)**.** *The capstone may claim formalization support only in one of the following forms:*

1. theorem-level *support, when the declared sources document exact public theorem or wrapper names;*
2. theorem-family or module-level *support, when the declared sources document only a family-level or module-level correspondence; or*
3. manuscript-specific audit *support, when a source manuscript states that its own theorem chain has been encoded in Lean and its exported paper-facing theorems are axiom-free, but does not expose the full public name surface in the retrieved documentation.*

No fourth, stronger form is admissible here.

Proof. Theorem-level support is the strongest form explicitly documented in *Foundational Closure of Admissibility and Standing* and *The Mathematics of Coherent Reasoning: A Bivalent Trajectory Theory*. Theorem-family or module-level support is also explicitly documented there when the paper-level theorem compresses several lower-level Lean results or infrastructure layers. The manuscript-specific audit form is the strongest support documented in *Non-Instantiability of Gödel Incompleteness in the Unique Admissible Interior* for the same-domain Gödel non-instantiability family. A stronger capstone-wide claim would require a dedicated unified Lean audit, which is not present in the declared source set. So no fourth, stronger form is admissible. □

Corollary 11.4 (Present capstone formalization posture)**.** *The present capstone is auditable in two exact ways and in no stronger one:*

1. *as an in-paper theorem chain whose proof burden is carried by the main sections and their declared source roles; and*
2. *as a layered manuscript whose major theorem families can be linked to documented Lean surfaces at theorem, theorem-family, or manuscript-audit level where such documentation exists.*

It is not licit on the present evidence to claim one unified capstone Lean development with a complete exported theorem surface.

Remark 11.5 (Formalization posture of the fixed floor)*.* Within the declared source set, the fixed floor is not treated as an omitted lower theorem waiting to be recovered by proof-assistant work. It is treated as a theorem-level boundary on what same-domain derivation could mean at all, while the downstream uniqueness, collapse, transport, irrecoverability, and non-instantiability families are exactly the parts for which theorem-level, theorem-family, or manuscript-audit Lean support is claimed. The formalization posture is therefore faithful to the proof order of the manuscript rather than weaker than it.

11.2 Downstream audit seal: axiom-necessity layer and closed-system closure

Definition 11.6 (Capstone axiom-necessity layer)**.** For audit purposes, collect the governing closure premises already used by Sections 3–8 under the name *capstone axiom-necessity layer*:

1. evaluated-fragment bivalence together with fail-closed governance;
2. finite same-scope compositional closure for candidate acts, histories, and continuations;
3. lawful redescription neutrality together with stable reference;
4. non-degeneracy of the evaluated reasoning regime;
5. irreversibility and the ban on retroactive repair of standing-bearing lineage;
6. no undeclared same-scope boundary selector, discriminator, or gate-multiplying authority;

7. identity discipline for same-scope operator or representative copies, so that label-only duplication cannot create standing-bearing difference; and
8. well-founded local denial-target discipline for hostile-reader audit and failure localization.

These are not new starting axioms of the capstone. They are the audit-facing names for the governing premises whose necessity is exposed by the completed theorem chain and by the witness package consolidated in *The Exhaustion Theorem* [21].

Theorem 11.7 (Axiom-necessity layer for non-degenerate reasoning)**.** *Each member of the capstone axiom-necessity layer is individually necessary for non-degenerate reasoning in the sense of this manuscript. More precisely:*

1. *weakening bivalent fail-closed governance reopens intermediate or competing standing classifications and destroys same-scope gate uniqueness;*
2. *weakening finite same-scope compositional closure reopens deferred admissibility and removes the forcing of pre-evaluation gate discipline;*
3. *weakening lawful redescription neutrality reopens presentation-dependent standing and non-factorizing same-scope discriminators;*
4. *weakening non-degeneracy collapses the regime into vacuous all-accept or all-reject form;*
5. *weakening irreversibility reopens retroactive repair, reinterpretation, and same-lineage reset;*
6. *weakening boundary closure reopens selectable competing gates, metadata-level authorizers, and undeclared same-scope scope drift;*
7. *weakening identity discipline for operator or representative copies reopens label-dependent splitting beneath unchanged standing content; and*
8. *weakening well-founded local denial-target discipline destroys unique minimal failure localization and with it the audit-level routing of objections.*

Hence none of these governing premises is dispensable if the capstone is to remain a non-degenerate same-scope closure theorem.

Proof. Sections 3 and 5 already force evaluated-fragment bivalence, fail-closed admissibility, and unique same-scope classification; allowing intermediate or fail-open standing statuses would therefore reinstall same-side splits or competing gates beneath the same fixed scope. Finite same-scope compositional closure is required because the capstone's gate theorems and irrecoverability claims are about lawful reuse, continuation, and recombination; if those candidate composites need not be formable, deferred admissibility reappears and pre-evaluation gate discipline is no longer forced. Lawful redescription neutrality and stable reference are necessary because otherwise standing can vary by presentation alone, which is exactly the representative-level non-factorization blocked by Section 5. Non-degeneracy is necessary because without at least one live admissible/inadmissible split the closure theorem becomes vacuous.

Irreversibility is necessary because otherwise same-act repair, retrospective reinterpretation, or lineage reset reopen, contradicting the no-repair family and the same-domain meta-closure results. Boundary closure is necessary because every same-scope selector, extra discriminator, metadata-level authorizer, or silently imported gate performs the very authority multiplication ruled out by the AMetric boundary and by Theorem 5.26. Identity discipline for operator or representative copies is necessary because otherwise label-only duplication or representative privileging can generate standing-bearing difference beneath unchanged quotient content, reopening the representative-smuggling routes blocked in Sections 5 and 6. Finally, well-founded local denial-target discipline is necessary for audit because the hostile-reader architecture of this capstone requires a unique minimal place at which a denial must land; without that discipline, failure localization blurs and the audit map ceases to route objections to a determinate theorem-bearing burden. The witness package of *The Exhaustion Theorem* records the short countermodel form of each of these failures, while the capstone sections named above show that none of them can be relaxed without reopening a forbidden route. □

Corollary 11.8 (Formal closed-system seal)**.** *Relative to the declared source set, the capstone is closed in the formal same-scope sense: every internal objection must do exactly one of the following:*

1. *deny one member of the capstone axiom-necessity layer;*
2. *deny a named theorem whose proof depends on that member; or*
3. *declare a scope change.*

No fourth same-scope internal route remains.

Proof. Theorem 11.7 removes the last "dispensable premises" objection by showing that each governing closure premise carries a corresponding failure mode when weakened. Section 5 closes residual dual-track governance and same-scope discriminator escape by Theorems 5.9 and 5.26. Section 8 then exhausts the same-domain meta-level rescue routes from above, especially in Theorems 8.23 and 8.27. So an objection that remains inside the same scope has no unclassified route left: it must identify the governing premise or theorem it denies. If it introduces new same-scope authority instead, the move is a scope change rather than an internal objection. □

Remark 11.9 (What the necessity layer does and does not do)*.* The present subsection does not convert the capstone into a free-standing axiom-starting paper. It does something narrower: it makes explicit that the governing closure premises already used by the manuscript are individually necessary for non-degenerate same-scope reasoning, so that the remaining hostile-reader move is forced from vague dissatisfaction into named premise denial or scope change.

11.3 Downstream dependency packaging

Theorem 11.10 (Final dependency register theorem)**.** *As a downstream audit consequence of the completed closure theorem, the capstone has one strict dependency order and no hidden parallel proof route:*

1. *Section 2 fixes the kernel floor and mechanization boundary.*

2. *Section 3 forces state-transition form, identity persistence, irreversible commitment, and governance-level bivalence.*
3. *Section 4 forces the AMetric boundary, relabeling-indifference collapse, no selector / no order / no deeper invariant, and singleton admissibility.*
4. *Section 5 fixes gate existence, standing emergence, standing–admissibility equivalence, same-scope factorization against standing-relevant discriminators, no repairs / no generators / no carriers, standing conservation, the standing quotient, the unique admissible interior, and the compressed no-alternative-architecture theorem for the same-domain interface.*
5. *Section 6 proves operator exhaustion and mathematics closure.*
6. *Section 7 proves physics closure and descriptive non-generativity.*
7. *Section 8 proves same-domain meta-closure and the non-instantiability of Gödel-style escape routes.*
8. *Section 9 proves two-spectrum exhaustion, closure parity, mirror classification, the prohibition package, and descriptive closure of admissible description.*
9. *Section 10 places the downstream consequence manuscripts without allowing them to backflow into the already closed theorem chain.*
10. *The present section adds no new foundational burden; it stabilizes the audit architecture of the completed chain, exposes the axiom-necessity layer for the governing closure premises, and states the formal closed-system corollary of the already completed theorem chain.*

No later section reopens an earlier burden without explicit revision of the declared source roles.

Proof. This is exactly the dependency discipline enforced by the capstone as rewritten. Each section depends only on the sections and declared sources activated before it, and Theorem 10.5 already rules out application backflow from Section 10. So the dependency order is strict and non-circular. □

Audit architecture principle. This section is archival, not generative. Its role is to make the theorem chain inspectable, not to supply missing proof content by packaging rhetoric.

A Proof-home and downstream consequence register

Register rule after Tier 3 hardening. Sections 2–8 are the local proof home for the core theorem spine. Sections 9–11 classify, audit, or route consequences of that spine. Downstream application manuscripts remain consequence-facing only and do not function as live proof sources for the local foundational arc.

Table 1: Final proof-home / downstream register for the hardened capstone

Capstone section	**Burden fixed there**	**Status in this manuscript**	**Downstream or audit note**
§ 1	title, scope, claim class, theorem arc, and source-role rule	local framing only	routes the reader; does not itself carry a theorem burden
§ 2	target class, kernel forcing, non-derivability, minimality, and mechanization boundary	**local proof home**	earlier source manuscripts remain provenance only for this layer
§ 3	modeling necessity, identity persistence, irreversible commitment, coherence meta-theorem, and finite fixed-scope trajectory geometry	**local proof home**	declared sources remain provenance and correspondence support only
§ 4	pre-fixational non-reference, relabeling indifference, equality-pattern completeness, AMetric collapse, no deeper invariant, and pre-boundary no-faithful-lower-generator closure	**local proof home**	declared sources remain provenance and correspondence support only
§ 5	gate existence, standing emergence, standing–admissibility equivalence, same-scope factorization, no repairs / generators / carriers, standing quotient, no stricter standing refinement, unique interior, and closure of the same-domain admissibility architecture	**local proof home**	declared sources remain provenance and correspondence support only

Capstone section	Burden fixed there	Status in this manuscript	Downstream or audit note
§ 6	standing-bearing carrier discipline, no same-scope totalization, conservative extension collapse, finite exhaustion of covered same-scope operator moves, interaction closure, and mathematics closure	**local proof home**	supporting operator manuscripts remain provenance and cross-check support only
§ 7	licensed metric description, physical non-generativity, role discipline, and compact application profiles	local classificatory section under declared support	this is downstream-facing in posture even though placed in the main body
§ 8	inferential-life exhaustion, inferential necessity of closure distinctions, code-role exhaustion, Gödel-threat alignment, translation filter, no same-domain diagonalization, rescue-route exhaustion, and non-instantiability	**local proof home**	the dedicated Gödel manuscript now functions primarily as provenance and audit support for this local section
§ 9	fixation split, two-spectrum exhaustion, closure parity, mirror classification, prohibition package, post-closure practice, and descriptive closure of admissible description	downstream classificatory synthesis	not a fresh foundational engine
§ 10	downstream consequence placement, claim-type discipline for applications, and no-backflow rule	downstream routing section	application manuscripts remain roadmap/profile sources only

Capstone section	Burden fixed there	Status in this manuscript	Downstream or audit note
§ 11	audit posture, formalization posture, axiom-necessity layer, and formal closed-system audit seal	audit layer	constrains claim type; does not reopen Sections 2–8

Remark A.1. The register is intentionally asymmetric. Sections 2–8 may be cited as the local theorem spine of the manuscript. Sections 9–11 may classify or audit consequences of that spine, but by Theorem 10.5 they do not function as new proof sources for it.

B Formalization map and theorem-register note

Claim discipline. This appendix states formalization posture only to the exact level documented in the declared sources. It does not enlarge the manuscript's proof claims, and it does not infer any unified public Lean surface where the sources themselves document only theorem families, wrapper surfaces, or manuscript-specific audits.

Documented package facts carried by the declared sources

Documented package facts.

- *Foundational Closure of Admissibility and Standing* documents a current Lean package `MaleyLean`, version `0.1.0`, on toolchain `leanprover/lean4:v4.28.0`, with public entry points `Main.lean` and `MaleyLean.lean`, theorem aggregation in `TheoremRegister.lean`, and paper-facing wrappers in `PaperStatements.lean`.
- *The Mathematics of Coherent Reasoning: A Bivalent Trajectory Theory* documents a public theorem surface exported through `TheoremRegister.lean` and `PaperStatements.lean`, with a dependency spine through `AmetricBoundary`, `Transport`, `Witness`, `NoRepair`, `Irrecoverability`, `StandingDeterminacy`, `Bivalence`, and `BivalenceUniqueness`.
- *Non-Instantiability of Gödel Incompleteness in the Unique Admissible Interior* states that its theorem chain has been encoded in Lean4 with the same proof order, that the exported paper-facing theorem ladder is axiom-free, and that the documented export surface includes the route-exhaustion theorems, Gödel-threat alignment, the translation filter for ordinary Gödel objections, and the main non-instantiability theorem, while assuming the same primitive floor rather than generating it internally.

Paper-to-Lean map at the exact documented level

Table 2: Paper-to-Lean map for the capstone, stated only to the level documented in the declared sources

Capstone burden	**Documented support, status, and boundary note**
Kernel floor, non-derivability, mechanization boundary (§ 2)	**Documented support.** *Foundational Closure of Admissibility and Standing* package facts and wrapper table. **Claim status here.** manuscript-level theorem-family documentation only; no separate public lower-layer theorem surface is claimed for kernel generation. **Boundary note.** The manuscript proves Section 2 locally. The formalization claim begins only downstream of the fixed kernel floor.
Bivalence, same-side split collapse, standing emergence, no same-act repair, no generators, no primitive carriers, uniqueness of admissible interior, standing conservation (§ 4–5)	**Documented support.** `PaperBivalenceInterfaceStatement`, `PaperSameSideSplitCollapseSummaryStatement`, `PaperStandingEmergenceStatement`, `PaperNoSameActRepairStatement`, `PaperNoGeneratorsStatement`, `PaperNoPrimitiveCarriersStandingFormStatement`, `PaperUniquenessOfAdmissibleInteriorStatement`, `PaperStandingConservationStatement`. **Claim status here.** theorem-level where documented in *Foundational Closure of Admissibility and Standing* for the fixed-kernel consequence layer; local proof surface in the manuscript itself for Sections 4–5. **Boundary note.** The manuscript now proves this layer locally and cites the documented Lean surface only as corroborative correspondence.
State-transition form, identity persistence, irreversible commitment, coherence meta-theorem, and finite fixed-scope trajectory geometry (§ 3)	**Documented support.** *The Mathematics of Coherent Reasoning: A Bivalent Trajectory Theory* package facts together with its documented theorem surface exported through `TheoremRegister.lean` and `PaperStatements.lean`. **Claim status here.** theorem-family and dependency-spine level for the modeling theorem family; theorem-level and theorem-family correspondence for the finite fixed-scope trajectory results where documented in *The Mathematics of Coherent Reasoning: A Bivalent Trajectory Theory*. **Boundary note.** The manuscript proves Section 3 locally. The documented Lean correspondence at this layer is only the trajectory-side surface explicitly recorded in *The Mathematics of Coherent Reasoning: A Bivalent Trajectory Theory*.
AMetric collapse, standing-region metric content, failure-region metric collapse, standing bivalence, bivalence uniqueness, irrecoverability family (§ 4–5)	**Documented support.** `AmetricCollapseTheorem`, `Paper_AmetricCollapseTheorem`, `Paper_StandingRegionCarriesAllAdmissibleMetricContent`, `Paper_FailureRegionSupportsNoAdmissibleMetricContent`, `Paper_IrrecoverabilityIsMetricInvariant`, `Paper_StandingBivalenceTheorem`, `Paper_BivalenceUniquenessTheorem`. **Claim status here.** theorem-level and theorem-family level in *The Mathematics of Coherent Reasoning: A Bivalent Trajectory Theory*. **Boundary note.** The manuscript proves the AMetric and interface-forcing layer locally; the Lean surface documented for this layer remains exactly the theorem-level / theorem-family surface stated in *The Mathematics of Coherent Reasoning: A Bivalent Trajectory Theory*, no more.

Capstone burden	Documented support, status, and boundary note
Operator exhaustion and mathematics closure (§ 6)	**Documented support.** none documented at the capstone level for the new same-scope operator-closure stack. **Claim status here.** text-level local proof surface from *Closure by Exhaustion for Same-Scope Operators under Admissibility*, *Clausura Ex Necessitate Structurae*, and *Fixed-Base Exhaustion for Internal Operators*, with post-closure classification support from *Eadem Natura, Duo Spectra.* **Boundary note.** No separate public Lean surface is claimed here.
Same-domain Gödel non-instantiability (§ 8)	**Documented support.** manuscript-specific Lean audit stated in *Non-Instantiability of Gödel Incompleteness in the Unique Admissible Interior*, including a documented paper-facing theorem ladder for inferential-life exhaustion, closure-relevance transmission, code-predicate role exhaustion, Gödel-threat alignment, the translation filter for ordinary Gödel objections, route exhaustion, and the main non-instantiability theorem. **Claim status here.** manuscript-audit level with documented exported theorem names for the Section 8 theorem ladder. **Boundary note.** The manuscript proves Section 8 locally and relies on the documented manuscript-specific audit only as stated there. No broader unified Lean surface is inferred.
Consequence manuscripts (§ 10)	**Documented support.** none in the present capstone. **Claim status here.** no capstone Lean claim. **Boundary note.** These remain downstream consequence roadmaps or profiles only.
Capstone-wide merged theorem register	**Documented support.** none supplied. **Claim status here.** not claimed. **Boundary note.** A dedicated unified capstone Lean audit would be required before such a claim becomes licit.

Interpretive rule. Where the declared sources give exact public theorem names, the manuscript may repeat them. Where they give only theorem families or module-level provenance, the manuscript must say so. Where they give only a manuscript-specific audit statement, the manuscript must stop there. No further formalization claim is licensed.

C Hostile-referee attack map

Use after Tier 3 hardening. For the present Zenodo manuscript, the core theorem spine is carried locally in Sections 2–8. This appendix therefore routes objections first to the in-manuscript theorem ladder. Corpus documents remain relevant only for downstream consequence layers, source provenance, or audit posture outside the local spine.

Table 3: Hostile-reader attack map for the hardened manuscript

To keep this objection alive, deny at least one of:	**Primary denial target**	**Why this is the right place to attack**
The target class is a protective narrowing rather than the minimal domain in which proposition-bearing reasoning remains assessable.	§ 2	The manuscript now carries the assessability-boundary argument locally, so this attack must engage the local theorem spine rather than a corpus-level dependency claim.
The kernel is merely a presentational starting point or a weak axiom block rather than a derived floor.	§ 2	Section 2 now bears the local forcing burden for kernel emergence, non-derivability, minimality, and no-lower-generator closure.
State-transition form is only a modeling preference.	§ 3	The modeling theorem, identity persistence, irreversible commitment, and coherence meta-theorem are now local; the attack must therefore deny the section's explicit forcing chain.
The AMetric boundary is assumed rather than derived.	§ 4	Section 4 now carries the local symmetry, equality-pattern, no-selector, no-order, no-metric, and no-deeper-invariant ladder.
Uniqueness of the admissible interior is merely announced.	§ 5	The standing-regime section now proves gate uniqueness, standing emergence, no repairs / no generators / no carriers, quotient closure, and unique interior locally.
Admissibility and standing remain distinct analytical dimensions or dual-track governance survives on the fixed domain.	§ 5	Section 5 now proves the same-scope equivalence and blocks residual dual-track governance locally.

To keep this objection alive, deny at least one of:	**Primary denial target**	**Why this is the right place to attack**
The same-domain discriminator space is not actually exhausted; some sixth same-scope route still survives.	§ 5	The local quotient and factorization theorems force same-scope standing-relevant discriminators to factor through the standing quotient; non-factorization is itself the illicit-authority witness.
Mathematics can still reopen the interior through forcing, completion, compactness, choice, or other operator families.	§ 6	The operator-exhaustion layer is now local and explicitly carries no same-scope totalization, conservative extension collapse, family-wise closure, and interaction closure.
Gödel, semantic truth, infinitary rules, or same-domain extension reopen the closure theorem from above.	§ 8	Section 8 now contains the explicit incompleteness-attack defeat, including code-role exhaustion, route exhaustion, and same-domain non-instantiability.
Physics still generates standing through descriptive or metric richness.	§ 7	This remains the softest claim layer of the local manuscript and is the correct place to attack if one rejects the physics-facing descriptive discipline.
Mathematics grounds physics, physics grounds mathematics, or a meta-level platform reinstalls privilege.	§ 9	Section 9 is now explicitly downstream and classificatory; an objector must therefore deny the synthesis as a post-closure classification result rather than as a hidden second foundation.

To keep this objection alive, deny at least one of:	**Primary denial target**	**Why this is the right place to attack**
The downstream application manuscripts are hidden premises of the foundational arc.	§ 10 together with Theorem 10.5	Section 10 fixes the one-way consequence posture and blocks downstream-to-foundation backflow explicitly.
The periodic Navier–Stokes roadmap is invalid because the official theorem would have to be proved by a standard constructive PDE continuation argument.	§ 10 together with Proposition 10.6 and *Carrier Universality and Global Smoothness for the Periodic 3D Navier–Stokes Equations* as theorem roadmap only	This objection now lands wholly in the downstream roadmap layer, not in the foundational spine.
The Yang–Mills roadmap overclaims universality or hides endpoint restrictions.	§ 10 together with the Yang–Mills roadmap	This is a downstream roadmap objection only and no longer bears on the local theorem spine.
The measurement consequence is only semantic redescription and not closure-relevant.	§ 10 together with *The Measurement Problem as a Problem of Reference*	This is also downstream consequence-facing only; the attack cannot now be used to reopen Sections 2–8.
The formalization claims are overstated.	§ 11 and Appendix B	The manuscript now separates local proof force from documented formalization posture more sharply; an objector must engage the exact documented level claimed.
The capstone still rests on dispensable closure premises that could be weakened without reopening anything essential.	§ 11 together with Theorem 11.7 and Corollary 11.8	The final audit section exposes premise-necessity and forces the critic either to deny a named premise, deny a theorem depending on it, or declare scope change.

Use of the map. For the hardened manuscript, slogan-level objections are no longer enough against the core theorem spine. A critic must now deny a local section theorem, or else move explicitly into the downstream consequence layers where import posture is openly declared.

D Downstream reading path and appendix guide

Downstream consequence-facing only. This appendix is archival packaging downstream of Sections 1–9 and Section 10. It records downstream consequence routing or audit posture only and does not function as a live proof source for the foundational arc.

Shortest authoritative reading path

1. Front matter: title page and abstract.
2. § 1 for claim class, scope posture, source-role discipline, and the theorem arc.
3. § 2 for the target class, kernel floor, non-derivability, and mechanization boundary.
4. § 3 for modeling necessity, identity persistence, irreversible commitment, and the coherence theorem family.
5. § 4 for the AMetric boundary, relabeling-indifference collapse, and the no-deeper-invariant theorem.
6. § 5 for gate existence, standing emergence, standing–admissibility equivalence, quotient / factorization closure against same-scope discriminators, unique interior, and the compressed no-alternative-architecture theorem.
7. § 6 for operator exhaustion and mathematics closure.
8. § 8 for same-domain meta-closure and the non-instantiability theorem.
9. § 9 for the two-spectrum synthesis and the descriptive-closure theorem.
10. § 10 only if downstream main-body placement matters to the reader's objection.
11. § 11 together with Appendices A, B, C, D, and E for the audit architecture, the axiom-necessity layer, and the formal closed-system seal.
12. Appendix F and the later atlas appendices only after the foundational arc is fixed.

This path verifies the capstone as one forced closure theorem before any downstream roadmap or consequence profile is read in detail.

Application-specific reading after the foundational arc

1. If the objection targets the periodic Navier–Stokes consequence, read § 10 and then *Carrier Universality and Global Smoothness for the Periodic 3D Navier–Stokes Equations* beginning with its page-1 claim-type block, same-object convention, compressed proof spine, theorem map, and denial map.
2. If the objection targets the Yang–Mills consequence, read § 10 and then *Vacuum-sector Mass Gap for the Local Gauge-Invariant Sharp-Local Yang–Mills Net* together with the proof-home

role splits recorded in *Companion I: Ultraviolet Gate and Route 1 Mass-Gap Chain*, *Companion II: Lane A Sharp-Local Construction*, and *Companion III: Reconstruction, Non-Triviality, and Faithful-Wilson Universality*.

3. If the objection targets the main-body measurement consequence, read § 10 and then *The Measurement Problem as a Problem of Reference* beginning with the fixed downstream substrate, the measurement-reference quotient theorem, the Everettian non-selection theorem, the fixation-commutation criterion, and the selector-exhaustion taxonomy.

4. If the objection targets a Zenodo-only appendix-atlas consequence, begin with Appendix F, then read the relevant appendix, and only then consult the cited source manuscript beginning from its own fixed-domain declaration and non-claims.

Appendix guide

Table 4: Downstream appendix guide for the 42 consequence-atlas version

Appendix	**What it does**
Downstream consequence register	States the one-directional theorem arc and the first downstream consumers of each main section.
Downstream formalization map	States exactly what Lean surfaces are documented and at what level.
Downstream hostile-reader attack map	Lists the named denial targets for foundational and downstream objections.
Downstream reading path	Gives the shortest authoritative route through the manuscript and the post-foundational application reading order.
Downstream source-role and proof-home note	Declares which documents remain live theorem sources, which remain consequence-facing only, and which are excluded from the present dependency chain.
Downstream appendix atlas	States the governing exhaustion schema and the no-backflow rule for every later consequence appendix.
Measurement appendix	Records the measurement-reference quotient, Everettian non-selection, fixation commutation, and theory-level confirmation payoff.
Entanglement appendix	Records the Bell-scope joint-standing-class reading and the elimination of hidden-variable, signaling, mechanism, and repair readings on the same scope.
Gauge / constraint appendix	Records the roadmap-level payoff for canonical GR and selector-bearing quantum structure relative to admissibility-derived rigidity.

Appendix	What it does
Interior-bounds appendices	Record the beginning-side L-metric boundary and the end-side exhaustion of admissible redescription without origin or termination narratives.
Baryogenesis appendix	Reclassifies matter–antimatter asymmetry as a standing-preserving identity consequence under irreversible cosmic evolution.
Black-hole appendix	Reclassifies horizons as fixation boundaries and dissolves the information paradox without changing exterior black-hole physics.
Life appendix	Records standing localization, life as minimal standing instantiation, and the constrained roles of intelligence and technology.
Mind appendix	Preserves the roadmap-level structural mind theorem without importing a second foundational proof ladder.
Meta-necessity appendix	Records the proof-of-proof corollaries on irreversibility, temporal asymmetry, and informational non-primitivity.
Dark-sector appendix	Records the meta–mystery and dark-sector eliminative constraints as late consequence diagnostics.
References	Bibliography for the final manuscript.

E Downstream source-role and proof-home note

Downstream consequence-facing only. This appendix is archival packaging downstream of Sections 1–9 and Section 10. It records source-role discipline only and does not function as a second proof engine for the foundational arc.

Final source-role rule. No document functions as a live proof source for the capstone unless its role is declared below. Consequence manuscripts, theorem roadmaps, companion proof homes, and provenance-only documents do not become live dependencies by proximity, familiarity, or later convenience.

Core proof-home rule. Sections 1–8 now carry the load-bearing theorem spine locally. The source-role freeze below therefore governs only (i) provenance and audit traceability for that local spine and (ii) downstream consequence typing beyond it. A cited source is not thereby a live premise unless this appendix says so.

Zenodo posture. The appendix atlas extends explanatory and consequence coverage only. It does not silently reopen the foundational burden of the main theorem chain.

Live proof sources for the foundational arc

- *Foundational Closure of Admissibility and Standing*: target class, kernel forcing, non-derivability, minimality, mechanization boundary, and the documented fixed-kernel Lean surface.
- *The AMetric Boundary: Necessity, Non-Parameterization, and Closure by Exhaustion*: canonical state-transition representation, coherence necessity, the coherence meta-theorem, governance-level bivalence, state-transition normal form of bivalence, pre-fixational non-reference, relabeling symmetry, equality-pattern completeness, no selector / no order / no metric beyond equality, boundary non-action, no deeper invariant, singleton meta-necessity, and the derived AMetric-boundary axiom.
- *The Mathematics of Coherent Reasoning: A Bivalent Trajectory Theory*: finite fixed-scope trajectory closure, prefix and composition geometry, lawful redescription invariance, irrecoverability of coherence failure, local fixed-substrate closure, and the documented theorem-level / theorem-family Lean correspondence.
- *Necessity of Admissibility in Non-Degenerate Compositional Systems*: the fixed-domain pre-boundary no-faithful-lower-generator hardening used in Section 4, together with gate existence, standing emergence, no same-act repair, no generators, no independent carriers, standing conservation, unique admissible interior, and the same-domain extension-stable no-lower-generator hardening used in Section 2.
- *Standing–Admissibility Equivalence*: the positive equivalence closure between admissibility and standing used in Section 5 to eliminate every residual dual-track governance form.
- *Anchor / Tensor / Skin: A Role Decomposition for Admissible Description*: the standing-quotient universal property, soundness/completeness of standing presentations, standing-quotient uniqueness, role exhaustiveness, and skin non-authority in derived form.
- *The Mathematics of Coherent Reasoning: A Bivalent Trajectory Theory* (in its fixed-substrate closure role): the no-stricter-refinement theorem showing that no admissibility-preserving standing generator survives beneath tensor content on the fixed substrate.
- *Unus Solus Possibilis Est*: the typed-domain closure packaging that makes discrete multiplicity elimination and apex uniqueness explicit for the admissible interior.
- *The Exhaustion Theorem*: typed boundary discipline, quotient-factorization against same-scope standing-relevant discriminators, and the necessity-layer witnesses by which any remaining internal objection must identify the governing premise it denies.
- *Closure by Exhaustion for Same-Scope Operators under Admissibility*: graph rigidity on the standing-bearing carrier, no same-scope totalization, the five covered mechanisms for apparent same-scope totalization, family-wise operator closure, interaction closure, and the global closure-by-exhaustion theorem for mathematics.
- *Clausura Ex Necessitate Structurae*: conservative extension collapse, interpretability collapse, the finite standing-extending-operator schema, mathematics closure, physics closure, fixation split, and the capstone closure engine.

- *Fixed-Base Exhaustion for Internal Operators*: predecessor and cross-check for the old-base no-internal-totalization hinge and the auxiliary-data dichotomy in fixed-base language.
- *Non-Instantiability of Gödel Incompleteness in the Unique Admissible Interior*: same-domain meta-closure, inferential-life exhaustion, code-role exhaustion, Gödel-threat alignment, the translation filter for ordinary Gödel objections, rescue-route exhaustion, and same-domain Gödel non-instantiability.
- *Eadem Natura, Duo Spectra*: two-spectrum synthesis, closure parity, mirror classification, prohibition package, and descriptive closure of admissible description.

Consequence-facing only under the present source-role freeze

- *The Measurement Problem as a Problem of Reference* is consequence-facing only. The capstone preserves its claim class and hardening results — measurement-reference quotient, equivalence collapse, Everettian non-selection, fixation commutation, selector exhaustion, and denial-condition matrix — without importing it as a second foundational proof spine.
- *Structura Ex Necessitate Standardis Modelis* is consequence-facing only. The capstone preserves its witness-normal-form and transport-certificate role without promoting its fixed-load ledger to a new foundational source.
- *Carrier Universality and Global Smoothness for the Periodic 3D Navier–Stokes Equations* is a theorem roadmap only. The capstone preserves its official endpoint, claim-class discipline, same-object convention, one-route dependency audit, and theorem-local denial map, but it does not import the Navier–Stokes manuscript as a premise of Sections 1–9.
- *Vacuum-sector Mass Gap for the Local Gauge-Invariant Sharp-Local Yang–Mills Net* is a theorem roadmap only, and *Companion I: Ultraviolet Gate and Route 1 Mass-Gap Chain*, *Companion II: Lane A Sharp-Local Construction*, and *Companion III: Reconstruction, Non-Triviality, and Faithful-Wilson Universality* remain proof homes behind that roadmap only. The capstone preserves theorem-level object, theorem map, endpoint hygiene, claim boundaries, and non-claims, but does not reactivate the companions as hidden premises of the foundational arc.
- *Quantum Entanglement as Boundary-Level Compatibility* is consequence-facing only. The capstone uses it to type Bell-scope correlations as joint-standing-class tensor compatibility and to eliminate hidden-variable, signaling, mechanism, and repair readings on that fixed scope; it does not use the manuscript as a premise of Sections 1–9.
- *Admissibility and the Structural Origin of Gauge and Constraint Dynamics* is roadmap-only in the 42 atlas. The rigidity theorem proved there is not imported as a new proof source. Only its downstream classificatory payoff for canonical GR and selector-bearing quantum formalisms is preserved.
- *Beginning of the Admissible Interior* and *End of the Admissible Interior* are consequence-facing only. The capstone preserves their licensing-side and exhaustion-side classifications of the admissible interior without importing either manuscript as a premise of the kernel, standing, or closure theorems.

- *Matter–Antimatter Asymmetry as a Standing Constraint: Baryogenesis Under Admissible Cosmic Evolution* is consequence-facing only. The capstone preserves its identity-persistence payoff under irreversible cosmic evolution without importing any microphysical mechanism claim into the foundational arc.
- *Black Holes Without Interior Facts* is consequence-facing only. The capstone preserves its fixation-boundary diagnosis of black-hole interiors without promoting it to a new quantum-gravity source.
- *Standing Localization and the Structural Inevitability of Life* is consequence-facing only. The capstone preserves its structural roles for life, intelligence, and technology without importing biochemical, probabilistic, or anthropic claims into the foundational arc.
- *Mind as a Structurally Rigid Interface of Admissible Evaluation* is roadmap-only in the 42 atlas. The capstone preserves its structural rigidity payoff without importing the source's full proof-granularity ladder as a second foundational theorem spine.
- *Meta-Necessity of Admissibility: A Proof-of-Proof for the Maley Corpus* is consequence-facing only. Its proof-of-proof corollaries on irreversibility, temporal asymmetry, and informational non-primitivity are carried over only as downstream consequences and not as replacements for the capstone's own kernel derivation.
- *The Meta–Mystery Theorem and the Dark Sector Eliminative Principle* is consequence-facing only. The capstone preserves its eliminative diagnostic for persistent mystery and dark-sector patch cascades without allowing it to backflow into Sections 1–9.

Not claimed as live proof sources

- No unified capstone Lean development or capstone-wide theorem register beyond what the declared sources document.
- No dormant or privately circulated manuscript promoted after the present freeze.
- No slides, oral explanations, or unpublished notes.
- No source-level hardening from a downstream manuscript may be treated as theorem-bearing for the foundational arc unless the source set is explicitly reopened and the theorem order rebuilt.

Interpretive consequence. The capstone is stable against theorem drift only if these role distinctions are kept literal. A proof home is not automatically a live premise. A theorem roadmap is not automatically part of the foundational arc. A consequence profile is not a hidden reconstruction of the main theorem chain. The Zenodo appendix atlas therefore extends explanatory coverage only by strict consequence typing, never by silent backflow into the locally proved spine.

F Downstream appendix atlas and claim-type discipline

Consequence-facing only. This appendix and every appendix that follows it are consequence-facing only. None functions as a proof source for Sections 1–9. Each is conditional on the fixed closure theorem of the main text, and deleting the entire atlas leaves the foundational arc unchanged.

Scope and dependency declaration

This appendix records the governing rule for the Zenodo-only atlas. It depends only on the claim-type discipline of Section 10, the AMetric and standing results of Sections 4–5, the operator and physics closure results of Sections 6–7, and the meta-closure results of Sections 8–9. It introduces no new primitives, no new admissibility layers, and no new same-scope selector.

Claim type

This is an architecture appendix. Its task is not to prove a new theorem family beyond the main text, but to type every later downstream appendix in advance so that no appendix can drift into hidden-proof-source status.

Definition F.1 (Downstream exhaustion schema)**.** Fix a same-scope downstream problem posed inside the closure regime of Sections 1–9. A proposed downstream resolution belongs to exactly one of the following classes:

1. a boundary selector or admissibility-relevant parameterization;
2. a same-scope carrier, auxiliary authorizer, or privileged continuation relation;
3. an operator or mechanism upgrade that treats standing-bearing compatibility as if it were generated later in the interior;
4. an explicit scope change by added witness load, new ontology, new dynamics, or new target gating; or
5. a bookkeeping-only redescription that preserves the already fixed same-scope standing-bearing content.

Theorem F.2 (Downstream exhaustion schema)**.** *For any fixed-domain downstream problem posed on the original scope, every same-scope proposed resolution falls under exactly one class of Definition F.1. On the fixed scope, only bookkeeping-preserving or quotient-forced consequence survives; the first four classes are either already blocked by Sections 4–9 or else constitute scope change.*

Proof. If a proposal acts before the standing-bearing content of the problem is fixed, it is a boundary selector or parameterization and is blocked by the AMetric boundary and the no-second-classifier results of Sections 4–5. If it adds a same-scope authorizer beneath or alongside the unique

admissibility gate, it is a carrier or privileged continuation relation and is blocked by the no-carriers, no-repair, and quotient-closure results of Section 5. If it treats quotient-level compatibility as a later mechanism, it violates the operator-exhaustion and role-discipline results of Sections 6–7. If it adds new witness load, new ontology, new dynamics, or new target gating, it changes scope rather than resolving the original problem. If it does none of these, then it preserves the already fixed same-scope standing-bearing content and is bookkeeping-only. No sixth same-scope class remains. □

Corollary F.3 (Atlas typing rule)**.** *Every appendix that follows must therefore do three things explicitly: name its fixed-domain object, state which clause of the downstream exhaustion schema eliminates the prohibited readings, and then state only the bookkeeping-preserving or quotient-forced consequence that survives on the original scope.*

Proof. Without a fixed-domain object, the later appendix would leave scope drift uncontrolled. Without an explicit elimination map, the appendix could be misread as freeform interpretation rather than closure consequence. And without a surviving same-scope remainder, the appendix would either become a hidden constructive proposal or a merely rhetorical gloss. The atlas typing rule therefore follows immediately from Theorem F.2 and Proposition 10.16. □

No-backflow clause for the atlas. The atlas records what the fixed closure theorem does to downstream domains. It does not repair, extend, or motivate the theorem chain of Sections 1–9, and it may be removed in its entirety without altering any upstream theorem or proof burden.

G Measurement records, referential fixation, and theory-level confirmation

Consequence-facing only. This appendix is a compact consequence profile for *The Measurement Problem as a Problem of Reference*. It is conditional on the fixed substrate of Sections 5, 7, and 8 and does not function as a proof source for Sections 1–9.

Scope and dependency declaration

The fixed-domain object is the class of admissible reference assignments for measurement records in a physically non-selective theory used to underwrite truth-apt diachronic discourse about memory, anticipation, deliberation, and theory-level confirmation. The appendix consumes the standing regime of Section 5, the physics-closure and descriptive-role discipline of Section 7, and the same-domain route-exhaustion results of Section 8.

Claim type

This appendix is a compact consequence profile with theorem-facing summary. It preserves the quotient theorem, equivalence collapse, Everettian non-selection, fixation commutation, and selector-exhaustion payoffs of *The Measurement Problem as a Problem of Reference* without reproducing its entire theorem ladder.

Proposition G.1 (Measurement-reference consequence profile)**.** *On the fixed downstream substrate, admissible reference assignments for measurement records factor through a unique standing-bearing quotient; bare Everettian branching, entanglement, and decoherence do not by themselves collapse the admissible class to one quotient class; and, absent such collapse, theory-level confirmation, memory, anticipation, and deliberation are assignment-relative rather than theory-relative.*

Proof. This is the exact fixed-domain payoff of *The Measurement Problem as a Problem of Reference*. The source manuscript proves a measurement-reference quotient theorem, a global equivalence-collapse theorem for same-scope content spaces, an Everettian non-selection theorem, a fixation-commutation hinge for theory-level confirmation, a diachronic inheritance theorem for memory, anticipation, deliberation, and agency, and a selector-exhaustion theorem for interpretation packages. The present appendix imports only that downstream consequence package. By Theorem 10.5, none of these results functions as a premise of Sections 1–9.[13] □

Corollary G.2 (Measurement-record stabilization trichotomy)**.** *Any discourse-committing completion of a physically non-selective measurement theory must do exactly one of three things on the target discourse: fix singular reference dynamically, fix it by added ontological privileging, or restrict admissible assignments strongly enough that they collapse into one standing-equivalence class. Refusing the target truth-apt discourse is the only remaining non-committing option.*

Proof. This is the selector-exhaustion and interpretation-taxonomy consequence carried by *The Measurement Problem as a Problem of Reference* once the fixed substrate of the capstone is in place. Dynamic selection, ontic privileging, semantic restriction, and discourse deflation exhaust the same-scope options for fixing singular reference to measurement records.[13] □

Exact non-claims. This appendix does not derive the Born rule, alter the Hilbert-space formalism, or claim that all interpretations of quantum mechanics are equivalent. Its narrower claim is that, on the target explanatory practice, the measurement problem first appears as a problem of referential determinacy for measurement records and only derivatively as a problem of outcome production.[13]

H Entanglement as boundary-level compatibility

Consequence-facing only. This appendix is a compact consequence profile for *Quantum Entanglement as Boundary-Level Compatibility*. It consumes the AMetric boundary, standing conservation, and uniqueness-of-interior results of Sections 4, 5, 7, and 9 and introduces no new dynamics, ontology, or hidden variables.

Scope and dependency declaration

The fixed-domain object is a Bell scope consisting of Bell-violating, no-signaling bipartite quantum correlations. The appendix assumes the imported substrate already fixed in the main text: stable reference, lawful redescription, irreversible commitment, AMetric boundary discipline, bivalent admissibility, standing as reuse-stable admissibility, uniqueness of the admissible interior on a fixed scope, and anchor/tensor/skin role discipline.

Claim type

This appendix is a compact consequence profile. It uses *Quantum Entanglement as Boundary-Level Compatibility* only for the fixed-scope interpretive payoff that survives once the Bell data are read through the capstone's closure theorems.

Proposition H.1 (Bell-scope consequence profile)**.** *On the fixed Bell scope, an entangled preparation is a joint standing class carrying tensor-level compatibility rather than two independently sufficient local standing states; a local measurement is an admissible skin selection within that already admitted joint class rather than a tensor-changing act; and same-scope hidden-variable, signaling, mechanism, and repair readings are thereby eliminated.*

Proof. This is the exact structural result stated in *Quantum Entanglement as Boundary-Level Compatibility*. The source reclassifies entangled preparations as joint standing classes whose load-bearing content is quotient-level tensor compatibility; treats local measurement as admissible skin selection rather than tensor change; preserves no-signaling by denying any same-scope transfer, repair, or carrier route; and then eliminates Bell-local hidden-variable completion, superluminal signaling, tensor-to-mechanism upgrades, and same-scope repair readings on the original Bell scope. The capstone imports only that downstream classification, not a new proof source for its own AMetric, conservation, or uniqueness theorems.[22] □

Corollary H.2 (Unique same-scope Bell reading)**.** *Once the Bell data and the imported substrate are fixed, the boundary-level compatibility reading of entanglement is not one admissible option among several same-scope readings; it is the unique same-scope admissible reading up to lawful redescription.*

Proof. The source manuscript makes the uniqueness point explicit: once the admissible interior is unique on the fixed scope, the surviving Bell reading is not merely preferred but forced. Any

purported rival same-scope reading is therefore either skin-level paraphrase of the same tensor content, hidden load, or scope change.[22] □

Exact non-claims. This appendix does not derive the Hilbert-space formalism, the Born rule, or the Tsirelson bound from below. It constrains only the admissible interpretation of Bell-type operational data once those data are placed inside the fixed closure substrate.[22]

I Gauge / constraint dynamics, canonical GR, and selector-bearing quantum structure

Roadmap-only appendix. This appendix uses *Admissibility and the Structural Origin of Gauge and Constraint Dynamics* as a theorem roadmap with split downstream payoff. The rigidity theorem proved there is not imported as a premise of Sections 1–9. Only its downstream classificatory payoff is used inside 42.

Scope and dependency declaration

The source manuscript has two logically ordered layers: a theorem-bearing rigidity core for constrained Hamiltonian systems and a downstream classificatory payoff for canonical GR and selector-bearing quantum formalisms. In 42, only the second layer is imported. This appendix depends on the standing and role-discipline results of Sections 5–7 and on the no-backflow rule of Section 10.

Claim type

This appendix is a theorem roadmap with two consequence-facing subparts. It does not inline the rigidity proof and does not promote the source manuscript into a live proof home for the foundational arc.

Canonical general relativity as a downstream bookkeeping projection

Proposition I.1 (Canonical GR roadmap consequence)**.** *Relative to the rigidity theorem of* Admissibility and the Structural Origin of Gauge and Constraint Dynamics*, canonical general relativity is to be read as a presentation-level realization of admissibility-derived constrained Hamiltonian structure rather than as the primitive source of that structure.*

Proof. The source manuscript derives coisotropic standing, characteristic redescription, local constraint generators, first-class closure, and multiplier-type Hamiltonian structure from admissibility conditions and only afterward reads canonical GR through that structure. In that downstream reading, the ADM variables, constraint generators, and lapse/shift freedom become bookkeeping realizations of the admissible standing/redescription architecture rather than primitive explanatory

inputs. The present appendix imports only that classificatory payoff.[23] □

Collapse, Everett weights, and selector-bearing quantum formalisms

Proposition I.2 (Selector-bearing quantum consequence)**.** *Within the same admissibility framework, collapse noise, Everettian branch weights, typicality measures, and related selector-bearing quantum formalisms do not acquire standing merely by being useful. Unless the selector structure is reified into physical state or proven quotient-invariantly forced, it remains meta-selection, added load, or bookkeeping.*

Proof. This is the exact downstream diagnosis carried by the latter sections of *Admissibility and the Structural Origin of Gauge and Constraint Dynamics*. The source proves a collapse trilemma — pure gauge, meta-selector, or reified noise — and an Everett-weight trilemma — pure bookkeeping, meta-selection, or reification — and then strengthens the Everett point by exhibiting explicit non-invariance of branch decompositions under factorization and coarse-graining choices. In the 42 architecture, those results are not independent foundational theorems; they are the quantum-side payoff of treating selector-bearing structure as inadmissible on the original scope unless physically reified or quotient-forced.[23] □

Exact non-claims. This appendix does not reproduce the rigidity derivation, does not claim to derive canonical GR from the capstone itself, and does not prove that all collapse or Everettian completions are impossible simpliciter. Its narrower claim is that, on the original scope, selector-bearing structures are bookkeeping or added load unless reified into state or forced as quotient invariants.[23]

J Beginning of the admissible interior

Consequence-facing only. This appendix is a compact consequence profile for *Beginning of the Admissible Interior*. It depends on the AMetric boundary and standing-conservation package already fixed in Sections 4, 5, and 7 and does not introduce an origin narrative, pre-metric dynamics, or causal emergence.

Fixed-domain object

The fixed-domain object is the first licensed metric regime inside admissibility: the minimal standing condition under which metric surrogates become valid within an already admitted interior.

Proposition J.1 (Beginning-side consequence profile)**.** *If metric description is licensed at all inside an admissible interior, then its first licensed regime is governed by an L-metric boundary, admits no proto-metric partial onset, admits no local metric islands, requires a globally closing metric circuit, and sits at maximal tensor stress so that the surviving admissible redescriptions appear non-classical when written in interior metric language.*

Proof. This is the exact structural payload of *Beginning of the Admissible Interior*. The source defines the L-metric boundary as the minimal admissible standing condition for metricity, proves that metric surrogates are co-licensed rather than sequentially switched on, excludes local patchwork onset, proves an initial closure requirement for metricity, and then identifies maximal tensor stress at that first licensed metric regime. The paper's final payoff is that, under maximal tensor stress, the admissible successor structure appears as discretization, global consistency, and barrier-penetration behavior when translated into interior metric language. The present appendix imports only that beginning-side classification.[24] □

Exact non-claims. This appendix does not describe how admissibility arises, how the AMetric boundary is established, or why an admissible interior exists. "Beginning" is used strictly as a licensing notion and not as a temporal origin, causal process, or pre-metric dynamical regime.[24]

K End of the admissible interior

Consequence-facing only. This appendix is a compact consequence profile for *End of the Admissible Interior*. It depends on the standing and physics-closure package of Sections 5 and 7 and introduces no new modal, thermodynamic, or cosmological primitives.

Fixed-domain object

The fixed-domain object is terminal exhaustion of admissible redescription under fixed admissibility constraints.

Proposition K.1 (End-side consequence profile)**.** *Admissibility constraints are necessary but not sufficient for the persistence of an admissible interior. An admissible interior exists if and only if at least one standing-preserving redescription remains available; exhaustion is the absence of any admissible successor configuration under fixed constraints; and such exhaustion is neither temporal process, thermodynamic depletion, nor evolving-constraint dynamics.*

Proof. This is the exact classification established in *End of the Admissible Interior*. The source defines the admissible interior in terms of the availability of standing-preserving admissible successors, formalizes exhaustion as the empty admissible successor set under fixed constraints, distinguishes realizability failure from failure of admissibility itself, and then reclassifies heat death and related cosmological narratives as terminal metric regimes rather than admissible exhaustion. The present appendix imports only that terminal consequence profile.[25] □

Corollary K.2 (No same-scope successor interior)**.** *Once admissible exhaustion obtains on the original scope, no same-scope successor interior and no post-interior physics narrative remains licensed.*

Proof. The source manuscript states this as a direct consequence of exhaustion: fixed admissibility constraints remain intact, but no standing-preserving continuation survives. Any purported successor interior would therefore require new load, new scope, or illicit modal necessity.[25] □

Exact non-claims. This appendix does not describe a dynamical end-of-time process, vacuum decay, or thermodynamic consumption of admissibility. It classifies only the structurally unremarkable absence of any further admissible configuration under fixed constraints.[25]

L Matter–antimatter asymmetry and admissible baryogenesis

Consequence-facing only. This appendix is a compact consequence profile for *Matter–Antimatter Asymmetry as a Standing Constraint: Baryogenesis Under Admissible Cosmic Evolution*. It depends on the irreversible-commitment architecture of Section 3, the standing-conservation results of Section 5, and the descriptive closure of Section 7. It introduces no specific baryogenesis mechanism and no modification of Standard Model dynamics.

Fixed-domain object

The fixed-domain object is persistent matter–antimatter asymmetry under irreversible cosmic evolution, with baryon identity treated as a bookkeeping label that nonetheless bears standing across cosmic history.

Proposition L.1 (Baryogenesis consequence profile)**.** *If cosmic evolution is irreversible in the admissibility sense and persistent material structure is to preserve standing across that history, then exact global matter–antimatter symmetry is inadmissible as a standing-bearing bookkeeping regime. Some mechanism establishing a net persistent matter asymmetry is therefore structurally required, and baryogenesis is the standard model-building name for that admissible resolution.*

Proof. This is the structural conclusion of *Matter–Antimatter Asymmetry as a Standing Constraint: Baryogenesis Under Admissible Cosmic Evolution*. The source treats baryon identity as an interaction-defined bookkeeping label, shows that exact global matter–antimatter symmetry would fail to preserve standing once irreversible loss of retrievable correlation structure occurs, and then reclassifies baryogenesis as the admissible resolution of persistent identity bookkeeping rather than as a merely optional dynamical episode. The present appendix imports only that consequence-level necessity and none of the source's microphysical non-claims.[26] □

Exact non-claims. This appendix does not select a specific baryogenesis mechanism, derive the baryon-to-photon ratio, require particular CP-violating parameters, or modify Standard Model dynamics. The point is structural: asymmetry is required for standing-preserving identity bookkeeping under irreversible evolution.[26]

M Black holes without interior facts

Consequence-facing only. This appendix is a compact consequence profile for *Black Holes Without Interior Facts*. It depends on the fixed-domain reference and fixation discipline already carried by Sections 5, 7, and 8 and introduces no new quantum-gravity dynamics.

Fixed-domain object

The fixed-domain object is the black-hole horizon as a fixation boundary for truth-apt physical claims.

Proposition M.1 (Black-hole fixation consequence profile)**.** *On the fixed closure substrate, event horizons are to be read as fixation boundaries rather than as access points to a globally referencible interior fact-domain. Once that reclassification is enforced, the information paradox, firewall necessity, interior microstate ontology, and strong holographic upgrades are dissolved as category errors on the original scope, while exterior general relativity, Hawking radiation, and boundary thermodynamics remain unchanged.*

Proof. This is the exact diagnostic payoff of *Black Holes Without Interior Facts*. The source identifies horizons as fixation boundaries, distinguishes fixation failure from ordinary epistemic ignorance, blocks diachronic identity claims for non-fixable interior referents, and then shows that global bookkeeping demands across the horizon presuppose precisely the interior facthood that the fixation analysis rules out. The present appendix imports only that paradox-dissolution claim. Exterior black-hole physics is preserved because the source explicitly leaves locally anchored exterior observations, boundary thermodynamics, and Hawking-radiation bookkeeping intact.[27] □

Exact non-claims. This appendix does not modify general relativity, quantum field theory, or any observationally anchored exterior black-hole physics. It dissolves only the inadmissible demand that a non-fixable black-hole interior nevertheless constitute a domain of truth-apt globally referencible facts.[27]

N Standing localization, life, intelligence, and technology

Consequence-facing only. This appendix is a compact consequence profile for *Standing Localization and the Structural Inevitability of Life*. It depends on the standing-conservation and physics-closure results of Sections 5 and 7 and introduces no biochemical, probabilistic, anthropic, or sociological primitives.

Fixed-domain object

The fixed-domain object is a non-degenerate admissible physical regime that permits persistent bounded subsystems with nonzero repair capacity.

Proposition N.1 (Standing-localization consequence profile)**.** *In any such regime, standing cannot remain uninstantiated and must localize within bounded repairable subsystems; life is the minimal physical instantiation of that localization; intelligence is a constrained optimization of standing efficiency rather than an inevitable outcome; and sustained technological standing is structurally fragile because nonlocal amplification outruns repair capacity.*

Proof. This is the exact consequence package stated in *Standing Localization and the Structural Inevitability of Life*. The source proves a standing-localization theorem for non-degenerate admissible physical regimes, identifies life as the minimal localized standing-preserving structure, treats intelligence as contingent optimization rather than necessity, and derives structural instability for long-duration technological standing from the mismatch between nonlocal coupling and local repair. The present appendix imports only that conditional consequence profile and none of the source's explicitly disclaimed biochemical, probabilistic, or anthropic claims.[28] □

Exact non-claims. This appendix does not predict molecular substrates, estimate frequencies, appeal to anthropic selection, or offer a sociological narrative of civilizations. It states only the structural roles of life, intelligence, and technology once admissibility and standing conservation are already in force.[28]

O Mind as a structurally rigid interface of admissible evaluation

Roadmap-only appendix. This appendix uses *Mind as a Structurally Rigid Interface of Admissible Evaluation* as a strong consequence roadmap. It does not import a second theorem spine into 42 and does not function as a proof source for Sections 1–9.

Scope and dependency declaration

The fixed-domain object is an admissible evaluation regime in which projection descends through quotient structure, standing has grounded origin conditions, symmetric branching blocks singular self-location without added selector structure, and anchor/tensor/skin stratification is already available. This appendix depends on Sections 5, 8, and the ATS-derived role discipline carried in the main text.

Claim type

This appendix is a theorem roadmap with consequence-facing summary. It records the exact downstream payoff of *Mind as a Structurally Rigid Interface of Admissible Evaluation* without

reproducing its full proof-granularity ladder.

Proposition O.1 (Mind roadmap consequence)**.** *Relative to the capstone substrate, mind is to be read as a structurally rigid interface of admissible evaluation: projection descends through quotient structure, persistence requires non-derivative standing bearers, symmetric branching blocks selector-free singular self-location, identity does not duplicate without invariant asymmetry or standing failure, and anchor/tensor/skin stratification is forced rather than optional vocabulary.*

Proof. This is the integrated payoff of *Mind as a Structurally Rigid Interface of Admissible Evaluation*. The source develops quotient descent, standing-origin necessity, symmetric self-location impossibility, escape exhaustion, non-duplication of identity, anchor–tensor–skin stratification, and structural rigidity from one admissible evaluation regime. For 42, the important point is not to restage that proof ladder, but to register its consequence-facing reading: once the closure substrate is fixed, mind is not a free-floating extra ontology but a rigid interface condition on admissible evaluation. By Theorem 10.5, the roadmap status preserves that payoff without importing a second foundational proof spine.[29] □

Exact non-claims. This appendix does not claim to derive consciousness from below, identify a neural substrate, or replace the capstone's own foundational arc. Its narrower role is to record the structural mind theorem as a downstream roadmap whose proofs remain in the source manuscript.[29]

P Meta-necessity, temporal asymmetry, and information non-primitivity

Consequence-facing only. This appendix is a cross-domain consequence profile for *Meta-Necessity of Admissibility: A Proof-of-Proof for the Maley Corpus*. It depends on the kernel, irreversibility, and same-domain closure results of Sections 2–8 and does not replace the capstone's own proof of the kernel floor.

Fixed-domain object

The fixed-domain object is any non-degenerate irreversible decision system with unrestricted composition.

Proposition P.1 (Meta-necessity consequence profile)**.** *Any such system must enforce an admissibility invariant sufficient to preserve standing under composition or else fail to define irreversibility itself. On that basis, temporal asymmetry is reclassified as an interior bookkeeping artifact of irreversible fixation, and informational quantities are reclassified as post-admissible bookkeeping rather than ontic primitives.*

Proof. This is the exact eliminative and meta-level payoff of *Meta-Necessity of Admissibility: A Proof-of-Proof for the Maley Corpus*. The source proves that irreversibility without a standing-

preserving invariant is undefined under unrestricted composition, then derives meta-necessity of admissibility, treats operational gate sequences as enforcement surfaces rather than architectural preferences, and adds the arrow-of-time and information corollaries: temporal asymmetry is an artifact of irreversible fixation inside the admissible interior, and informational quantities presuppose fixation, reference stability, and encoding rather than grounding them. The present appendix imports only that corollary package and does not replace the capstone's own derivation of the kernel floor.[30] □

Exact non-claims. This appendix does not assert the corpus as semantically true, does not introduce a new primitive information ontology, and does not ground an ontic arrow of time. It classifies only what follows if irreversibility is to remain well-defined under unrestricted composition.[30]

Q Meta–mystery and dark-sector elimination

Consequence-facing only. This appendix is an eliminative constraint profile for *The Meta–Mystery Theorem and the Dark Sector Eliminative Principle*. It depends on the descriptive-closure and reference-stability results of Sections 7–9 together with the meta-necessity appendix immediately preceding it.

Fixed-domain object

The fixed-domain object is any persistent "mystery" generated by a framework whose alleged core referent fails invariance under standing-preserving redescription, together with any dark-sector posit introduced to stabilize the resulting bookkeeping imbalance.

Proposition Q.1 (Meta–mystery consequence profile)**.** *Persistent deep mysteries are a diagnostic of upstream admissibility failure: if a framework repeatedly generates explanatory pressure around a putative entity whose referent is not invariant under standing-preserving redescription, then the mystery is not ontological depth but bookkeeping instability. Dark-sector posits introduced to absorb the residuals of such instability are therefore non-ontic unless they independently acquire redescription-invariant reference.*

Proof. This is the exact eliminative principle stated in *The Meta–Mystery Theorem and the Dark Sector Eliminative Principle*. The source proves a meta–mystery theorem according to which persistent mysteries signal non-invariant core referents, then defines dark-sector posits as auxiliary structures introduced to absorb residuals, stabilize parameters, patch closure, or launder scope. Its theorem and corollaries show that patch cascades and dark-sector repairs are bookkeeping artifacts rather than ontology unless a redescription-invariant referent is fixed independently. The present appendix imports only that consequence-facing diagnostic.[31] □

Exact non-claims. This appendix does not deny empirical adequacy of locally useful representational regimes, and it does not claim that every difficult scientific problem is ill-posed. Its narrower claim is that persistent mystery plus patch-cascade behavior is evidence of non-invariant reference, not of hidden ontological depth.[31]

R References

References

[1] Amos Jay Maley, *Unified Capstone Manuscript: Unified Capstone Manuscript: Construction Skeleton and Drafting Blueprint*, version 1.0, 13 April 2026.

[2] Amos Jay Maley, *Foundational Closure of Admissibility and Standing: Non-Derivability, Minimality, and the Kernel of Non-Degenerate Reasoning*, companion manuscript, 2026.

[3] Amos Jay Maley, *The AMetric Boundary: Necessity, Non-Parameterization, and Closure by Exhaustion*, companion manuscript, 2026.

[4] Amos Jay Maley, *Necessity of Admissibility in Non-Degenerate Compositional Systems: AMetric Boundary, Bivalence, and the Unique Admissible Interior*, companion manuscript, 2026.

[5] Amos Jay Maley, *The Mathematics of Coherent Reasoning: A Bivalent Trajectory Theory*, companion manuscript, 2026.

[6] Amos Jay Maley, *Unus Solus Possibilis Est: Corpus Closure and Uniqueness of the Admissible Interior*, companion manuscript, 2026.

[7] Amos Jay Maley, *Anchor / Tensor / Skin: A Role Decomposition for Admissible Description*, companion manuscript, 2026.

[8] Amos Jay Maley, *Closure by Exhaustion for Same-Scope Operators under Admissibility*, manuscript, March 2026.

[9] Amos Jay Maley, *Fixed-Base Exhaustion for Internal Operators*, companion manuscript, 2026.

[10] Amos Jay Maley, *Clausura Ex Necessitate Structurae*, companion manuscript, 2026.

[11] Amos Jay Maley, *Eadem Natura, Duo Spectra: On the Post-Closure Relation Between Mathematics and Physics*, companion manuscript, 2026.

[12] Amos Jay Maley, *Structura Ex Necessitate Standardis Modelis*, companion manuscript, 2026.

[13] Amos Jay Maley, *The Measurement Problem as a Problem of Reference*, manuscript, 2026.

[14] Amos Jay Maley, *Non-Instantiability of Gödel Incompleteness in the Unique Admissible Interior*, audit edition manuscript, 2026.

[15] Amos Jay Maley, *Carrier Universality and Global Smoothness for the Periodic 3D Navier–Stokes Equations: A Closure-Induced Impossibility Proof of Fefferman's Periodic Statement (B)*, manuscript, 2026.

[16] Amos Jay Maley, *Vacuum-sector Mass Gap for the Local Gauge-Invariant Sharp-Local Yang–Mills Net*, manuscript, 2026.

[17] Amos Jay Maley, *Companion I: Ultraviolet Gate and Route 1 Mass-Gap Chain*, companion manuscript, 2026.

[18] Amos Jay Maley, *Companion II: Lane A Sharp-Local Construction*, companion manuscript, 2026.

[19] Amos Jay Maley, *Companion III: Reconstruction, Non-Triviality, and Faithful-Wilson Universality*, companion manuscript, 2026.

[20] Amos Jay Maley, *Standing–Admissibility Equivalence: Meta-Necessity Closure*, manuscript, 2026.

[21] Amos Jay Maley, *The Exhaustion Theorem: Standing, Admissibility, and the Determination of Consequence under Typed Boundary Discipline*, manuscript, 2026.

[22] Amos Jay Maley, *Quantum Entanglement as Boundary-Level Compatibility: A Non-Mechanistic Derivation from Admissibility, Standing Conservation, and the AMetric Boundary*, manuscript, April 2026.

[23] Amos Jay Maley, *Admissibility and the Structural Origin of Gauge and Constraint Dynamics*, manuscript, 8 April 2026.

[24] Amos Jay Maley, *Beginning of the Admissible Interior*, companion manuscript, February 2026.

[25] Amos Jay Maley, *End of the Admissible Interior*, companion manuscript, February 2026.

[26] Amos Jay Maley, *Matter–Antimatter Asymmetry as a Standing Constraint: Baryogenesis Under Admissible Cosmic Evolution*, manuscript, 2026.

[27] Amos Jay Maley, *Black Holes Without Interior Facts: Fixation Boundaries and the Dissolution of the Information Paradox*, manuscript, 2026.

[28] Amos Jay Maley, *Standing Localization and the Structural Inevitability of Life*, manuscript, 26 January 2026.

[29] Amos Jay Maley, *Mind as a Structurally Rigid Interface of Admissible Evaluation: Expanded Proof-Granularity Version*, manuscript, 2026.

[30] Amos Jay Maley, *Meta-Necessity of Admissibility: A Proof-of-Proof for the Maley Corpus*, manuscript, 6 January 2026.

[31] Amos Jay Maley, *The Meta–Mystery Theorem and the Dark Sector Eliminative Principle*, manuscript, 2026.

[32] Kurt Gödel, Über formal unentscheidbare Sätze der *Principia Mathematica* und verwandter Systeme I, *Monatshefte für Mathematik und Physik* **38** (1931), 173–198.

[33] Alfred Tarski, Der Wahrheitsbegriff in den formalisierten Sprachen, *Studia Philosophica* **1** (1936), 261–405.